# Traditional versus Shadow Banking in a Monetary Economy*

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**Abstract**

Why do some depositors fund traditional banks while others fund shadow banks? This paper develops a monetary model of banking to provide an information-based explanation for the coexistence of traditional and shadow banks. Depositors can either monitor banks ex ante, at a cost that differs across households, or rely on costly ex post verification. Shadow banks attract depositors who monitor, while traditional banks attract depositors who rely on verification. The key mechanism links money demand to monitoring incentives: monitoring depositors earn a higher deposit rate and hold more money, which affects aggregate liquidity and the size of the shadow banking system, which in turn affects the incentive to monitor. In symmetric pure-strategy equilibria, this feedback rules out shadow banking when monitoring costs are identical. With heterogeneous monitoring costs, monitoring decisions are strategic complements when both types of banks are active and strategic substitutes when only shadow banks are active. A monetary tightening expands the shadow banking system at low nominal interest rates, but contracts it at high nominal interest rates. Because households ignore their effect on aggregate liquidity and on verification costs, the shadow banking system is too large relative to its efficient size at low nominal interest rates and too small at high interest rates.

*Key words*: shadow banks, traditional banks, asymmetric information, costly state verification, monitoring, monetary policy, household heterogeneity

*JEL* Classification: E41, E44, E52, G21, G23.

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# 1 Introduction

The global financial crisis of 2007-2009 underscored the importance of shadow banks in the financial system, and has since attracted sustained attention from researchers and policymakers (Gorton and Metrick, 2012; Pozsar et al., 2013). Shadow banks remain a major component of the global financial system, accounting for roughly half of global financial assets (Financial Stability Board, 2025). Although traditional and shadow banks perform similar intermediation functions and compete for the same borrowers (Jiang et al., 2020), they rely on different sources of funding. Traditional banks rely primarily on insured deposits from small and relatively uninformed households and firms, whereas shadow banks rely on short-term wholesale funding from large and informed depositors, such as institutional investors, corporate treasurers, and even traditional banks (Pozsar et al., 2010; Huang and Ratnovski, 2011). This suggests that each type of bank attracts a distinct type of depositor, and that heterogeneity among depositors may help explain why the two coexist. Understanding this coexistence matters not only for understanding the financial system, but also for understanding how the composition of the banking system interacts with monetary policy.

This composition is not fixed, and it responds to monetary policy. Evidence from the United States (Nelson et al., 2018) and China (Chen et al., 2018) suggests that contractionary monetary policy shocks expand the shadow banking system. Drechsler et al. (2017) show that higher policy rates lead to deposit outflows from traditional banks, while Xiao (2020) documents that depositors shift from traditional banks to money market funds. Drechsler et al. (2017) attribute this reallocation to traditional banks' market power, while Xiao (2020) links it to differences in depositors' yield sensitivity. However, the response of the shadow banking system is not monotone. For example, Hodula and Libich (2023) document that monetary easing can also lead to growth of the shadow banking system as depositors search for yield. These explanations leave open a more fundamental question: why do different depositors choose different types of intermediaries in the first place?

Existing theories explain the existence of shadow banking mainly from the asset side or through regulatory arbitrage: shadow banks arise to avoid costly regulation (Acharya et al., 2013; Plantin, 2015; Lyonnet and Chretien, 2026) or to produce relatively safe assets (Gennaioli et al., 2013; Moreira and Savov, 2017). These theories do not explicitly account for differences across depositors or for why the size of the shadow banking system responds to monetary policy. To answer these questions, I develop a complementary theory in which the coexistence of traditional and shadow banks arises from differences in depositors' ability to acquire information. Depositors differ in the cost of monitoring a bank. Financially sophisticated depositors, for whom monitoring is relatively cheap, choose shadow banks, while unsophisticated depositors, for whom monitoring is costly, choose traditional banks. Monetary policy affects the incentive to monitor, and thereby the size of the shadow banking system. The theory thus provides a common explanation for the endogenous segmentation of depositors and its interaction with monetary policy.

I formalize this idea in a monetary model of banking with investment, aggregate risk, and asymmetric information. Banks collect deposits and lend to households that receive a risky investment opportunity but face liquidity constraints. The return on bank loans is subject to aggregate risk, and depositors cannot observe its realization. Depositors can mitigate this asymmetric information friction with one of two technologies: they can ex ante monitor, at a utility cost, or rely on ex post verification, which destroys part of the bank's loan income. I associate the first technology with shadow banking and the second with traditional banking. The underlying idea is that shadow bank depositors are informed and monitor continuously, whereas traditional bank depositors intervene only when distress arises. Accordingly, shadow bank deposits are conditioned on the bank's realized return, whereas traditional bank deposits are standard debt contracts with costly verification if banks default.

The central mechanism operates through the interaction between monitoring and money demand. Households decide whether to monitor before they learn whether they will have an investment opportunity, and at the same time as they choose their money holdings. By monitoring, households avoid the consequences of the asymmetric information friction on their deposit decisions and earn the expected loan rate on their deposits, rather than a deposit return reduced by the expected cost of verification. Monitoring households therefore have a stronger incentive to carry money and make larger deposits. As a consequence, in equilibrium, monitoring entrepreneurs no longer borrow and instead self-finance their investment. Information acquisition through monitoring thus changes not only the return on deposits, but also how much liquidity households carry. As more households monitor, aggregate liquidity and the size of the shadow banking system increase. Aggregate liquidity, in turn, determines the private return to monitoring. The composition of the banking system and aggregate liquidity provision are therefore jointly determined. Each household, however, takes aggregate liquidity as given and does not internalize how its own money and monitoring decisions affect aggregate liquidity. This externality makes monitoring decisions strategic.

Throughout the paper, I focus on symmetric pure-strategy equilibria, in which households with the same monitoring cost make the same decisions: they make the same monitoring choice, carry the same money balances, and make the same deposit and borrowing decisions. In a mixed banking equilibrium, where both types of banks are active, monitoring decisions are strategic complements. When more households monitor, shadow banks fund a larger share of loans, and unsophisticated households carry less money and borrow more as entrepreneurs. Relying on traditional banks then becomes more costly, and the private gain from monitoring rises. In a pure shadow banking equilibrium, unsophisticated households no longer carry money and this margin disappears. Additional monitoring households only add to the supply of loans, which lowers the loan rate and reduces the difference between the investment of sophisticated and that of unsophisticated entrepreneurs. This reduces the private gain of monitoring, making monitoring decisions strategic substitutes. The complementarity can generate multiple equilibria (Hellwig and Veldkamp, 2009): a single mixed banking equilibrium, a single pure shadow banking equilibrium, or two mixed banking equilibria together

with a single pure shadow banking equilibrium. An equilibrium without monitoring households and without shadow banks always exists, because a single household gains nothing from monitoring when there are no shadow banks. Which banking system prevails is therefore partly a matter of coordination, and monetary policy shapes which equilibria exist.

The same externality explains why heterogeneity in monitoring costs matters. When nobody monitors, there are no shadow banks, and a household gains nothing from monitoring. When all households monitor, liquidity is abundant and the private value of monitoring disappears. Hence, with a common monitoring cost, no symmetric pure-strategy equilibrium with shadow banks exists. The existence of shadow banking requires heterogeneity in monitoring costs. Heterogeneity also determines who monitors, so the model endogenously segments depositors between the banking systems. Households with low monitoring costs monitor and deposit in shadow banks, while those with high costs rely on traditional banks. Depending on the verification cost and the nominal interest rate, the economy can also support a pure shadow banking equilibrium, in which traditional banks are inactive.

Monetary policy determines not only the size of the shadow banking system, but also whether the size is efficient and whether shadow banking raises expected welfare relative to an economy with only traditional banks. Monetary policy enters through the cost of holding money. Households need money to finance investment, but holding money is costly due to inflation. The nominal interest rate therefore determines the opportunity cost of money and thus how much liquidity households carry. Near the Friedman rule, holding money is cheap, households carry enough liquidity to finance large investment, and traditional banks rarely default. The private gain from monitoring then mainly arises from cheaper financing: sophisticated entrepreneurs self-finance their investment, whereas unsophisticated entrepreneurs borrow at the loan rate. As the nominal interest rate rises, this financing advantage grows, and the share of monitoring households, and hence the size of the shadow banking system, increases. This is in line with evidence that monetary tightening expands the shadow banking system (Nelson et al., 2018; Chen et al., 2018; Xiao, 2020). For high interest rate, an increase in the nominal interest rate only reduces the money balances on which the benefits of monitoring are earned, and the share of sophisticated households reduces with the interest rate. The model thus predicts a non-monotone response of the shadow banking system to monetary policy.

Whether the size of the shadow banking system is optimal depends on monetary policy. For each nominal interest rate, the efficient size of the shadow banking system is the share of monitoring households that maximizes expected welfare, which is reached when the private gain of monitoring equals the social gain. Because households do not internalize the effect of their monitoring decisions on aggregate liquidity and on the incentives of other households to monitor, the private value of monitoring differs from the social one. Two wedges drive this difference. The first, a financing wedge, arises because the private gain from monitoring reflects a household's own financing cost, whereas the social gain reflects the real cost of capital. The second, a verification

wedge, arises because households do not internalize that, by leaving the traditional banking system, they reduce the expected cost of verification. Near the Friedman rule, traditional banks rarely default and liquidity is abundant. The verification wedge is then negligible, while the financing wedge makes households overvalue monitoring: the lower cost of financing is predominantly a transfer through the loan market. The shadow banking system is too large. At higher nominal interest rates, the financing wedge changes sign, and both wedges make households undervalue monitoring, so the shadow banking system is too small.

Next, I analyze whether shadow banking raises expected welfare at all. For a given nominal interest rate, I compare welfare in an equilibrium with both traditional and shadow banks to an equilibrium with only traditional banks. When nominal rates are low, the opportunity cost of holding money is small and liquidity is relatively abundant. The additional liquidity generated by shadow banks has limited social value, while monitoring remains costly. The equilibrium share of monitoring households is then inefficiently high, and an economy with both types of banks generates lower expected welfare than an economy with only traditional banks. At higher nominal interest rates, liquidity becomes more valuable and the additional liquidity through the shadow banking system outweighs the cost of monitoring, generating a welfare gain. For intermediate rates, the shadow banking system can be too large and still raise welfare, because the monitoring households are those with the lowest monitoring costs.

Finally, the mechanism does not depend on traditional bank deposits being uninsured. The verification technology is equivalent to an actuarially fair, risk-adjusted deposit insurance scheme in which the insurer has a limited ability to recover the return on banks' loans. Deposit insurance therefore removes neither the cost of bank failures nor the incentive of sophisticated households to monitor. Other insurance schemes, such as a subsidized or a flat-rate premium, would raise the return on traditional bank deposits and lower the private gain from monitoring, but would leave the main mechanism linking monitoring and aggregate liquidity unchanged.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 presents the model, and Section 4 characterizes the equilibria. Finally, before concluding, Section 5 analyzes expected welfare and the efficient size of the shadow banking system.

## 2 Related Literature

First, this paper contributes to the literature that explains the coexistence of traditional and shadow banks. Following the financial crisis, Pozsar et al. (2010) provide one of the first comprehensive descriptions of the shadow banking system and define it as credit intermediation outside the regulatory scope. Most theories link the emergence of shadow banking to regulatory arbitrage. Farhi and Tirole (2021) and Plantin (2015) show how regulatory constraints on traditional banks can result in an increase in shadow banking activities. Begenau and Landvoigt (2022) quantify the trade-off between regulation and the size of the shadow banking system in a general equilibrium model. Huang

(2018) models shadow banks as off-balance-sheet vehicles backed by implicit guarantees, whose leverage is limited by enforcement problems. Ordonez (2018) focuses on an asymmetric information friction between regulators and banks that makes regulation inefficient, while Lyonnet and Chretien (2026) explain coexistence through a trade-off between regulatory costs and access to deposit insurance. A second strand of literature does not rely on differences in regulation. For example, Gennaioli et al. (2013) show how demand for safe assets drives securitization, while Moreira and Savov (2017) model shadow banks as producers of liquidity that is fragile in times of uncertainty. Hanson et al. (2015) link the existence of both banking systems to differences in the funding structure that are intertwined with different strategies: traditional banks rely on stable deposit funding and follow a hold-to-maturity strategy, while shadow banks are funded with runnable short-term debt and invest in highly liquid assets. In these papers, the distinction between the two banking systems arises from regulation, beliefs, or the asset side of the balance sheet. I contribute to this literature by explaining coexistence through differences between depositors: shadow banks attract depositors with low monitoring costs, while traditional banks attract depositors with high monitoring costs. I show that, in symmetric pure-strategy equilibria, heterogeneity in monitoring costs is necessary for shadow banks to emerge.

Second, the paper contributes to the literature on asymmetric information between banks and their depositors. It is well established that banks are inherently opaque institutions. Dang et al. (2017) show that banks may optimally be opaque, so that their liabilities are information-insensitive and depositors do not need to acquire additional information. Other papers argue that depositors may monitor their bank. For example, Calomiris and Kahn (1991) show that demandable debt gives depositors an incentive to monitor banks and restrict bank risk taking. Huang and Ratnovski (2011) analyze this mechanism in a model with wholesale funding and public signals. They show that the presence of noisy public signals can hinder the incentive of wholesale investors to monitor. Ari et al. (2017) show that when depositors can discipline traditional banks, shadow banking emerges endogenously through runs and liquidation. In contrast, in my model depositors choose ex ante whether to become informed, and this choice determines which type of bank they use. Depositors who monitor acquire information and deposit in shadow banks. This is consistent with evidence from money market funds. Gallagher et al. (2020) show that investors selectively acquire information about the money market fund. The decision to monitor is thus closely related to strategic interaction in information acquisition, studied, for example, in Grossman and Stiglitz (1980) and Hellwig and Veldkamp (2009). In Grossman and Stiglitz (1980), information acquisition decisions are strategic substitutes, while in Hellwig and Veldkamp (2009) they are strategic complements. In my monetary model, monitoring decisions interact and can be either strategic complements or strategic substitutes depending on aggregate liquidity and monetary policy. To model unsophisticated depositors, I use the costly state verification framework of Townsend (1979) and Gale and Hellwig (1985), introduced into macroeconomic models by Williamson (1986, 1987) and popularized by Bernanke et al. (1999). This literature mostly studies the relationship between borrowing

firms and lenders. I instead apply the framework to the relationship between depositors and banks.

Third, the model builds on the monetary model by Lagos and Wright (2005); see e.g. Williamson and Wright (2010) and Nosal and Rocheteau (2011) for surveys. In particular, it extends the banking model of Berentsen et al. (2007), in which banks arise to reallocate idle money balances to agents with consumption opportunities. Other banking models in this framework include Williamson (2012) and Gu et al. (2013). Compared with those models, I introduce aggregate risk, asymmetric information, and two distinct forms of banking that affect the reallocation of liquidity. As in Berentsen et al. (2014) and Park (2020), a pecuniary externality is central to my results: households do not take into account how their decisions affect aggregate liquidity provision. In Berentsen et al. (2014) and Park (2020), this externality works through asset prices, while in my model it works through the price of money.

Finally, the paper relates to the literature on monetary policy and shadow banking. Xiao (2020) shows that shadow bank deposits grow when the central bank raises interest rates because shadow bank depositors are more yield-sensitive than traditional bank depositors, who value liquidity services. Similarly, Nelson et al. (2018) and Chen et al. (2018) find that monetary tightening expands shadow banking in the United States and China, respectively. Moreover, Hodula and Libich (2023) find that, in the euro area, both monetary easing and tightening can be associated with growth of the shadow banking system. I abstract from the liquidity services of deposits and instead show that monetary policy affects the composition of the banking system through the cost of holding money and aggregate liquidity provision. In addition, I show that whether the response of the shadow banking system is efficient depends on the nominal interest rate.

# 3 Model

## 3.1 Environment

Time is discrete and infinite, with periods indexed by $t = 0, 1, 2, \ldots$ Each period consists of two consecutive subperiods: a decentralized market (DM), which encompasses a competitive banking market and a competitive capital market, followed by a centralized market (CM), which is a frictionless consumption goods market. There is a single perishable consumption good, $x$, and a single investment good, capital $k$, which depreciates fully each period. The consumption good is produced and consumed during the CM, while the capital good is traded and invested during the DM. The economy is populated by two types of agents: a unit mass of households and a representative capital producer.[1] All agents discount across, but not within, periods, with a common discount factor $\beta$.

Both households and the capital producer are risk-neutral and consume the con-

[1]The linear production function of the capital producer and the competitive market make the mass of producers irrelevant (see, e.g., Rocheteau et al. (2018) for a similar set-up).

sumption good. The consumption good is produced by households at unit cost or through their investment projects. Each household is characterized by a permanent monitoring cost $\gamma$, which is drawn independently across households at $t = 0$ from a distribution $G(\gamma)$ on $(0, \Gamma)$. In each CM, a household decides whether to monitor or not. By incurring this utility cost, a household can observe the success of risky investment projects and access the shadow banking system, as discussed in detail below. Heterogeneity in monitoring costs captures differences in households' ability to acquire information and understand financial information. Because monitoring is a choice, whether a households is informed is endogenous. I refer to households who monitor as financially sophisticated and to those who do not as unsophisticated.

At the beginning of each period, both sophisticated and unsophisticated households face an idiosyncratic shock independent of their monitoring choice. With probability $\sigma_1 \geq \frac{1}{2}$, a household has access to an investment opportunity and becomes an entrepreneur, with probability $\sigma_2$ a household has access to a banking technology and becomes a banker, and with the remaining probability $\sigma_3 = 1 - \sigma_1 - \sigma_2$ a household has no access to either technology and is idle.

During the DM, entrepreneurs have access to a risky investment opportunity, which requires the capital good $k$ as input and has a random return in the subsequent CM: with probability $\delta$ the project returns $f(k)$ units of the consumption good, where $f(k)$ is homogeneous of degree $\alpha \in (0, 1)$ and satisfies the Inada conditions, while with probability $1 - \delta$ the project is unsuccessful and returns nothing. The probability of success is itself random and is i.i.d. distributed on $(0, 1)$ over time according to a twice continuously differentiable distribution $H(\delta)$, with density $h(\delta)$. The law of large numbers then implies that the realized fraction of successful projects across entrepreneurs is $\delta$. As is standard in contract theory, I assume that the hazard rate, $\mathcal{H}(\delta) = \frac{h(\delta)}{1-H(\delta)}$, is weakly increasing. The outcome of the project is privately observed by entrepreneurs and by sophisticated households. Monitoring is an ongoing activity, and I assume that an entrepreneur cannot monitor while running their project, for example because the project absorbs their full attention. Therefore, each entrepreneur observes only whether their own project succeeded or failed, while financially sophisticated idle households and bankers observe the success of every entrepreneur. Consequently, an entrepreneur who invests is uninformed about the return on banks' loan portfolio and cannot deposit in a shadow bank, regardless of whether they monitored in the previous CM.[2] The linear preferences in the CM ensure that aggregate quantities do not reveal the realization of $\delta$. The capital good is produced by the representative capital producer at unit cost.

As in Lagos and Wright (2005), I assume that in the competitive capital market households are anonymous and cannot commit. Therefore, the capital producer requires immediate compensation from entrepreneurs to sell capital. This generates a need for

[2]This assumption implies that, for a household that becomes an entrepreneur, the monitoring cost is sunk. For entrepreneurs, the pay-off monitoring then only enters through their real money balances and not through access to the shadow banking system. This assumption is crucial for the coexistence of the shadow banking system. Alternatively, I would require to introduce frictions on the loan size, such as limited pledgeability. As this paper focuses on frictions on the deposit side, I adopt this assumption instead.

a medium of exchange, namely fiat money. Fiat money is an intrinsically worthless asset supplied by the central bank. Households thus carry real money balances across periods to finance their investment in the DM. Only entrepreneurs require liquidity in the DM, while bankers and idle households have no direct need for liquidity. Therefore, reallocation of liquidity creates potential welfare gains (see, e.g. Berentsen et al. (2007) and Ferraris and Watanabe (2008)). The efficiency of this reallocation depends on the information friction present in the banking system.

To allow for credit, I follow Berentsen et al. (2007) and assume a record-keeping technology that allows banks to track financial histories within the period. In addition, banks can operate a verification technology. This technology allows banks to observe the realization of entrepreneurs' projects and allows other agents to verify the value of the bank's assets in the CM. I refer to banks operating with this technology as traditional banks and to those operating without it as shadow banks.

As households make their monitoring decision before the idiosyncratic shock, bankers who have monitored are already informed and have no use for the verification technology. In contrast, bankers who have not monitored can operate as a bank only by obtaining this technology. I assume that bankers who have monitored set up shadow banks, whereas bankers who have not monitored set up traditional banks. Although this assumption may appear restrictive, it can be derived naturally from the timing of the model and a cost associated with the verification technology. Suppose that the technology is costly. Then, bankers who have monitored are already informed when they become bankers and therefore have no reason to incur the additional cost. They thus choose to operate as shadow banks, while bankers who have not monitored operate as traditional banks. The two types of banks then arise endogenously from the interaction between the ex ante monitoring decision and the subsequent idiosyncratic shock. For any positive cost, bankers who have monitored strictly prefer to operate without the costly technology. Therefore, my assumption corresponds to the limiting case in which the cost goes to zero.

Both types of banks use their real money balances to issue risky loans to entrepreneurs in the DM. I assume that banks can only seize the output of the financed project and not force entrepreneurs to work. Consequently, only entrepreneurs with a successful project repay their loans. The record-keeping technology also allows them to raise deposits from idle households. However, because unsophisticated idle households cannot observe the realization of $\delta$, they cannot observe the bank's loan income either. Consequently, there is an asymmetric information friction between the bank and its unsophisticated depositors. To overcome this friction, unsophisticated idle households rely on traditional banks and their ex post verification technology.

Following Townsend (1979), depositors can request verification of the bank's assets in the CM. Verification is costly and destroys a share, $\zeta$, of the bank's loan income. One interpretation is the cost of a bankruptcy procedure in court, which is time-consuming and costly. I impose the following upper bound on the verification cost:

**Assumption 1.** *The verification cost, $\zeta$, is bounded from above*

$$\zeta < \bar{\zeta} = \min\left[1, \frac{1}{\bar{\delta}\mathcal{H}(\bar{\delta})}\right]$$

*where $\bar{\delta}$ is given by:*

$$\frac{\sigma_2}{\sigma_2 + \sigma_3}\mathbb{E}[\delta] = \int_{\bar{\delta}}^{1}(\delta - \bar{\delta})dH(\delta)$$

This assumption rules out the possibility of credit rationing in equilibrium (see Williamson (1987)). Throughout the paper, I restrict attention to deterministic verification strategies and symmetric contracts.[3] With risk-neutral agents and deterministic verification, the optimal contract between the traditional bank and its depositors takes the form of a standard debt contract (Gale and Hellwig, 1985; Williamson, 1986). This contract has two components: the bank promises a fixed repayment, and, if the bank has insufficient funds to repay the fixed amount, the depositors verify and share the bank's remaining assets. In Appendix B, I show that the verification technology is equivalent to a government operating an actuarially fair deposit insurance scheme, where the government is less able to collect loan income. Specifically, in case of bankruptcy a share $\zeta$ of loan income is destroyed.

Because shadow banks lack the verification technology, unsophisticated households cannot enforce truthful reporting and therefore deposit only in traditional banks. In contrast, financially sophisticated idle households can observe the realization of $\delta$. I assume that, under full information, payments contingent on $\delta$ are enforceable, so that the shadow bank cannot misreport its loan income to its monitoring depositors. Additionally, I assume that the shadow bank contract and payments are bilateral and private, so an unsophisticated household cannot observe what the bank pays to its other depositors. Because $\delta$ is an aggregate shock, any public signal would unravel the traditional bank deposit contract and would allow depositors to free-ride on the information of monitoring depositors. Sophisticated households have no need for the costly verification technology offered by traditional banks and, to avoid the cost of verification, they optimally deposit in shadow banks. Thus, monitoring determines access to the two intermediation technologies. The monitoring decision is made before the household's role in the DM is realized, while the banking system is only active in the DM.

The central bank controls the fiat money supply by injecting money into or withdrawing money from the economy during the CM through lump-sum transfers to households. Let $M_t$ denote the aggregate stock of money at the beginning of period $t$, $m_t$ an individual household's money holdings, $\phi_t$ the price of money in terms of period

[3] In a costly state verification framework, Mookherjee and Png (1989) show that stochastic verification generally dominates deterministic verification. However, Krasa and Villamil (2000) show that, in an environment in which households cannot commit to their verification strategy, the optimal verification strategy is deterministic. Second, because contracts are symmetric, a household that deposits in a traditional bank accepts the same contract as idle households and therefore does not run a project. This implies that an entrepreneur either runs the project or deposits and acts as an idle household.

$t$'s consumption goods, and $\mu_t$ the gross growth rate of money. Throughout, I restrict attention to $\mu_t > \beta$ and consider the limit $\mu \to \beta$ as the Friedman rule. Let $z_t = \phi_t m_t$ denote the real money balances, in terms of the consumption good, during period $t$. I focus on stationary, non-contingent policy rules under which real money balances are constant, so that $\phi_{t-1} M_{t-1} = \phi_t M_t$ and $\mu = \phi_t / \phi_{t+1} = 1 + \pi$, where $\pi$ denotes inflation. Let $T_t$ denote the nominal lump-sum transfer. As there is a unit mass of households, the central bank's budget constraint, in nominal terms, is:

$$T_t = M_{t+1} - M_t = (\mu - 1) M_t. \tag{1}$$

I conjecture (and later verify) that, absent a state-contingent monetary policy, the equilibrium price of money is independent of the realization of $\delta$. With a state-contingent price, the role of money would change, as the price could itself reveal the state of the aggregate economy to unsophisticated households. This would significantly complicate the model, and I therefore abstract from state-contingent monetary policy.

Following Williamson and Wright (2010), I define $i$ as the nominal interest on a hypothetical illiquid one period bond that can be bought in the CM, but not traded in the DM. As households are risk neutral, the real interest rate satisfies $1 + r = \frac{1}{\beta}$, and the Fisher equation gives:

$$1 + i = \frac{\mu}{\beta} = (1 + \pi)(1 + r)$$

By choosing the growth rate of the money supply, the central bank implicitly pins down the nominal interest rate $i$, which is also the opportunity cost of holding money. Throughout the paper, I use the subscript $+1$ to denote the next period and omit other time indices. All proofs are provided in the appendix. Finally, Figure 1 illustrates the timing of the model.

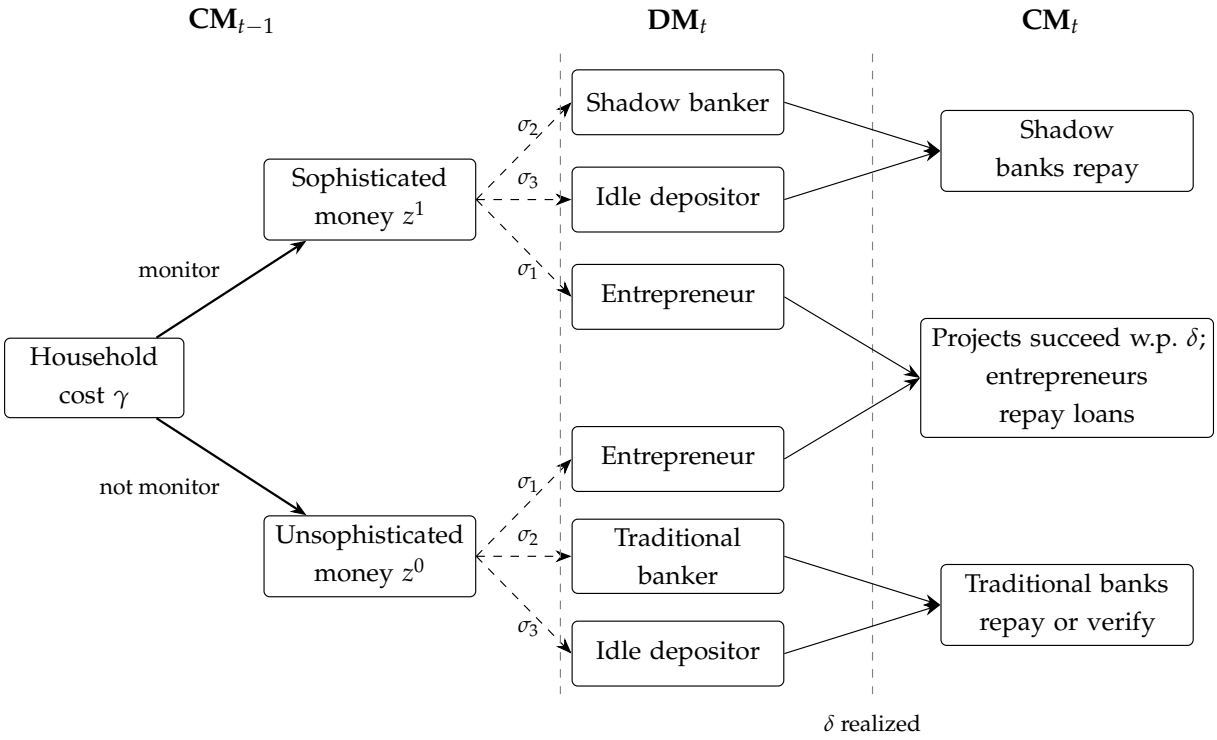



Figure 1: Time-Line.

### 3.2 Capital Producer

I start by looking at the decisions of the capital producer. The producer does not value real money balances, $z_t$, in the decentralized market. Because carrying money across periods is costly, the producer never carries money across periods. In the centralized market, the producer uses all its real money balances to buy consumption goods. Let $W^c(z)$ denote the producer's value function in the CM:

$$W^c(z) = x^c + \beta V^c_{+1}, \text{ s.t. } x^c = z$$

where the constraint states that the producer consumes its entire wealth. This implies that the producer's value function is linear in its money holdings. In the capital market, the producer produces capital at unit cost and then sells it at a price $p$. The producer's problem is then simply

$$V^c = \max_{k^c} \left\{ -k^c + W^c(pk^c) \right\}.$$

A positive and finite supply of capital requires the following price:

$$p = 1. \tag{2}$$

### 3.3 Households

Next, I turn to the decisions of households. I solve for the decisions of a household with monitoring cost $\gamma$ by backward induction, starting with their decisions in the CM and then analyzing those in the DM.

### 3.3.1 Centralized Market

In the CM, I can distinguish between different types of households depending on their role in the DM and their previous monitoring decision. Let the superscript $J \in \{0,1\}$ denote whether a household monitored in the previous period, $J = 1$, or not $J = 0$. With this superscript, I need to distinguish only three roles: entrepreneurs, bankers, and idle households. Each household chooses their net consumption $x^J \in \mathbb{R}$, how much money to bring to the next period $z^J_{+1}$, and whether to monitor $J^{+1} \in \{0,1\}$.

An entrepreneur enters the CM with real money balances $z^J$, capital invested $k^J$, and loans $l^J$. An entrepreneur has either a successful or an unsuccessful project. An entrepreneur with a successful project receives the return $f(k^J)$ and repays the outstanding debt $(1+i_l)l^J$, where $i_l$ is the loan rate. An entrepreneur with an unsuccessful project has no capital income and defaults on the loan. The budget constraint of an entrepreneur is

$$x^J + \mu z^J_{+1} = \tau + z^J + \mathcal{S} \times \left( f(k^J) - (1+i_l)l^J \right)$$

where $\mathcal{S}$ denotes the entrepreneurs project state: $\mathcal{S} = 1$ if the project succeeded and zero otherwise and $\tau$ the real lump-sum government transfer. The value function of an entrepreneur with monitoring cost $\gamma$ is then given by:

$$\begin{aligned} W^{J,e}(z^J, k^J, l^J | \delta; \gamma; \mathcal{S}) = \max_{x^J, z^{J^{+1}}_{+1}, J^{+1} \in \{0,1\}} & \; x^J + J^{+1} \left( -\gamma + \beta \mathbb{E}[V^1_{+1}(z^1_{+1})] \right) \\ & + (1 - J^{+1}) \beta \mathbb{E}[V^0_{+1}(z^0_{+1})] \end{aligned}$$

subject to the non-negativity constraint, $z_{+1} \geq 0$ and the budget constraint, where $V^{J^{+1}}_{+1}(z^J_{+1})$ is the value function in the next period's DM, which depends on the household's current monitoring decision. Using the budget constraint, the entrepreneur's value function becomes:

$$W^{J,e}(z^J, k^J, l^J | \delta; \gamma; \mathcal{S}) = \mathcal{S} \times (f(k^J) - (1+i_l)l^J) + z^J + W(0)$$

where

$$\begin{aligned} W(0) = \tau + \max_{J \in \{0,1\}, z^J_{+1}} & \Big\{ J(-\mu z^1_{+1} - \gamma + \beta \mathbb{E}[V^1_{+1}(z^1_{+1})]) \\ & + (1 - J) \left( -\mu z^0_{+1} + \beta \mathbb{E}[V^0_{+1}(z^0_{+1})] \right) \Big\} \end{aligned}$$

Before characterizing the CM decisions, I describe the CM problems of bankers and idle households. Bankers enter the CM with bank equity $e^J$, which pays a state-contingent gross return $\omega^J(\delta)$, and real money balances $z^J$. Their budget constraint is $\omega^J(\delta)e^J + z^J + \tau = x^J + \mu z^{J^{+1}}_{+1}$, and their value function is

$$W^{J,b}(z^J, e^J | \delta; \gamma) = \omega^J(\delta)e^J + z^J + W(0) \tag{3}$$

Lastly, an idle household enters the CM with real money balances $z^J$ and bank deposits

$d^J$, which pay a state-contingent return $\theta^J(\delta)$. For unsophisticated households, the verification strategy is determined by the deposit contract in the DM (see Lemmas 1 and 2). Verification occurs for certain realizations of $\delta$ and destroys a share $\zeta$ of the bank's loan income. This lowers the amount depositors receive in those states. I therefore let $\theta^0(\delta)$ denote the return on traditional bank deposits net of verification costs. The budget constraint of an idle household is $\theta^J(\delta)d^J + z^J + \tau = x^J + \mu z^{J+1}_{+1}$ and the CM value function is:

$$W^{J,i}(z^J, d^J|\delta;\gamma) = \theta^J(\delta)d^J + z^J + W(0) \tag{4}$$

As is standard in Lagos-Wright models, a household's choice of money balances is independent of their past role in the DM. The choices of a household with a given monitoring cost are given by:

$$W(0) = \tau + \max_{J\in\{0,1\},\, z^J_{+1}} \left\{ J(-\mu z^1_{+1} - \gamma + \beta\mathbb{E}[V^1_{+1}(z^1_{+1})]) + (1-J)\left(-\mu z^0_{+1} + \beta\mathbb{E}[V^0_{+1}(z^0_{+1})\right) \right\}$$

The choice of real money balances satisfies:

$$\frac{\mu}{\beta} = 1 + i \geq \mathbb{E}[V^{J+1}_{z,+1}(z^{J+1}_{+1})] \tag{5}$$

which holds with equality if $z^{J+1}_{+1} > 0$. Due to the linear preferences, all households with the same monitoring decision exit the CM with the same amount of real money holdings.

Finally, a household with monitoring cost $\gamma$ monitors if the expected utility of monitoring exceeds that of not monitoring. Let $W^1(\gamma)$ and $W^0$ denote the expected utility of monitoring and of not monitoring. For a household with monitoring cost $\gamma$, these are given by:

$$W^1(\gamma) = \tau - \gamma - \mu z^1_{+1} + \beta\mathbb{E}[V^1_{+1}(z^1_{+1})] \tag{6}$$
$$W^0 = \tau - \mu z^0_{+1} + \beta\mathbb{E}[V^0_{+1}(z^0_{+1})] \tag{7}$$

A household, therefore, monitors if $W^1(\gamma) > W^0$, that is, if their monitoring cost is below the following threshold:

$$\hat{\gamma} = -\mu(z^1_{+1} - z^0_{+1}) + \beta\mathbb{E}[V^1_{+1}(z^1_{+1}) - V^0_{+1}(z^0_{+1})] \tag{8}$$

This threshold endogenously separates monitoring from unsophisticated households. Depending on their monitoring cost, households thus decide whether to monitor. Monitoring is an ex ante decision and it determines whether the household can access the shadow banking system and affects the value of carrying liquidity into the DM. The monitoring choice thus depends on the expected utility of being financially sophisticated versus being unsophisticated as well as on the amount of liquidity carried.

### 3.4 Decentralized Market

At the beginning of each period, households face an idiosyncratic shock. With probabilities $\sigma_1, \sigma_2$ and $\sigma_3$, they become entrepreneurs, bankers, or idle, respectively. Consequently, the expected DM value function is

$$\mathbb{E}[V^J(z^J)] = \sigma_1 \mathbb{E}[V^{J,e}(z^J)] + \sigma_2 \mathbb{E}[V^{J,b}(z^J)] + \sigma_3 \mathbb{E}[V^{J,i}(z^J)]. \tag{9}$$

First, consider entrepreneurs. I conjecture that in equilibrium the expected loan rate is above the value of real money balances $\mathbb{E}[\delta](1+i_l) > 1$. Entrepreneurs cannot both invest and deposit in banks. In equilibrium, they prefer investing their money balances to depositing them in either bank. Entrepreneurs thus decide only how much to invest in capital and how much to borrow. Because all capital must be financed with money balances, the entrepreneur is constrained by their liquidity position, consisting of both loans and their own real money holdings. Using the fact that an entrepreneur's project succeeds with probability $\delta$, the entrepreneur faces the following problem:

$$\begin{aligned} V^{J,e}(z^J) &= \max_{k^J \geq 0, l^J \geq 0} \left\{ \mathbb{E}\left[ W^{J,e}(z^J + l^J - k^J, k^J, l^J) \right] \right\} \\ &= \max_{k^J \geq 0, l^J \geq 0} \left\{ \mathbb{E}[\delta]\left( f(k^J) - (1+i_l) l^J \right) + z^J + l^J - k^J \right\} + W(0) \end{aligned}$$

subject to the liquidity constraint

$$z^J + l^J \geq p k^J \tag{10}$$

For a given loan interest rate $1 + i_l$, the entrepreneur's capital investment is given by:

$$k^J = \begin{cases} k^*, & \text{if } z^J \geq p k^*, \\ \frac{z^J}{p}, & \text{if } \bar{z} \leq z^J < p k^* \\ \frac{z^J + l^J}{p}, & \text{if } z^J < \bar{z} \end{cases} \tag{11}$$

where $k^*$ is the efficient amount of capital invested, given by $\mathbb{E}[\delta] f'(k^*) = 1$, and $l^J$ and $\bar{z}$ are given by

$$f'\left(\frac{\bar{z}}{p}\right) = 1 + i_l \tag{12}$$

$$f'\left(\frac{z^J + l^J}{p}\right) = 1 + i_l. \tag{13}$$

My conjecture that in equilibrium $\mathbb{E}[\delta](1+i_l) > 1$ implies that $k < k^*$. If the marginal product of capital financed by their own real money balances is below the loan interest rate, the entrepreneur does not borrow and invests only their own money balances. Otherwise, they borrow until the expected marginal value of investment equals the expected cost of borrowing. This also implies that the entrepreneur's liquidity constraint

always binds. Therefore, the value function for an entrepreneur is:

$$V^{J,e}(z^J) = \mathbb{E}[\delta](f(k^J) - (1+i_l)l^J) + W(0) \tag{14}$$

Second, I jointly analyze the decisions of bankers and idle households (depositors). Sophisticated bankers set up shadow banks and attract sophisticated idle households as depositors.[4] Similarly, unsophisticated bankers and idle households form the traditional banking system. I solve both systems separately starting with the shadow banking system. Both systems lend to entrepreneurs through a common competitive loan market at the loan rate $i_l$.

**Shadow Banking System**

In the DM, both bankers and idle households enter with real money balances $z^1$. For notational convenience, let $e^1 = z^1$ denote the money balances of a shadow banker, where I anticipate that bankers use all their money balances as equity, and $z^1$ the money balances of an idle household. In the banking market, the shadow bank chooses a deposit contract and its portfolio of loans and real money balances. As $\mathbb{E}[\delta](1+i_l) > 1$, it is profitable to issue loans to entrepreneurs rather than to keep its own money balances as reserves. Therefore, a banker always invests all their money balances in loans. Consider a banker who does not raise any additional deposits and therefore invests only their own equity in loans. Using Equation 3, the expected utility of this banker is then given by:

$$\mathcal{U}^{1,b} = \mathbb{E}[\delta](1+i_l)e^1 + W(0), \tag{15}$$

Any deposit contract must leave the bank with at least the same expected utility, and Equation 15 is thus the bank's outside option. I assume that, in a competitive market, banks share the demand for deposits equally. As there is a mass $\sigma_2 G(\hat{\gamma})$ of shadow banks and $\sigma_3 G(\hat{\gamma})$ of sophisticated idle households, each bank attracts $\eta = \frac{\sigma_3}{\sigma_2}$ depositors. Next, consider a bank that attracts $\eta d^1$ units of deposits. Its expected utility is:

$$V^{1,b}(z^1) = \mathbb{E}[\delta](1+i_l)(e^1 + \eta d^1) - \mathbb{E}[\theta^1(\delta)]\eta d^1 + W(0)$$

Idle households deposit in shadow banks only if the expected gross return on deposits exceeds the value of money. Using Equation 4, the expected value of depositing is:

$$\begin{aligned} V^{1,i}(z^1) &= \mathbb{E}[W^{1,i}(z^1 - d^1, d^1)] \\ &= \mathbb{E}[\theta^1(\delta) - 1]d^1 + z^1 + W(0) \end{aligned}$$

The expected utility of depositors is increasing in repayments $\theta^1(\delta)d^1$, while the expected utility of the banker is decreasing in repayments. The competitive deposit

[4] Monitoring bankers set up shadow banks by assumption, while for idle households it is optimal to deposit in the shadow bank. Otherwise, they are exposed to potentially costly verification. Non-monitoring bankers can only set up a traditional bank, while non-monitoring idle households are only able to deposit in the traditional bank. Otherwise, they are unable to mitigate the information friction.

market implies that the bank's participation constraint always binds. Consequently, $\mathbb{E}[\delta](1+i_l) = \mathbb{E}[\theta^1(\delta)] > 1$ and idle households deposit all their money balances. The shadow bank's repayment function must satisfy a set of feasibility constraints and the participation constraint:

$$\eta z^1 \theta^1(\delta) \leq \delta(1+i_l)\left(\eta z^1 + e^1\right), \forall \delta \in [0,1], \tag{16}$$

$$\mathbb{E}[\delta](1+i_l)\left(\eta z^1 + e^1\right) - \mathbb{E}\left[\theta^1(\delta)\right]\eta z^1 = \mathbb{E}[\delta](1+i_l)e^1. \tag{17}$$

There is a continuum of repayment functions that satisfy these constraints. For example, the simplest repayment function is given by

$$\theta^1(\delta) = \delta(1+i_l). \tag{18}$$

Under this repayment scheme, both the depositor and the bank receive a share of the loan income proportional to their investment. This fully contingent contract resembles the contract offered by a money market fund with a floating net asset value, which offers its depositors a share price based on the performance of the underlying securities. An alternative contract could pay a fixed amount when feasible and the entire loan income in the event of default. This contract resembles a fixed net asset value money market fund that breaks the buck in the event of default. None of my results relies on a specific contractual form, and they extend to all admissible repayment functions. I therefore do not impose a specific contractual form.

Consequently, both shadow bankers and their depositors (indirectly) invest all their money balances in loans. Therefore, the value functions of a sophisticated idle household and a shadow banker are

$$V^{1,i}(z^1) = \mathbb{E}[\theta^1(\delta)]z^1 + W(0) = \mathbb{E}[\delta](1+i_l)z^1 + W(0) \tag{19}$$

$$V^{1,b}(z^1) = \mathbb{E}[\delta](1+i_l)z^1 + W(0) \tag{20}$$

**Traditional Banking System**

Next, I turn to the decisions of traditional banks and their depositors. A traditional bank raises deposits from unsophisticated idle households who cannot observe the realized return on the bank's loan portfolio. To overcome this information friction, traditional bank depositors can verify the traditional bank's assets ex post. Specifically, I use the costly state verification approach introduced by Townsend (1979). I depart from most of the literature along two dimensions.

First, the competitive banking market implies that the contract maximizes the expected utility of the lender rather than that of the borrower. In my set-up, this means that the contract maximizes the expected utility of depositors rather than that of banks, subject to the participation constraint of bankers.

Second, whereas the standard costly state verification setting takes the borrower's investment technology as fixed, here banks choose their investment strategy from a continuum of different investment projects. Because the standard debt contract is

optimal for any given portfolio, this generalization does not affect the contract's form.

As the bank maximizes the expected utility of its depositors, it takes its share of depositors, $\eta$, as given. I assume that banks offer symmetric contracts and that each bank attracts an equal share of depositors. As there is a mass $\sigma_2(1 - G(\hat{\gamma}))$ of banks and a mass $\sigma_3(1 - G(\hat{\gamma}))$ of unsophisticated idle households, $\eta = \frac{\sigma_3}{\sigma_2}$. I show that, even with these changes, the optimal contract still resembles a standard debt contract.

For a given investment strategy, let $y = \delta(1 + i_l)L^0 + R^0$ denote the random variable that captures the return on the bank's asset portfolio, with distribution $H_y(y)$. Formally, a contract with verification is characterized by a deposit size $d^0$, an investment strategy $\{L^0, R^0\}$, a repayment function $\mathcal{R}(\cdot)$, a message sent by the bank after observing its realized income, and a verification strategy chosen by depositors. This contract is admissible if it satisfies the non-negativity constraints and the bank's balance sheet.

Lemma 1 formally states that the standard debt contract is optimal. The proof is in the appendix and builds on Williamson (1986) and Winton (1995).

**Lemma 1.** *Any optimal contract has the following properties:*

1. *The bank repays a fixed amount, independent of the realization of the random variable.*
2. *If the bank has insufficient funds to repay the fixed amount, the depositors verify, the bank declares bankruptcy, and the remaining assets are shared among depositors.*

Reserves held by the bank earn the same return as money held directly by the household. However, unlike reserves held by the bank, those held by the household are not exposed to the cost of bankruptcy. Consequently, banks never invest any deposits in reserves, and I can restrict attention to contracts in which banks invest exclusively in loans.

Using Lemma 1, I can analyze the optimal deposit size and the repayment function. The repayment function $\theta^0(\delta)$ is:

$$\theta^0(\delta) = \begin{cases} (1 + i_d), & \text{if } \delta \geq \hat{\delta} \\ \frac{\delta(1+i_l)(1-\zeta)(e^0+\eta d^0)}{\eta d^0}, & \text{if } \delta < \hat{\delta} \end{cases} \tag{21}$$

where $1 + i_d$ is the fixed gross repayment per unit of deposit, and $\hat{\delta}$ is the break-even point, given by:

$$\hat{\delta} = \frac{(1 + i_d)\eta d^0}{(1 + i_l)(\eta d^0 + e^0)} \tag{22}$$

Using this repayment function, the expected utility of this deposit contract for an unsophisticated idle household is:

$$V^{0,i}(z^0) = (1 - H(\hat{\delta}))(1 + i_d)d^0 + \int_0^{\hat{\delta}} \left( \frac{\delta(1 + i_l)(1 - \zeta)(\eta d^0 + e^0)}{\eta} \right) dH(\delta) \tag{23}$$
$$+ z^0 - d^0 + W(0)$$

The bank's participation constraint is:

$$\int_{\hat{\delta}}^{1} \left( \delta(1+i_l)(\eta d^0 + e^0) - (1+i_d)\eta d^0 \right) dH(\delta) \geq \mathbb{E}[\delta](1+i_l)e^0 \tag{24}$$

The bank chooses the deposit size and the deposit interest rate to maximize the expected utility of its depositors (Equation 23), subject to its own participation constraint (Equation 24), the idle household's liquidity constraint $d^0 \leq z^0$, and non-negativity constraints.

Lemma 2 characterizes the optimal deposit interest rate and the deposit size:

**Lemma 2.** *Suppose that* $\mathbb{E}[\delta](1+i_l) > 1$. *Then the optimal deposit contract* $\{d^0, i_d\}$ *is given by:*

$$d^0 = \begin{cases} z^0, & \text{if } z^0 < d^* \\ d^*, & \text{if } z^0 \geq d^* \end{cases}$$
$$1 + i_d = \frac{\hat{\delta}(1+i_l)(\sigma_3 d^0 + \sigma_2 e^0)}{\sigma_3 d^0}$$

*where the optimal deposit size and break-even point* $\hat{\delta}$ *are given by*

$$1 = (1+i_l)\left[ \mathbb{E}[\delta] - \zeta \left( \int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta} \int_{\hat{\delta}}^{1} (\delta - \hat{\delta}) dH(\delta) \right) \right] \tag{25}$$

$$\frac{\sigma_2 e^0}{\sigma_2 e^0 + \sigma_3 d^0} \mathbb{E}[\delta] = \int_{\hat{\delta}}^{1} (\delta - \hat{\delta}) dH(\delta) \tag{26}$$

Lemma 2 characterizes the optimal traditional bank deposit contract. The expected return on loans is strictly higher than the marginal value of money. The risk associated with loans triggers costly verification by depositors in bad states. As a result, depositors require a higher promised return on deposits to compensate for their risk of incurring the verification cost. If the idle household has sufficient money holdings, they deposit until the marginal expected return on depositing equals the marginal value of money. Otherwise, the household is constrained by their money balances and deposits all of them.

Using Lemma 2 and the bank's binding participation constraint, the value functions of traditional banks and of unsophisticated idle households are given by:

$$V^{0,b}(z^0) = \mathbb{E}[\delta](1+i_l)z^0 + W(0) \tag{27}$$

$$V^{0,i}(z^0) = \left( (1 - H(\hat{\delta}))\hat{\delta} + (1-\zeta) \int_0^{\hat{\delta}} \delta dH(\delta) \right) (1+i_l) \frac{\sigma_2 e^0 + \sigma_3 d^0}{\sigma_3} + z^0 - d^0 + W(0) \tag{28}$$

The first term of $V^{0,i}(z^0)$ is the expected return on the depositor's share of loans, net of expected verification costs, while the second term is the value of their remaining money balances. Because the bank's participation constraint binds and $e^0 = z^0$, a traditional

banker earns the same expected return as if they lent their own money balances directly. The expected cost of verification is therefore fully internalized by unsophisticated idle households.

## 4 Equilibrium

Having described the decisions of households and the capital producer, I can now define a symmetric stationary monetary equilibrium. I first state the market clearing conditions. Given the monitoring cost threshold $\hat{\gamma}$, there is a share $G(\hat{\gamma})$ of sophisticated households and a share $1 - G(\hat{\gamma})$ of unsophisticated households.

Let $K$ denote the aggregate capital produced by the capital producer. The market-clearing condition for the capital market is:

$$K = \sigma_1 G(\hat{\gamma}) k^1 + \sigma_1 (1 - G(\hat{\gamma})) k^0 \tag{29}$$

where the RHS is the total demand for capital from sophisticated and unsophisticated entrepreneurs, respectively. Given the producer's linear cost function, the supply of capital is perfectly elastic and adjusts to clear the market at the level determined by demand.

Market clearing in the loan market gives:

$$\sigma_1 (1 - G(\hat{\gamma})) l^0 + \sigma_1 G(\hat{\gamma}) l^1 = (\sigma_2 + \sigma_3) G(\hat{\gamma}) z^1 + (1 - G(\hat{\gamma}))(\sigma_2 z^0 + \sigma_3 d^0) \tag{30}$$

The LHS is the demand for loans coming from a mass $\sigma_1 G(\hat{\gamma})$ of sophisticated entrepreneurs and a mass of $\sigma_1 (1 - G(\hat{\gamma}))$ unsophisticated entrepreneurs. The supply of loans comes from both banking systems. There is a mass of $\sigma_2 G(\hat{\gamma})$ shadow banks that invest their own money balances and those of their depositors in loans, while there is a mass of $\sigma_2 (1 - G(\hat{\gamma}))$ of traditional banks that invest their own balances fully and the deposits they raise.

Finally, the money market clears when:

$$\phi M = G(\hat{\gamma}) z^1 + (1 - G(\hat{\gamma})) z^0 \tag{31}$$

where the left-hand side is the real money supply and the right-hand side is the demand for real money balances of monitoring and unsophisticated households, respectively.[5] As is standard in monetary models, I restrict attention to symmetric pure-strategy equilibria (SSM):

**Definition 1.** *For a given nominal interest rate $i$, a pure-strategy symmetric stationary monetary equilibrium consists of prices $\{p, \phi\}$, capital allocations $\{k^1, k^0\}$, entrepreneurs' loans $\{l^1, l^0\}$, deposits $\{d^1, d^0\}$, interest rates $\{i_l, i_d, \theta^1(\delta)\}$, bank equity $\{e^1, e^0\}$, real money balances $\{z^1, z^0\}$, CM allocations $\{x^1, x^0\}$, a break-even point $\{\hat{\delta}\}$, and a monitoring threshold $\{\hat{\gamma}\}$, such that the choices of households and the capital producer are optimal, the central bank's budget constraint holds, and all markets clear.*

[5] The consumption-goods market then clears by Walras' law.

In this economy, the distribution of monitoring costs plays a crucial role, as monitoring determines which intermediation technology a household can use. Consequently, in what follows I discuss the pure-strategy symmetric stationary monetary equilibria that result from different distributions of monitoring costs. It is important to note that in this economy the equilibrium is not necessarily unique. As explained in detail below, because the value of monitoring to an individual household depends on how many other households monitor, a coordination problem arises, and multiple equilibria generally exist.

First, I characterize expected welfare in the general economy.[6] Expected aggregate welfare is determined by the expected net return on investment and the expected cost of mitigating the information friction:

$$(1-\beta)\mathcal{W} = \sigma_1 G(\hat{\gamma})\beta\left(\mathbb{E}[\delta]f(k^1)-k^1\right)+\sigma_1(1-G(\hat{\gamma}))\beta\left(\mathbb{E}[\delta]f(k^0)-k^0\right) \tag{32}$$
$$-\int_0^{\hat{\gamma}}\gamma dG(\gamma)-\beta(1-G(\hat{\gamma}))(\sigma_2 z^0+\sigma_3 d^0)\zeta f'(k^0)\int_0^{\hat{\delta}}\delta dH(\delta)$$

The first two terms are the expected net income from the investment projects of sophisticated and unsophisticated entrepreneurs, the third term is the expected cost of monitoring that is incurred in the previous period, and the last term is the expected cost of verification.

To isolate the role of the information friction and the two different technologies, I first characterize a benchmark economy in which all households can monitor at no cost. With a slight abuse of terminology, I refer to this economy as the full information economy.[7]

### Full-information Benchmark

As a benchmark, consider the case in which all households can monitor at no cost. Proposition 1 characterizes this equilibrium.

**Proposition 1** (Full-Information)**.** *Suppose that $\gamma = 0$ for all households. Then for any $i > 0$, there exists an SSM equilibrium in which all households monitor. The equilibrium is given by:*

$$1+i = \mathbb{E}[\delta]f'(k) \tag{33}$$
$$\sigma_1 k = z \tag{34}$$

In the benchmark, all households monitor and only shadow banks are active.[8] In this equilibrium, all idle money balances are reallocated through the banking market to entrepreneurs. The banking system allows liquidity to be reallocated efficiently across the idiosyncratic states of households. The equilibrium coincides with the perfect

[6] I define expected welfare here as the expected life-time utility of a household before drawing its monitoring costs. This coincides with expected welfare defined as the equally-weighted sum of households' expected life-time utilities. As a consequence, higher welfare does not make every household better off.

[7] Formally, the asymmetric information friction is still present. However, it is costless to mitigate the friction.

[8] As households are indifferent between monitoring and not monitoring, when $\gamma = 0$, the equilibrium without monitoring of Proposition 2 also exists.

enforcement equilibrium of Berentsen et al. (2007) and banks improve the allocation by efficiently reallocating liquidity. As is standard in monetary models, the only remaining distortion is the cost of holding money, which disappears at the Friedman rule. This benchmark therefore isolates the potential consequences of the information friction, which I introduce in the remaining sections.

### 4.1 Traditional Banking Equilibrium

I first show that there always exists an equilibrium in which households do not monitor:

**Proposition 2.** *For any $i > 0$ there exists a SSM equilibrium with only traditional banks. There exists a threshold $\bar{i}$, such that the equilibrium allocation is given by:*

- *for $i \leq \bar{i}$, idle households deposit only part of their money balances, $d < z$, and the allocation is:*

$$i = (1-\sigma_3)(\mathbb{E}[\delta]f'(k) - 1) \tag{35}$$

$$1 = f'(k)\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)\right] \tag{36}$$

$$\sigma_1 k = (\sigma_1 + \sigma_2)z + \sigma_3 d \tag{37}$$

$$\mathbb{E}[\delta]\sigma_2 z = (\sigma_2 z + \sigma_3 d)\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta) \tag{38}$$

- *for $i > \bar{i}$, idle households deposit all their money balances and the allocation is:*

$$1 + i = \mathbb{E}[\delta]f'(k) - \sigma_3\zeta f'(k)\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right) \tag{39}$$

$$\sigma_1 k = z \tag{40}$$

$$z = d \tag{41}$$

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2 + \sigma_3} = \int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta) \tag{42}$$

*where the threshold is given by:*

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2 + \sigma_3} = \int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)$$

$$\bar{i} = \frac{(1-\sigma_3)\zeta\left[\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right]}{\mathbb{E}[\delta] - \zeta\left[\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right]}$$

In this equilibrium the information friction is resolved through ex post verification and only traditional banks are active. The intuition for this result is simple: monitoring can only be valuable if others monitor. If a household believes that no one else monitors, there is no private gain from monitoring. As a sophisticated idle household, there is no shadow bank to deposit in, and depositing in a traditional bank yields the same return as

without monitoring, as a sophisticated banker, the household earns the same expected return, and as an entrepreneur, monitoring plays no role. Consequently, the marginal value of money is the same as without monitoring, such that sophisticated households bring the same real money balances to the DM and obtain the same expected utility, but have to incur the monitoring cost. Monitoring is then worthless to an individual household, and the belief that nobody monitors is self-fulfilling. Households would thus only monitor if there is a private gain from monitoring.

Proposition 2 shows the allocation in a traditional banking economy. In the traditional banking economy, the information friction enters through the deposit contract. Because unsophisticated idle households cannot observe $\delta$, the contract problem is characterized by asymmetric information and costly verification. Idle households deposit partially when the interest rate is low and fully when the interest rate is high. For $i \leq \bar{i}$, carrying money is relatively cheap, and the expected gain from reallocation is relatively low. Idle households deposit only part of their money balances, such that the expected marginal return on deposits equals the value of money. In this case, idle households exit the DM with positive real money balances and banks are not able to reallocate all money balances. For $i > \bar{i}$, carrying money is relatively costly and the gains from reallocation are high. In this case, the net expected return on depositing is higher than the value of money and idle households deposit all their money balances.

Compared to the benchmark, the information friction and costly state verification introduce three distortions. First, there is the direct expected cost of verifying. Second, households carry fewer real money balances as the marginal value of money for idle households is lower. Third, the banking system no longer efficiently reallocates liquidity for $i < \bar{i}$.

Nevertheless, expected welfare is not necessarily decreasing in the verification cost $\zeta$. Corollary 1 shows that if $\delta$ is uniformly distributed, welfare is increasing in $\zeta$ for low interest rates.

**Corollary 1.** *Suppose that $\delta \sim U(0,1)$, then for $i < \bar{i}$ expected welfare is increasing in the cost of verification.*

For low interest rates, idle households keep each additional unit as real money balances, so the verification cost does not affect the marginal value of money or the amount of capital invested (Equation 35). An increase in the verification cost affects only the deposit decisions of idle households, but does not affect the marginal value of money for idle households. Therefore, the amount of capital invested is independent of the verification cost, while both the break-even point and the total loans of traditional banks decrease. There are thus opposing effects on expected welfare: a negative direct effect, as verification becomes more costly, and two positive indirect effects, as the likelihood of default and the amount of loans exposed to verification decrease. For a uniform distribution, the indirect effects dominate.

**Corollary 2.** *For any $i > 0$ and $\zeta > 0$, investment and expected welfare in the traditional banking equilibrium are strictly below the benchmark.*

As expected, Corollary 2 shows that investment and expected welfare in the traditional banking economy are always lower than in the full information benchmark. The information friction always reduces the amount of capital invested through two different channels. First, if $i \leq \bar{i}$, idle households keep part of their money balances outside the banking system. This reduces the amount of money invested in the DM directly. Second, for any interest rate the marginal value of money for idle households is lower; they either do not deposit fully or the expected cost of verification lowers the expected return on deposits. This decreases the marginal value of money and lowers the money demand and thus the amount of capital invested. As expected welfare is determined completely by the amount of capital invested and the expected cost of mitigating the information friction, expected welfare is always strictly lower in the traditional banking equilibrium. Both channels through which the information friction reduces capital and expected welfare interact with monetary policy: the friction only distorts expected welfare and investment because carrying money is costly.

This clearly shows that ex post verification affects the economy beyond the direct cost of mitigating the information friction. As verification takes place after households have chosen their money balances, the technology changes the optimal decision of households and affects their money demand. Monitoring affects the economy in a fundamentally different way. Monitoring is chosen before the household's DM role is realized. If all households monitored, the DM allocation would be undistorted by the information friction as sophisticated households are fully informed. This raises the question: if monitoring improves the allocation, why would households not simply monitor when the monitoring cost is sufficiently small? In the next sections, I answer this question by first focusing on a homogeneous monitoring cost, before turning to heterogeneous monitoring costs.

### 4.2 Homogeneous Monitoring Cost

As hinted at in the previous section, given that monitoring mitigates the distortion of the information friction in the DM, one would expect households to monitor when monitoring costs are low. Proposition 3 shows, however, that such an equilibrium does not exist if all households face the same monitoring cost.

**Proposition 3.** *Suppose that all households have the same monitoring cost $\gamma > 0$. Then for any $i > 0$, there exists no SSM equilibrium in which all households monitor.*

Behind this impossibility result lie a pecuniary externality and a free-riding problem. When all households monitor, liquidity is abundant and the expected loan rate equals the cost of carrying money across periods. A household, therefore, receives no surplus from carrying additional money. By choosing not to monitor, the household can rely on the liquidity supplied by other monitors while saving the monitoring cost and the cost of bringing real money balances themselves.[9] The individual household

[9]The core of the argument does not rely on the fact that a deviating household carries no money, but that in equilibrium the return of money equals its cost. Suppose that entrepreneurs had to finance a share of their investment with their own money balances. As money earns exactly its cost when all households monitor, the deviator could

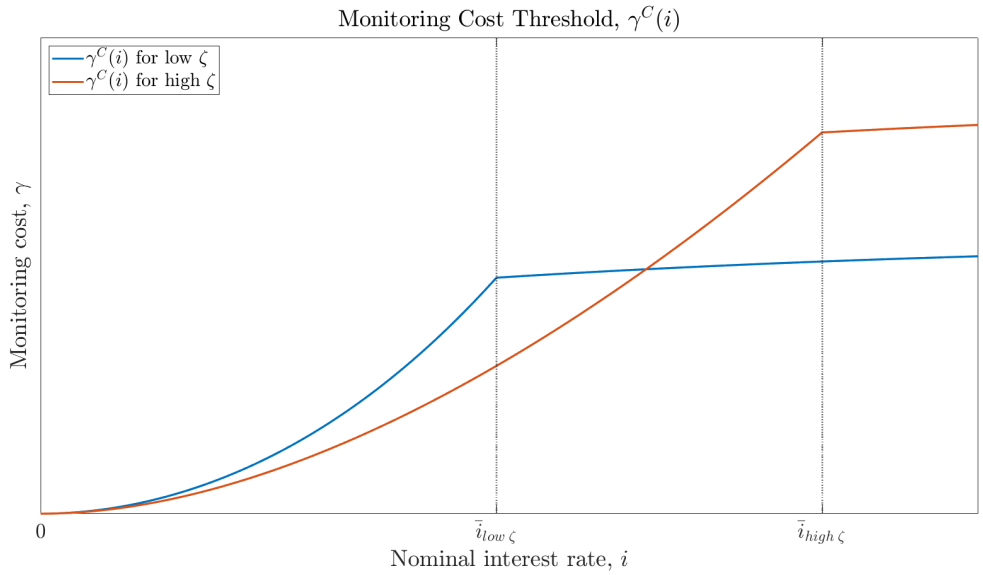



Figure 2: Monitoring cost threshold

The figure shows the threshold $\gamma^C(i)$ for two values of the verification cost. The red line shows the threshold for a high verification cost, $\zeta = 0.6$, and the blue line for a low verification cost, $\zeta = 0.4$. The dotted lines indicate the thresholds $\bar{i}$ above which idle households deposit all their money balances. The figure is created with $\sigma_1 = 0.6$, $\sigma_2 = 0.1$, $\sigma_3 = 0.3$, $\beta = 0.96$, and $f(k) = 10k^{1/3}$, while $\delta \sim U(0,1)$.

does not internalize the effect of their real money balances on the aggregate liquidity provision. Monitoring can thus be socially valuable without being privately optimal. Households prefer others to incur the monitoring cost and supply the liquidity they need as entrepreneurs.

**Corollary 3.** *Suppose that all households have the same monitoring cost $\gamma$ and can commit to monitoring. Then, for any $i > 0$, there is a threshold $\gamma^C(i) > 0$, such that expected welfare in an economy in which all households monitor is higher than in a traditional banking economy if and only if $\gamma < \gamma^C(i)$.*

Corollary 3 formalizes the result that an economy in which all households monitor can be socially valuable. By monitoring, the economy benefits from higher capital investment and expected return and from saving on the expected cost of verification, at the cost of monitoring. For each nominal interest rate, there is a threshold below (above) which monitoring is welfare improving (reducing). Figure 2 illustrates this threshold $\gamma^C(i)$ for two different verification costs. However, as there is no private gain from bringing additional liquidity, households want to deviate and rely on the abundant liquidity supplied by the banking system. Monitoring is thus only privately valuable if it allows households to earn a premium over not monitoring, while it is socially valuable if the monitoring cost is sufficiently low.

Lastly, combining Propositions 2 and 3 gives the following result:

**Corollary 4.** *Suppose that all households have the same monitoring cost $\gamma > 0$, then for any*

meet this requirement by carrying just the necessary money balances at no loss in expected utility, and would still save the monitoring cost.

*$i > 0$, the unique SSM equilibrium is the traditional banking equilibrium of Proposition 2.*

With identical monitoring costs, the economy relies entirely on the verification technology even when monitoring would yield a higher aggregate welfare. In the next section, I explore the case with heterogeneous monitoring costs.

### 4.3 Heterogeneous Monitoring Costs

Suppose now that $\gamma$ is continuously distributed on $(0, \Gamma)$. Households with sufficiently low costs may then find it privately optimal to monitor, while households with higher monitoring costs remain unsophisticated.

**Lemma 3.** *In any SSM equilibrium with $G(\hat{\gamma}) > 0$ and $i > 0$, the following two results hold:*

1. *Only financially unsophisticated entrepreneurs borrow: $l^0 > l^1 = 0$.*
2. *Financially sophisticated households carry more real money balances than unsophisticated ones: $z^1 > z^0$.*

Lemma 3 provides two preliminary results linking monitoring and liquidity provision. As the information friction no longer distorts their DM choices, sophisticated idle households receive a strictly higher deposit rate. Since all households face the same cost of carrying money, sophisticated households bring more real money balances. Monitoring therefore not only changes the information available to a household, but also the amount of liquidity carried into the DM.

**Proposition 4.** *If $\gamma$ is continuously distributed on $(0, \Gamma)$ and $\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$, then for any $i > 0$ and $\zeta \in (0, \bar{\zeta})$ there exists an SSM equilibrium with sophisticated households. This equilibrium is either*

1. *a mixed banking equilibrium, in which both traditional and shadow banks exist.*
2. *a pure shadow banking equilibrium, in which only shadow banks exist.*

Proposition 4 establishes the existence of an equilibrium with monitoring. There are two types of equilibria with sophisticated households: a mixed banking equilibrium in which both traditional banks and shadow banks are active and a pure shadow banking equilibrium in which only shadow banks are active. The difference depends solely on the liquidity balances of unsophisticated households. If unsophisticated households do not bring any money, only shadow banks exist, while if unsophisticated households bring money, both types of banks exist.

The money balances of unsophisticated households depend on how many households monitor. If many households monitor, shadow banks fund most loans and unsophisticated households need to bring little money. The composition of the banking system is thus endogenous and depends on the private value of monitoring. Let $\Xi(\hat{\gamma})$ denote the private gain from monitoring when the monitoring threshold is $\hat{\gamma}$. A household then monitors whenever

$$\Xi(\hat{\gamma}) \geq \gamma$$

From Equation 8, the private monitoring gain is:

$$\Xi = \beta \left[ \mathbb{E}[V^1(z^1)] - \mathbb{E}[V^0(z^0)] - (1+i)(z^1 - z^0) \right]$$

where the first two terms are the difference in expected utility in the DM, and the last term is the difference in the cost of carrying money balances. From Lemma 3, sophisticated households carry larger real money balances. This additional liquidity is supplied to unsophisticated entrepreneurs through the loan market and the shadow banking system, which in turn changes the equilibrium allocation of credit. Consequently, monitoring changes the liquidity provision, which in turn changes loan market conditions, and these changes feed back into the private gain of monitoring. The private gain of monitoring thus consists of several components.

First, by monitoring, an idle household receives the expected loan rate on their money balances, whereas the expected deposit rate for unsophisticated households is lower due to the expected cost of verification. Money is therefore more valuable to a sophisticated household. This difference ensures that sophisticated households bring larger money balances, refrain from borrowing, and invest more. This gives the second positive component: by monitoring, a household carries more liquidity across periods. As an entrepreneur, they invest more capital, while as an idle household or banker, they lend more at the expected loan rate. Third, sophisticated entrepreneurs finance their projects entirely with their own money, whereas unsophisticated entrepreneurs borrow. Consequently, the net benefit of investing for unsophisticated entrepreneurs is lower. However, carrying money balances is costly due to inflation, which reduces the private monitoring gain.

To summarize, the private monitoring gain reflects how costly it is to rely on traditional banks, while the monitoring cost reflects the cost of accessing the shadow banking system. The marginal household is indifferent between monitoring and not monitoring, so in equilibrium the private monitoring gain equals the monitoring cost threshold. Conditional on this threshold, the equilibrium can be characterized separately for a mixed and a pure shadow banking equilibrium.

**Proposition 5** (Mixed banking equilibrium)**.** *In a mixed banking equilibrium, the allocation is given by:*

- *for $i \leq \bar{i}$, unsophisticated idle households deposit only part of their money balances, $d^0 < z^0$,*

*and the allocation is:*

$$i = (1-\sigma_3)\left(\mathbb{E}[\delta]f'(k^0) - 1\right) \tag{43}$$

$$1 = f'(k^0)\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta\, dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta)\right)\right] \tag{44}$$

$$1 + i = \mathbb{E}[\delta]\left[\sigma_1 f'(z^1) + (\sigma_2 + \sigma_3)f'(k^0)\right] \tag{45}$$

$$\sigma_1 k^0 = (\sigma_1 + \sigma_2)z^0 + \sigma_3 d^0 + \frac{G(\hat{\gamma})}{1 - G(\hat{\gamma})}(\sigma_2 + \sigma_3)z^1 \tag{46}$$

$$\mathbb{E}[\delta]\sigma_2 z^0 = (\sigma_2 z^0 + \sigma_3 d^0)\int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta) \tag{47}$$

- *for $i > \bar{i}$, unsophisticated idle households deposit all their money balances and the allocation is:*

$$1 + i = f'(k^0)\left(\mathbb{E}[\delta] - \sigma_3\zeta\left(\int_0^{\hat{\delta}} \delta\, dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta)\right)\right) \tag{48}$$

$$1 + i = \mathbb{E}[\delta]\left[\sigma_1 f'(z^1) + (\sigma_2 + \sigma_3)f'(k^0)\right] \tag{49}$$

$$\sigma_1 k^0 = z^0 + \frac{G(\hat{\gamma})}{1 - G(\hat{\gamma})}(\sigma_2 + \sigma_3)z^1 \tag{50}$$

$$z^0 = d^0 \tag{51}$$

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2 + \sigma_3} = \int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta) \tag{52}$$

*where the monitoring threshold is given by:*

$$\begin{aligned}\hat{\gamma} = {} & \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]\left[f(z^1) - f(k^0)\right] \\ & - \beta\zeta f'(k^0)\sigma_3 d^0\left(\int_0^{\hat{\delta}} \delta\, dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta)\right) \\ & + \beta\zeta f'(k^0)(\sigma_2 z^0 + \sigma_3 d^0)\int_0^{\hat{\delta}} \delta\, dH(\delta).\end{aligned} \tag{53}$$

Proposition 5 characterizes a mixed banking equilibrium.[10] The key property of the mixed equilibrium is that investment by both types of entrepreneurs is independent of the share of sophisticated households. Unsophisticated entrepreneurs borrow until the marginal value of investment equals the expected cost of borrowing, which pins down the loan rate independently of the share of sophisticated households. As sophisticated money balances are determined by the loan rate, their investment is also independent of their share. The share of sophisticated households thus only determines the relative size of the banking systems. Moreover, the amount of capital invested by unsophisticated entrepreneurs is the same as in the traditional banking equilibrium.

[10] Propositions 5 and 6 are obtained directly from the proof of Proposition 4.

**Proposition 6** (Pure shadow banking equilibrium)**.** *In a pure shadow banking equilibrium, unsophisticated households carry no money,* $z^0 = 0$*, and the allocation is given by:*

$$1 + i = \mathbb{E}[\delta] \left[ \sigma_1 f'(z^1) + (\sigma_2 + \sigma_3) f'(k^0) \right] \tag{54}$$

$$\sigma_1 k^0 = \frac{G(\hat{\gamma})}{1 - G(\hat{\gamma})} (\sigma_2 + \sigma_3) z^1 \tag{55}$$

*where the monitoring threshold is given by:*

$$\hat{\gamma} = \beta \sigma_1 (1 - \alpha) \mathbb{E}[\delta] f(z^1) \left[ 1 - \left( \frac{k^0}{z^1} \right)^{\alpha} \right]. \tag{56}$$

As shown in Proposition 6, once unsophisticated households no longer carry money, the share of sophisticated households directly determines the capital investment of both types of entrepreneurs. As unsophisticated entrepreneurs rely fully on the shadow banking system for funding of their investment, the aggregate money balances of sophisticated households determine the expected loan rate and the real money balances of an individual sophisticated household depend on the share of sophisticated households. Therefore, the pure shadow banking equilibrium differs fundamentally from the mixed banking equilibrium. This difference determines how the private monitoring gain responds to the decisions of other households.

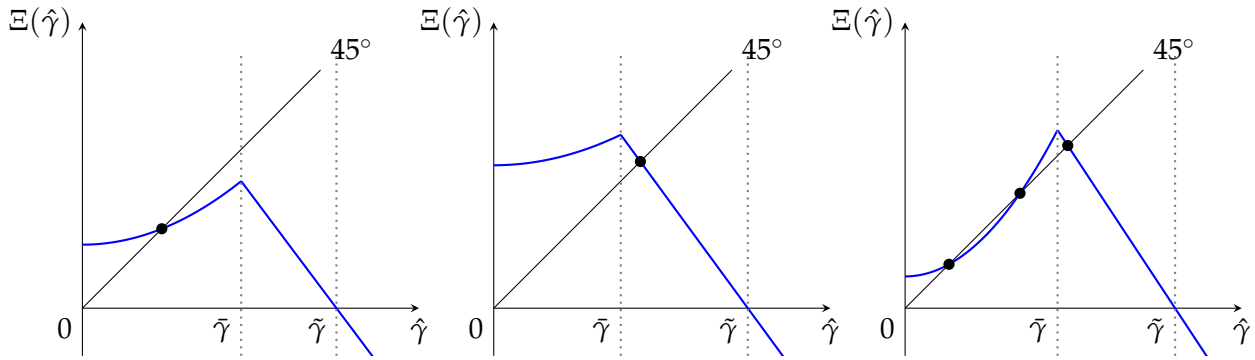
In both equilibria, the private monitoring gain corresponds to the RHS of Equations 53 and 56, from which it is clear that the private gain depends on the behavior of other households. When more households monitor, the shadow banking system funds a larger share of loans, which puts downward pressure on the expected loan rate. In a mixed banking equilibrium, unsophisticated households carry lower real money balances into the DM, which has four effects on the private monitoring gain. First, the cost of carrying money lowers the private monitoring gain, because sophisticated households bring relatively more money. Second, traditional banks lend less, which reduces the expected cost of verification and thereby the benefit of not having to verify. Third, unsophisticated entrepreneurs invest the same amount but finance a larger part of it with loans, which decreases their expected utility and increases the private monitoring gain. Finally, unsophisticated bankers and idle households hold smaller real money balances, which reduces their expected utility. These last two effects dominate until unsophisticated households no longer hold any real money balances. Once unsophisticated households no longer carry money, this margin disappears. Additional sophisticated households then increase the supply of loans by shadow banks, which lowers the loan rate and raises the investment of unsophisticated entrepreneurs. The difference in investment decreases and the private monitoring gain decreases with the share of sophisticated households. Therefore, monitoring decisions are strategic complements in a mixed banking equilibrium, while they are strategic substitutes in a pure shadow banking equilibrium.

These forces determine the share of the sophisticated households as shown in the following corollary:

**Corollary 5.** *The private monitoring gain $\Xi(\hat{\gamma})$ is continuous on $(0, \Gamma)$ and depends on $\hat{\gamma}$ only through $\frac{G(\hat{\gamma})}{1-G(\hat{\gamma})}$. Suppose that $\frac{g(\hat{\gamma})}{1-G(\hat{\gamma})}$ is increasing. Then $\Xi(\hat{\gamma})$ has the following properties:*

- $\lim_{\hat{\gamma}\to 0^+} \Xi(\hat{\gamma}) > 0$
- *$\Xi(\hat{\gamma})$ is linear in $\frac{G(\hat{\gamma})}{1-G(\hat{\gamma})}$, strictly increasing, and convex on $(0, \bar{\gamma})$*
- *$\Xi(\hat{\gamma})$ is strictly decreasing on $(\bar{\gamma}, \tilde{\gamma})$, with $\Xi(\tilde{\gamma}) = 0$.*

*where $\tilde{\gamma} = G^{-1}(\sigma_1)$ and $z^0 > 0$ if and only if $\hat{\gamma} < \bar{\gamma}$.*




(a) Unique mixed banking equilibrium

(b) Unique pure shadow banking equilibrium

(c) Two mixed and one pure shadow banking equilibrium

Figure 3: The private gain of monitoring and its fixed points. (Illustrative)

As illustrated in Figure 3, the equilibrium with monitoring is a fixed point of the monitoring threshold. The strategic complementarity of monitoring decisions in the mixed banking equilibrium can generate multiple equilibria. Specifically, there are three possible scenarios for an SSM equilibrium with sophisticated households:[11]

1. there is a single mixed banking equilibrium
2. there is a single pure shadow banking equilibrium
3. there are two mixed banking equilibria and a single pure shadow banking equilibrium.

**Proposition 7.** *Suppose that $\gamma$ is continuously distributed on $(0, \Gamma)$ with an increasing hazard rate and $\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$. Then there exists a $\zeta^*$ such that*

1. *if $\zeta < \zeta^*$, the SSM equilibrium with monitoring is a unique mixed banking equilibrium for all $i > 0$,*
2. *if $\zeta \geq \zeta^*$, there exists a non-empty set of nominal interest rates $\mathcal{I}(\zeta)$, such that*
   - *if $i \in \mathcal{I}(\zeta)$, there exists a single pure shadow banking equilibrium and zero or two mixed banking equilibria,*
   - *if $i \notin \mathcal{I}(\zeta)$, the SSM equilibrium with monitoring is a mixed banking equilibrium.*

[11] There always exists a pure traditional banking equilibrium in which no household monitors.

From Proposition 7, the fixed point depends on both the nominal interest rate and the verification cost. The nominal interest rate and the verification cost affect monitoring decisions of households through different channels. The nominal interest rate is the opportunity cost of carrying money across periods, while the verification cost affects the return on idle money balances. The expected cost of verification affects the marginal value of money for unsophisticated idle households. Sophisticated idle households are fully informed and deposit in the shadow bank earning the full expected loan rate on their deposits. For given real money balances, a sophisticated household therefore has a higher marginal value of money. In equilibrium, sophisticated households thus carry larger money balances (see also Lemma 3). The difference in the return on idle money balances thus determines how much more money sophisticated households carry, while aggregate liquidity determines the size of these money balances.

When the nominal interest rate is low, the expected loan rate is low, and the difference in the return on idle money balances is small, regardless of the verification cost. Unsophisticated idle households deposit until the expected marginal return on deposits equals the value of money in the DM. A higher verification cost reduces how much they deposit, but does not affect the marginal value of money. The differences in money balances are small and the private monitoring gain is low. When the nominal interest rate rises, the expected loan rate rises and idle unsophisticated households increase their relative deposit size. This increases the wedge between the return on idle money balances of sophisticated and unsophisticated households. If the verification cost is high, the marginal value of money for unsophisticated households falls below the cost of carrying money balances. In this case, unsophisticated households no longer carry money and rely on the liquidity provided by the sophisticated households. If the verification cost is low, the wedge is insufficient for unsophisticated households to stop carrying real money balances. For interest rates above $\bar{i}$, liquidity becomes more costly. The advantage of monitoring is then earned on smaller money balances, the private monitoring gain falls, and the liquidity provided by the shadow banking system decreases. These forces imply that the size of the shadow banking system responds non-monotonically to changes in the nominal interest rate as described in the following corollary:

**Corollary 6.** *Suppose that $\gamma$ is continuously distributed on $(0, \Gamma)$ with an increasing hazard rate and $\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$, and $\zeta < \zeta^*$. Then*

1. *if $h(0) > 0$, there exists an $i^1 > 0$, such that for $i \in (0, i^1)$, $\frac{d\hat{\gamma}}{di} > 0$*
2. *for $i > \bar{i}$, $\frac{d\hat{\gamma}}{di} < 0$*

At the Friedman rule, nobody monitors, and the share of sophisticated households rises with the nominal interest rate as the difference in capital investment grows. Above $\bar{i}$, the nominal interest rate no longer affects the break-even point and unsophisticated idle households fully deposit. Therefore, an increase in the nominal interest rate only reduces the money balances on which the advantage of monitoring is earned. The response of the shadow banking system to monetary policy depends on the level of

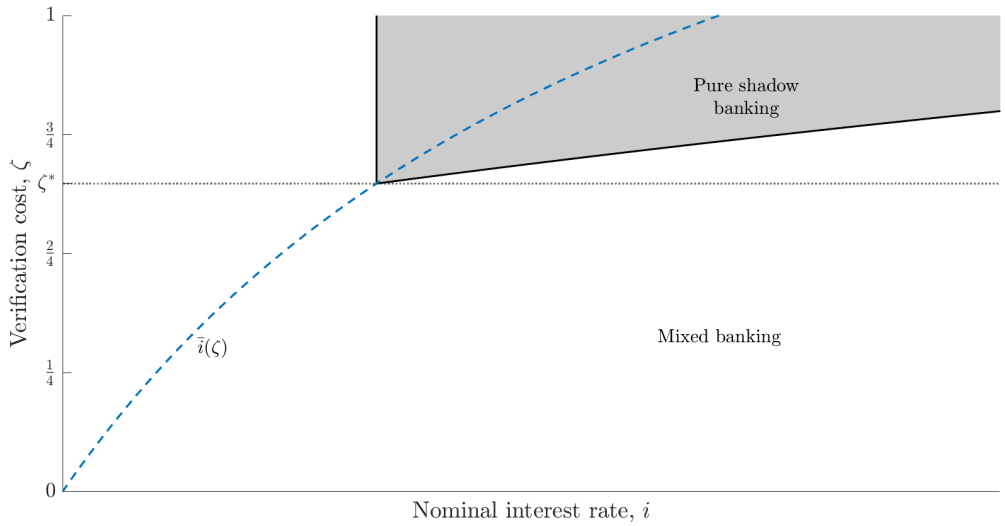



Figure 4: Parameter Space

The figure shows the parameter space in the $\{i, \zeta\}$ plane, where a pure shadow banking equilibrium exists and the threshold $\zeta^*$. The figure is created with $\sigma_1 = 0.6$, $\sigma_2 = 0.1$, $\sigma_3 = 0.3$, $\beta = 0.96$, and $f(k) = 10k^{0.4}$, while $\delta \sim U(0,1)$ and $\gamma \sim U(0, \frac{1}{2})$.

the nominal interest rate. For low rates, a tightening expands the shadow banking system, which is consistent with the empirical evidence of Nelson et al. (2018) and Chen et al. (2018). Specifically, Financial Stability Board (2023) shows that during the 2022 tightening, which started at the zero lower bound, money market funds grew in size. However, for high interest rates the model predicts that the shadow banking system contracts after a monetary policy tightening.

The value of their own money balances for unsophisticated households rises in a pure shadow banking equilibrium. As the interest rate rises further, the value of money for unsophisticated households rises until they start carrying money again. Figure 4 gives a graphical representation of the existence region.[12] For low nominal interest rates or low verification costs, only a mixed banking equilibrium can exist, while for intermediate interest rates and high verification costs a pure shadow banking equilibrium exists. Proposition 7 identifies a sufficient condition, an upper bound on the verification cost, under which the equilibrium with sophisticated households is a mixed banking equilibrium for all nominal interest rates. This upper bound might lie outside the economically relevant range.

**Corollary 7.** *Suppose that $\gamma$ is uniformly distributed on $(0, \Gamma)$. There exists a threshold $\bar{\Gamma}$ such that if $\Gamma \geq \bar{\Gamma}$, then for any $\zeta \in (0, \bar{\zeta})$ and $i > 0$ the equilibrium with sophisticated households is a mixed banking equilibrium.*

Corollary 7 shows that for a uniform distribution of $\gamma$, if the upper bound is sufficiently high, the equilibrium with monitoring is unique and a mixed banking equilibrium for all $\zeta \in (0, \bar{\zeta})$ and nominal interest rates. As shown in Corollary 5, the private

[12] Figure 4 shows a closed interval for the set $\mathcal{I}(\zeta)$. Analytically, $\mathcal{I}(\zeta)$ is closed and bounded, but it may consist of a union of disjoint closed intervals.

monitoring gain is bounded from above. If the share of households with low monitoring cost is low, the liquidity supplied by the shadow banking system is insufficient to sustain a pure shadow banking equilibrium.

# 5 Expected Welfare & Banking Size

My analysis shows how nominal interest rates and monitoring decisions determine the composition of the banking system. In this section, I examine the welfare implications of shadow banks by comparing expected welfare in an equilibrium with shadow banks to expected welfare in the equilibrium with only traditional banks. Because the traditional banking equilibrium exists for all parameters, this comparison measures the welfare effect of access to a monitoring technology at a given interest rate. The welfare results should therefore be interpreted as comparisons across equilibria for given parameters, rather than as statements about optimal monetary policy.

I distinguish between two questions. The first is whether shadow banking raises welfare relative to an economy with only traditional banks. The second is whether the equilibrium share of sophisticated households, and hence the size of the shadow banking system, is efficient. The answers differ because the first question compares the social gain against the average monitoring cost of households who monitor, whereas the second compares it with the cost of the marginal household that is indifferent between monitoring and not monitoring. Shadow banking can therefore improve welfare relative to traditional banking even when the equilibrium share of sophisticated households is not socially optimal.

## 5.1 Private vs. Social Value of Monitoring

Consider first the mixed banking equilibrium, in which some households monitor and deposit through shadow banks, while the remaining households rely on traditional banks. In a mixed banking equilibrium, the capital of unsophisticated entrepreneurs, the break-even point, and the relative deposit share are the same as in the traditional banking equilibrium. Consequently, the presence of shadow banks does not change the capital investment of unsophisticated entrepreneurs. The following lemma shows a useful decomposition of expected welfare in the mixed banking equilibrium.

**Lemma 4.** *In a mixed banking equilibrium, expected welfare is given by:*

$$(1-\beta)\mathcal{W}^M = (1-\beta)\mathcal{W}^T + G(\hat{\gamma})\left(\beta S - \mathbb{E}[\gamma|\gamma \leq \hat{\gamma}]\right) \tag{57}$$

*with:*

$$\begin{aligned} S = {} & \sigma_1 \left[\mathbb{E}[\delta] f(z^1) - z^1 - \left(\mathbb{E}[\delta] f(k^0) - k^0\right)\right] \\ & + \zeta f'(k^0) \left(\sigma_2 + \sigma_3 \frac{d^0}{z^0}\right) \int_0^{\hat{\delta}} \delta dH(\delta) \left(\frac{\sigma_1 k^0 + (\sigma_2+\sigma_3) z^1}{\sigma_1 + \sigma_2 + \sigma_3 \frac{d^0}{z^0}}\right), \end{aligned} \tag{58}$$

*where $S$ is independent of $\hat{\gamma}$.*

Expected welfare in the mixed banking equilibrium is thus expected welfare in the traditional banking equilibrium plus the expected net gain of sophisticated households. The social gain of monitoring, $S$, consists of two components: the additional net output from the larger investment of sophisticated entrepreneurs, and the reduction in the expected cost of verification. As it is independent of the share of sophisticated households, every additional sophisticated household adds the same social gain.

In equilibrium, the marginal household is indifferent: the private gain equals its cost of monitoring, $\Xi(\hat{\gamma}) = \hat{\gamma}$. Consequently, an additional sophisticated household is welfare improving if $\beta S > \Xi(\hat{\gamma})$ and welfare reducing if $\beta S < \Xi(\hat{\gamma})$. Welfare is maximized if all households with a monitoring cost below the social gain monitor, that is, at the efficient threshold $\hat{\gamma}^* = \beta S$. This gives the following conditions on the size of the shadow banking system:

1. If $\beta S < \Xi(\hat{\gamma})$, the private gain exceeds the social gain, too many households monitor, and the shadow banking system is too large.

2. If $\beta S > \Xi(\hat{\gamma})$, the social gain exceeds the private gain, too few households monitor, and the shadow banking system is too small.

The difference between the private and social gains of monitoring consists of two wedges. The first, which I refer to as the financing wedge, comes from investment. The social gain from additional liquidity is the increase in expected output net of the real cost of capital, while the private gain reflects the difference between expected output and the household's own financing cost. Near the Friedman rule, liquidity is abundant, and the investment of both sophisticated and unsophisticated entrepreneurs is close to its efficient level. The extra investment of an additional sophisticated household then has little social value, whereas its private gain also includes a lower cost of financing. As an entrepreneur, a sophisticated household does not need to borrow and avoids paying the loan rate. This lower financing cost is largely a transfer between households through the loan market rather than additional output, so the financing wedge is positive. At high interest rates, liquidity is scarce and the investment gap between sophisticated and unsophisticated entrepreneurs is large. An additional sophisticated household then raises output substantially, but its private gain is smaller, because the real cost of financing its investment exceeds the real cost of capital. The financing wedge becomes negative.

The second, the verification wedge, arises because monitoring reduces the use of traditional banks and the expected cost of verification. A sophisticated household carries larger real money balances, so the shadow banking system originates more loans. These loans replace loans that would otherwise be funded by traditional banks and be subject to costly verification. An individual household, however, internalizes only the effect of monitoring on its own financing and investment. An unsophisticated household chooses its money balances such that the marginal value of money equals the cost of carrying money. As an idle household, the marginal return of depositing reflects

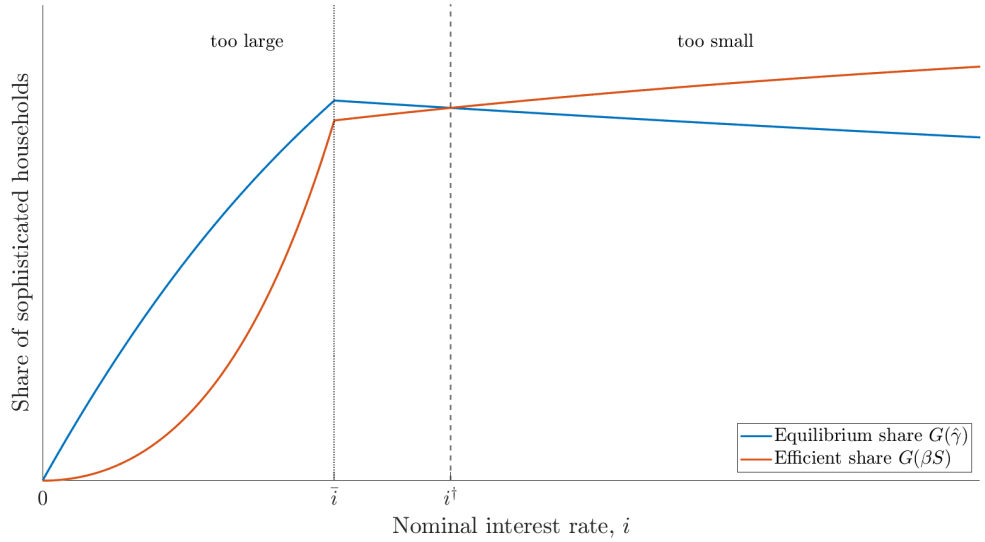



Figure 5: Equilibrium versus Efficient Share of Sophisticated Households

The figure shows the equilibrium share of sophisticated households, $G(\hat{\gamma})$, and the efficient share, $G(\beta S)$, as functions of the nominal interest rate $i$ in a mixed banking equilibrium. The shadow banking system is too large for $i < i^\dagger$ and too small for $i > i^\dagger$, where $i^\dagger$ denotes the interest rate at which the private and social gains of monitoring are equal. The figure is created with $\zeta = 0.2 < \zeta^*$, $\sigma_1 = 0.6$, $\sigma_2 = 0.1$, $\sigma_3 = 0.3$, $\beta = 0.96$, and $f(k) = 10k^{0.4}$, while $\delta \sim U(0,1)$ and $\gamma \sim U(0, \frac{1}{2})$.

the marginal cost of verification, whereas the return itself reflects only the average cost, which is lower. An idle unsophisticated household therefore earns a surplus on its deposits, which it gives up by monitoring. Both effects make the private gain smaller than the social gain, and the size of the verification wedge increases with the probability of verification.

Close to the Friedman rule, the break-even point is low and traditional banks hardly default. The verification wedge is then negligible, the financing wedge dominates and too many households monitor. In other words, the shadow banking system is too large. At higher interest rates, the probability of verification rises and both wedges work in the same direction, so the shadow banking system is too small. Figure 5 illustrates this by plotting the private and social gains from monitoring against the nominal interest rate. The equilibrium size of the shadow banking system can therefore be inefficient in either direction.

### 5.2 Welfare in the Mixed Banking Equilibrium

To compare welfare in the mixed and the traditional banking equilibrium, the social gain must be compared with the expected monitoring cost of the households who monitor, $\mathbb{E}[\gamma|\gamma \leq \hat{\gamma}]$, rather than the private gain of the marginal household. Proposition 8 gives the following two results:

**Proposition 8.** *Suppose that $\gamma$ is continuously distributed on $(0, \Gamma)$ with an increasing hazard rate, $\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$, and $\zeta < \zeta^*$. Then*

1. *If* $h(0) > 0$ *and* $g(0) > 0$*, then for* $i \approx 0$*, expected welfare in the mixed banking equilibrium is lower than in the traditional banking equilibrium.*

2. *For* $i > \bar{i}^W(\zeta)$*, expected welfare in the mixed banking equilibrium is higher than in the traditional banking equilibrium.*

Close to the Friedman rule, the opportunity cost of holding money is low, and households carry enough liquidity to finance a relatively large investment. The loan rate is relatively low, and traditional bank depositors deposit only part of their money balances. The break-even point is therefore low, traditional banks rarely default, and the expected cost of verification is small. Both components of the social gain are small. More precisely, the social gain is of second order in the nominal interest rate: $S$ and its derivative are zero at $i = 0$. The private gain, by contrast, is of first order because of the financing wedge, so the share of sophisticated households, and with it the total monitoring cost incurred, rises at first order in the nominal interest rate. The aggregate monitoring cost therefore exceeds the social gain and expected welfare in the mixed banking equilibrium is lower than in the traditional banking equilibrium.

**Corollary 8.** *Suppose that all conditions of Proposition 8 hold. Then there exists a non-empty interval of nominal interest rates on which* $\beta S < \Xi(\hat{\gamma})$ *and expected welfare in the mixed banking equilibrium is higher than in the traditional banking equilibrium.*

This changes as the nominal interest rate increases. Carrying money becomes more costly, and liquidity becomes more valuable, so the social gain of additional liquidity supplied by shadow banks rises. At the same time, up to $\bar{i}$, traditional bank depositors deposit a larger share of their money balances, the break-even point rises, and traditional banks are more likely to default. The expected cost of verification increases, and so does the social gain from monitoring. Above $\bar{i}^W(\zeta)$, the social gain exceeds the private gain of the marginal household, $\beta S > \Xi(\hat{\gamma})$. Since every sophisticated household has a cost below $\hat{\gamma}$, each of them adds more to welfare than it costs, and the mixed banking equilibrium generates higher expected welfare than the traditional banking equilibrium. Combining the two comparisons gives three regions of interest rates, summarized in Table 1. The welfare difference is positive when $\beta S > \mathbb{E}[\gamma|\gamma \leq \hat{\gamma}]$, while the size of the shadow banking system is efficient when $\beta S = \hat{\gamma}$. Since the average cost of monitoring is below the marginal cost, there is an intermediate range of interest rates in which the shadow banking system is too large but still raises welfare. This is formalized in Corollary 8.

With uniformly distributed monitoring costs, as in Figure 6, $\mathbb{E}[\gamma|\gamma \leq \hat{\gamma}] = \frac{1}{2}\hat{\gamma}$, so the welfare difference is positive whenever $\beta S > \frac{1}{2}\Xi(\hat{\gamma})$. Figure 6 illustrates the welfare difference between the mixed and traditional banking equilibria as a function of the nominal interest rate. The welfare difference is negative at low interest rates and positive at higher interest rates. The figure also marks the interest rate at which the private and social gains from monitoring coincide, which lies above the interest rate at which the welfare difference turns positive, highlighting the distinction between the welfare comparison across equilibria and the efficiency of the equilibrium monitoring decision.

| | Low Rates | Intermediate Rates | High Rates |
|---|---|---|---|
| Comparison with marginal cost | $\beta S < \Xi(\hat{\gamma})$ | $\beta S < \Xi(\hat{\gamma})$ | $\beta S > \Xi(\hat{\gamma})$ |
| Size of Shadow Banking | too large | too large | too small |
| Comparison with average cost | $\beta S < \mathbb{E}[\gamma \vert \gamma \leq \hat{\gamma}]$ | $\beta S > \mathbb{E}[\gamma \vert \gamma \leq \hat{\gamma}]$ | $\beta S > \mathbb{E}[\gamma \vert \gamma \leq \hat{\gamma}]$ |
| Welfare Mixed vs. Traditional | lower | higher | higher |

Table 1: Size and expected welfare of the shadow banking system in a mixed banking equilibrium (Proposition 8 and Corollary 8). That each region is a single interval holds in the numerical example of Figure 6.

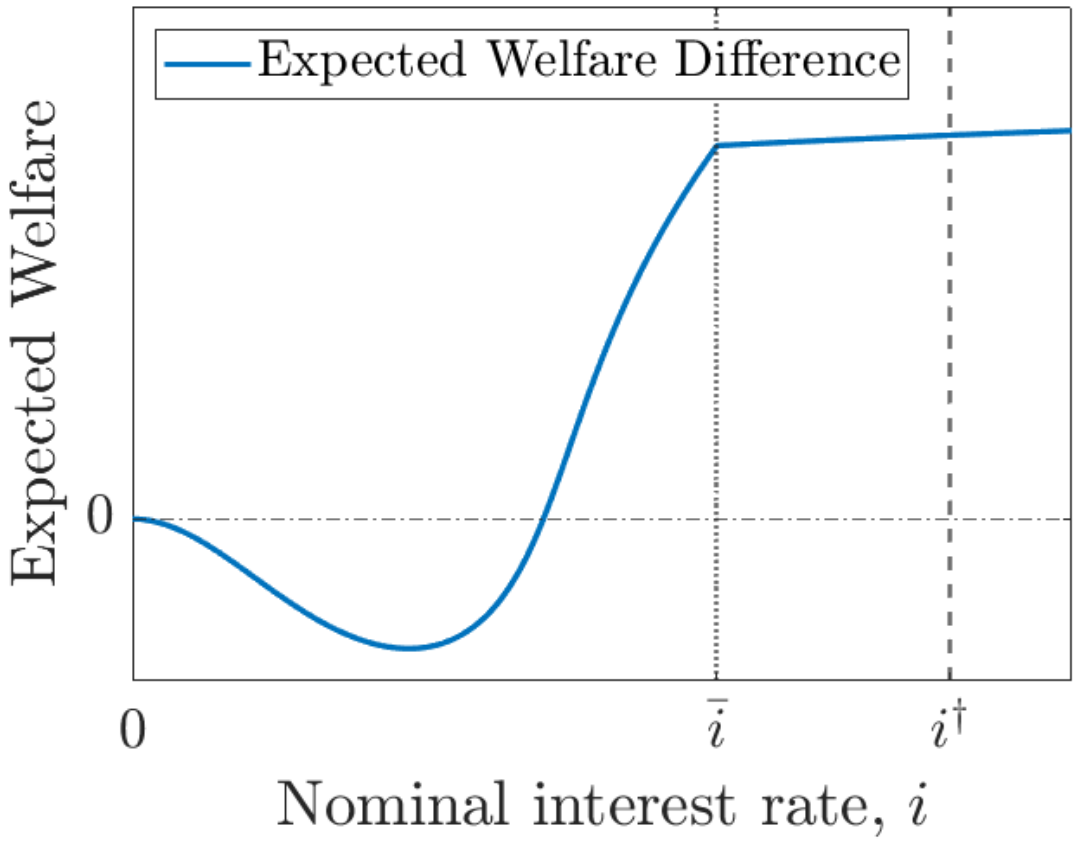



Figure 6: Expected Welfare Difference

The figure shows the expected welfare difference between a mixed banking and a traditional banking equilibrium for the nominal interest rate $i$. On the x-axis it shows both $\bar{i}$, where the equilibrium moves from partial to full depositing, and $i^\dagger$, where the social and private gain of monitoring are equal ($\beta S = \Xi(\hat{\gamma})$, so that the equilibrium share of sophisticated households is the efficient one at that interest rate. The figure is created with $\zeta = 0.2 < \zeta^*$, $\sigma_1 = 0.6$, $\sigma_2 = 0.1$, $\sigma_3 = 0.3$, $\beta = 0.96$, and $f(k) = 10k^{0.4}$, while $\delta \sim U(0,1)$ and $\gamma \sim U(0, \frac{1}{2})$.

The interest rate at which the private and social gains coincide is not necessarily the interest rate at which the welfare difference between the two equilibria is largest. The former identifies when the equilibrium monitoring decision is efficient at a given interest rate, while the latter compares the welfare levels of two banking systems. Neither

determines the interest rate that maximizes overall welfare. Although the welfare advantage of the mixed banking equilibrium can be largest at a positive interest rate, overall welfare is maximized in the limit at the Friedman rule. For any $i > 0$, expected welfare in every equilibrium is below the full-information welfare of Proposition 1. At the Friedman rule, carrying real money is costless, households can fully self-insure against their idiosyncratic shock, and entrepreneurs invest efficiently without relying on either banking system.

The same forces determine how the size of the shadow banking system responds to monetary policy. At the Friedman rule, nobody monitors in the mixed banking equilibrium, and the share of sophisticated households rises with the interest rate as the financing advantage of monitoring grows. At high interest rates, carrying money becomes so costly that the advantage of monitoring is earned on smaller money balances. With monitoring costs uniformly distributed on $(0, \frac{1}{2})$, as in the numerical example, $G(\hat{\gamma})$ is proportional to $\hat{\gamma} = \Xi(\hat{\gamma})$, so the private gain in Figure 5 is proportional to the share of sophisticated households. It rises up to $\bar{i}$ and declines afterwards. As stated in Corollary 6, the size of the shadow banking system does not respond monotonically to the nominal interest rate. For low nominal interest rates, the size of the shadow banking system rises with the nominal interest rate.

### 5.3 Pure Shadow Banking Equilibrium

A similar comparison applies when the economy supports a pure shadow banking equilibrium. In this case, households who do not monitor carry no money, so traditional banks are inactive, and unsophisticated entrepreneurs finance their investment entirely with loans from shadow banks. Corollary 3 characterizes the welfare gain $\gamma^C(i)$ per household that would arise if all households could commit to monitoring. Without commitment, such an equilibrium cannot be sustained with homogeneous monitoring costs. With heterogeneous monitoring costs, a pure shadow banking equilibrium can emerge because households with sufficiently low monitoring costs choose to monitor.

The resulting allocation differs from the commitment allocation as only a share of households monitor and investment differs across entrepreneurs according to their sophistication. Nevertheless, there is no verification, so no resources are lost through the traditional banking technology. Sophisticated entrepreneurs invest more than in the equilibrium with commitment, while unsophisticated entrepreneurs invest at least as much as in the traditional banking equilibrium. The welfare gain per sophisticated household is therefore at least $\gamma^C(i)$. Since every sophisticated household has a cost below that of the marginal household, the pure shadow banking equilibrium raises welfare if the marginal cost is below $\gamma^C(i)$. In a pure shadow banking equilibrium, $\hat{\gamma} \geq \bar{\gamma}$ and $\Xi(\hat{\gamma})$ is decreasing on $(\bar{\gamma}, \tilde{\gamma})$, so the marginal cost of monitoring satisfies $\hat{\gamma} = \Xi(\hat{\gamma}) < \Xi(\bar{\gamma})$. This yields a sufficient condition, independent of the equilibrium threshold, under which shadow banking is welfare improving relative to the traditional banking equilibrium. Proposition 9 formalizes this result.

**Proposition 9.** *Suppose that $\gamma$ is continuously distributed on $(0, \Gamma)$ with an increasing hazard*

*rate,* $\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$, $\zeta \geq \zeta^*$, *and* $i \in \mathcal{I}(\zeta)$. *Then expected welfare in the pure shadow banking equilibrium is higher than in the traditional banking equilibrium if* $\Xi(\bar{\gamma}) < \gamma^C(i)$.

This result complements the analysis of the mixed banking equilibrium. In both cases, the relative welfare consequences of shadow banking depend on the benefit of reducing the information friction relative to the cost of monitoring. However, the relative allocations differ across the two equilibria, which yields different criteria for the relative welfare results. Together, these results show that monetary policy, by affecting the value of liquidity, monitoring incentives, and the expected cost of verification, determines whether shadow banking improves welfare.

## 6 Conclusion

Why do some depositors fund traditional banks, while others fund shadow banks? This paper proposes an information-based explanation. In a monetary model of banking with risky investment and asymmetric information, depositors can mitigate the information friction either by monitoring banks ex ante or by relying on ex post verification, which I associate with shadow and traditional banking, respectively. In symmetric pure-strategy equilibria, homogeneous monitoring costs lead to an equilibrium in which only traditional banks operate. Heterogeneity in monitoring costs is what allows the two banking systems to coexist, and it endogenously segments depositors across them by their monitoring cost.

The mechanism operates through the interaction between information acquisition and aggregate liquidity provision. Sophisticated households earn the full expected loan rate on deposits rather than a deposit return reduced by the expected cost of verification, so they carry more money and deposit more. This changes the amount of liquidity supplied through shadow banks, which in turn changes the private return to monitoring. The composition of the banking system is therefore determined jointly with the supply of liquidity, and both depend on the monetary environment. This makes monitoring decisions strategic complements in a mixed banking equilibrium and strategic substitutes in a pure shadow banking equilibrium, so several equilibria can coexist.

Because money demand is crucial for the mechanism, monetary policy shapes the banking system. An increase in the nominal interest rate expands the shadow banking system for low rates, but contracts at high nominal interest rates. The model thus predicts that the response of shadow banking to monetary policy depends on the level of the nominal interest rate. Monetary policy also determines whether the size of the shadow banking system is optimal. Households do not internalize the effects of their decisions on aggregate liquidity provision, and two wedges drive the difference between the private and social gains of monitoring. The private gain includes a lower cost of financing, which is largely a transfer through the loan market, while households ignore that, by leaving traditional banks, they reduce the expected cost of verification. Close to the Friedman rule, liquidity is abundant and traditional banks rarely default.

The shadow banking system is then too large, and expected welfare is lower than with only traditional banks. At higher interest rates, liquidity becomes more valuable and verification more likely, the shadow banking system is too small and shadow banks improve welfare. The same monetary conditions that make shadow banking grow thus also determine whether its growth is desirable.

These results raise several questions for future research. First, it would be interesting to see how banking regulation interacts with the efficient size of the shadow banking system. The appropriate regulation of traditional banks, for example through capital requirements, reserve requirements, or an interest rate on reserves, may depend on the monetary policy stance. Second, shadow bank deposits are less liquid than traditional bank deposits and generally cannot be used as a medium of exchange (Xiao, 2020). Allowing for inside money would make it possible to evaluate the robustness of the information mechanism and whether it survives when banks also provide liquidity survives. Third, shadow bank funding is runnable (Gorton and Metrick, 2012), and in the model the sophisticated depositors are precisely those households who would run. Combining monitoring with the possibility of runs would show whether the information that makes shadow banks attractive also makes them fragile, providing insights into the workings of the shadow banking system.

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# Appendix

## A Proofs

*Proof of Lemma 1.* The proof follows Williamson (1986) and Winton (1995).

For a given investment portfolio, the bank's income is given by $y = \delta(1 + i_l)L^0 + R^0$. After the realization of $\delta$, the bank observes its income and sends a message to its depositors. Following Winton (1995), and using the revelation principle, the set of messages can be restricted to the domain of $y$. The contract states that for $m \in \mathcal{C}$ the depositor verifies, while if $m \in \mathcal{C}^c$ the depositor does not verify. Let $R(y)$ and $\hat{R}(y)$ denote the payment functions in the case of non-verification and verification, respectively.

First, I show that the bank pays a fixed amount, independent of the realization of $\delta$. Suppose the bank sends a message that does not result in verification. If, in this case, the repayment is state-dependent, the bank will choose the message that corresponds to the minimum repayment possible. As the depositor will not verify, this is a feasible strategy. Hence if the bank sends a message such that the depositors do not verify, it pays a fixed amount to its depositors. Let $R(y) = \bar{R}$ denote this fixed repayment per depositor. If $y \geq \eta\bar{R}$, the bank has sufficient funds to repay its deposits, while if $y < \eta\bar{R}$, the bank has insufficient funds. Consequently, $\eta\bar{R}$ is not only the total fixed repayment but also the break-even point.

If $L^0 = 0$, the bank's income is risk-free and it pays the fixed repayment in every state. Next, I claim that if $L^0 > 0$, then for $y < \eta\bar{R}$, the bank defaults, and the depositors verify and share all the bank's remaining income. This implies that the repayment function per depositor in the case of verification is $\hat{R}(y) = (1 - \zeta)\frac{y}{\eta}$, where $\eta$ is the number of depositors per bank.

Suppose this claim does not hold, and there exists an admissible contract $\bar{R}, \hat{R}(y)$, such that on a set of incomes with positive probability, the bank holds on to some funds $\hat{R}(y) < \frac{(1-\zeta)y}{\eta}$. The expected utility for a depositor under this contract is:

$$U = \int_{y<\eta\bar{R}} \hat{R}(y) dH_y(y) + \int_{y\geq\eta\bar{R}} \bar{R} dH_y(y)$$

while the expected profit of the bank is:

$$\Pi = \int_{y\leq\eta\bar{R}} (1 - \zeta)y - \eta\hat{R}(y) dH_y(y) + \int_{y\geq\eta\bar{R}} y - \eta\bar{R} dH_y(y)$$

Substituting the expected utility into the expected profit of the bank gives:

$$\Pi = \mathbb{E}[y] - \zeta \int_{y<\eta\bar{R}} y \, dH_y(y) - \eta U$$

Now consider an alternative admissible contract, $\{\bar{R}', \hat{R}'(y)\}$ with $\hat{R}'(y) = \frac{(1-\zeta)y}{\eta} \geq \hat{R}(y)$, where the inequality is strict on a set of positive probability. Because depositors' expected utility is continuous and increasing in $\bar{R}$ , there exists an $\bar{R}' < \bar{R}$ such that

depositor receive the same expected utility $U$ for the same investment portfolio. The bank's expected profit under this contract is:

$$\Pi' = \mathbb{E}[y] - \zeta \int_{y<\eta\bar{R}'} y\, dH_y(y) - \eta U$$

and the difference in expected profits is:

$$\Pi' - \Pi = \zeta \left[ \int_{y<\eta\bar{R}} y\, dH_y(y) - \int_{y<\eta\bar{R}'} y\, dH_y(y) \right]$$

As $\bar{R}' < \bar{R}$, this is strictly positive. Consequently, if the claim does not hold, there always exists an alternative contract that increases the expected profit of a banker without changing the expected utility of depositors. Therefore, the optimal repayment function in the case of verification is $\hat{R}(y) = \frac{(1-\zeta)y}{\eta}$. Finally, verifying when $y \geq \eta\bar{R}$ only destroys income without changing what depositors (maximally) receive, so verification occurs exactly when $y < \eta\bar{R}$. □

*Proof of Lemma 2.* First, I rewrite the problem in terms of $d^0$ and $\hat{\delta}$ by substituting $\hat{\delta}$ and the bank's balance sheet into the bank's participation constraint and the objective function:

$$J(\hat{\delta}, d^0) = \max_{\hat{\delta}, d^0} \left[ (1 - H(\hat{\delta}))\hat{\delta} + (1-\zeta) \int_0^{\hat{\delta}} \delta dH(\delta) \right] (1 + i_l) \frac{(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3} + z^0 - d^0 + W(0)$$

subject to:

$$\mathbb{E}[\delta]\sigma_2 e^0 \leq (\sigma_2 e^0 + \sigma_3 d^0) \int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta) \tag{A1}$$

$$0 \leq d^0 \leq z^0 \tag{A2}$$

Given Assumption 1, I show that the participation constraint of the banker always binds. Fix $d^0 \in [0, z^0]$. The RHS of equation A1 is strictly decreasing in $\hat{\delta}$, so the equation holds if and only if $\hat{\delta} \leq \bar{\delta}(d^0)$, where $\bar{\delta}(d^0)$ solves Eq. A1 with equality. The function $\bar{\delta}(d^0)$ is increasing in $d^0$, with $\bar{\delta}(0) = 0$. In equilibrium, bankers and unsophisticated idle households carry the same money balances, $d^0 \leq z^0 = e^0$. Moreover, $\bar{\delta}(z^0)$ is equivalent to $\bar{\delta}$ from Assumption 1. Therefore, I find:

$$\int_{\bar{\delta}(d^0)}^1 (\delta - \hat{\delta}) dH(\delta) = \frac{\sigma_2 e^0}{(\sigma_2 e^0 + \sigma_3 d^0)} \geq \frac{\sigma_2}{\sigma_2 + \sigma_3} = \int_{\bar{\delta}}^1 (\delta - \bar{\delta}) dH(\delta)$$

Consequently $\bar{\delta}(d^0) \leq \bar{\delta}$. Differentiating the objective with respect to $\hat{\delta}$ for a given $d^0$

gives:

$$J_{\hat{\delta}} = (1+i_l)\frac{(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3}\left[(1-H(\hat{\delta})) - \hat{\delta}h(\hat{\delta}) + (1-\zeta)\hat{\delta}h(\hat{\delta})\right]$$
$$= (1-H(\hat{\delta}))(1-\zeta\hat{\delta}\mathcal{H}(\hat{\delta}))(1+i_l)\frac{(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3}$$

As the hazard rate is increasing, $\delta\mathcal{H}(\hat{\delta})$ is increasing, and assumption 1 implies that $\zeta\hat{\delta}\mathcal{H}(\hat{\delta}) < 1$ and $J_{\hat{\delta}} > 0$ on $[0, \bar{\delta}(d^0)]$. Consequently, for any deposit size, depositors strictly prefer a higher break-even point, or a higher fixed repayment. A bank that earns excess profits can therefore always raise $i_d$ to the benefit of depositors. Hence, the participation constraint binds and there is no credit rationing as in Williamson (1987):

$$\mathbb{E}[\delta]\sigma_2 e^0 = (\sigma_2 e^0 + \sigma_3 d^0)\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)$$

As the RHS is decreasing from $\mathbb{E}[\delta](\sigma_2 e^0 + \sigma_3 d^0)$ to zero, the binding participation constraint gives $\hat{\delta}$ as a function of $d^0$. Implicitly differentiating gives:

$$\frac{\partial\hat{\delta}}{\partial d^0} = \frac{\sigma_3\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)}{(1-H(\hat{\delta}))(\sigma_2 e^0 + \sigma_3 d^0)}$$

Consequently, the problem can be rewritten as choosing only $d^0$:

$$J(d) = \max_{d^0\in[0,z^0]}\left\{\left[(1-H(\hat{\delta}))\hat{\delta} + (1-\zeta)\int_0^{\hat{\delta}}\delta dH(\delta)\right]\frac{(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3} + z^0 - d^0\right\} + W(0)$$

Substituting the binding participation constraint into the objective gives:

$$J(d) = \max_{d^0\in[0,z^0]}(\mathbb{E}[\delta](1+i_l) - 1)d^0 - \zeta\int_0^{\hat{\delta}}\delta dH(\delta)(1+i_l)\frac{\sigma_2 e^0 + \sigma_3 d^0}{\sigma_3} + z^0 + W(0)$$

Ignoring the box constraint on $d$, the F.O.C. and S.O.C. are:

$$J'(d) = \mathbb{E}[\delta](1+i_l) - 1 - \zeta(1+i_l)\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)$$
$$J''(d) = -\zeta(1+i_l)\left(\mathcal{H}'(\hat{\delta})\hat{\delta} + \mathcal{H}(\hat{\delta})\right)\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\frac{\partial\hat{\delta}}{\partial d}$$

As the hazard rate $\mathcal{H}(\hat{\delta})$ is increasing, the problem is concave. As at $d = 0$, $\hat{\delta} = 0$ and

$J'(0) > 0$, the deposit size is given by:

$$1 + \lambda = (1 + i_l) \left[ \mathbb{E}[\delta] - \zeta \left( \int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta} \int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta) \right) \right]$$

$$\mathbb{E}[\delta]\sigma_2 e^0 = (\sigma_2 e^0 + \sigma_3 d^0) \int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)$$

$$\lambda(z^0 - d^0) = 0$$

□

*Proof of Proposition 1.* If $\gamma = 0$, all households monitor. This implies that $G(\hat{\gamma}) = 1$. Dropping the superscript, $J = 1$, I can reduce an equilibrium to the tuple $\{k, z\}$ using:

1. Equation 2: $p = 1$
2. Equation 10: $l = k - z$
3. Banks invest all money balances: $e = z$
4. Shadow bank deposit repayment function satisfies Equation 17:

$$\frac{\sigma_3}{\sigma_2} z\theta(\delta) \leq \delta(1 + i_l) \left( \frac{\sigma_3}{\sigma_2} z + e \right), \forall \delta \in [0, 1],$$

$$\mathbb{E}[\delta](1 + i_l) \left( \frac{\sigma_3}{\sigma_2} z + e \right) - \mathbb{E}\left[\theta(\delta)\right] \frac{\sigma_3}{\sigma_2} z = \mathbb{E}[\delta](1 + i_l)e$$

5. The gross growth rate of money, $\mu$, the initial money supply $M_0$, and market clearing in money market Equation 31: $\phi = \frac{z}{\mu^t M_0}$

Given the conjecture $\mathbb{E}[\delta](1 + i_l) > 1$, banks invest in loans. Consequently, for the loan market to clear, entrepreneurs have to borrow. Therefore, the optimal loan size is given by Equation 13, which using $p = 1$ and $z + l = k$ becomes:

$$f'(k) = 1 + i_l$$

Before characterizing the full equilibrium, I derive the marginal value of money for households in the DM. First, entrepreneurs take $i_l$ and $p$ as given. Using that $pk = z + l$ and $f'(k) = 1 + i_l$, from Equation 14 the marginal value of money for an entrepreneur is:

$$V_z^e(z) = \mathbb{E}[\delta] f'(k)$$

From Equations 20 and 19, the marginal value of money for a shadow banker and an idle household is:

$$V_z^b(z) = \mathbb{E}[\delta](1 + i_l)$$
$$V_z^i(z) = \mathbb{E}[\delta](1 + i_l)$$

Combining according to Equation 9 and substituting into the optimal choice equation gives:

$$1 + i = \mathbb{E}[\delta] f'(k)$$

Using loan market clearing (Equation 30) gives $\sigma_1 k = z$.

Finally, I verify my three conjectures: i) the price of money is independent of the current realization of $\delta$, ii) households have no incentive to deviate and not monitor, and iii) $\mathbb{E}[\delta](1 + i_l) > 1$.

First, $z$ is independent of the realization of $\delta$, so $\phi = \frac{z}{\mu^t M_0}$ is also independent of $\delta$.

Second, households take prices and contracts as given. Consider a household who deviates and does not monitor. If they become an entrepreneur, they face the same problem as any other entrepreneur given their money balances. If they become a banker, they can set up a traditional bank and receive their participation constraint $\mathbb{E}[\delta](1 + i_l)$. If they become idle, no traditional bank exists, so they can only keep their money balances. The choice of money balances is:

$$1 + i \geq \sigma_1 \mathbb{E}[\delta] f'(k) + \sigma_2 \mathbb{E}[\delta](1 + i_l) + \sigma_3$$

A deviating unsophisticated household's value of money is lower than that of sophisticated households. Therefore, given the nominal interest rate, the optimal money choice is $z = 0$ and $l = k$. The utility of deviating is then given by:

$$\hat{W} = \tau + \sigma_1 \beta \mathbb{E}[\delta] \left( f(k) - (1 + i_l)k \right) + \beta W(0)$$

where I used the properties of a homogeneous function. The expected utility of monitoring:

$$\begin{aligned} W &= \tau - \gamma - \mu z_{+1} + \beta \left( \sigma_1 \mathbb{E}[\delta] \left( f(k) - f'(k) l \right) + (\sigma_2 + \sigma_3) \mathbb{E}[\delta](1 + i_l) z \right) + \beta W(0) \\ &= \tau - \gamma + \sigma_1 \beta \mathbb{E}[\delta] \left( f(k) - (1 + i_l)k \right) + \beta W(0) \end{aligned}$$

where I used that $\frac{\mu}{\beta} = 1 + i = \mathbb{E}[\delta](1 + i_l)$ and that $\sigma_1 k = z$. Therefore the difference is:

$$W - \hat{W} = -\gamma$$

Consequently, a household has no incentive to deviate if $\gamma = 0$.

Third, suppose that $\mathbb{E}[\delta] f'(k) \leq 1$. Then shadow banks would not invest in loans. Therefore, the supply of loans is zero and $l = 0$, $k = z$. Entrepreneurs invest at least until $\mathbb{E}[\delta] f'(k) = 1$, so if $\mathbb{E}[\delta] f'(k) > 1$, each entrepreneur strictly prefers to borrow, which contradicts the zero loan supply. Suppose instead that $\mathbb{E}[\delta] f'(k) = 1$, which implies that $k = k^*$. Then the marginal value of money equals 1 for every household, and Equation 5 requires $i = 0$, which contradicts $i > 0$. □

*Proof of Proposition 2.* I solve for the equilibrium under the conjecture that households do not monitor and then check the validity of this conjecture. This implies that $G(\hat{\gamma}) = 0$.

Dropping the superscript, $J = 0$, I can reduce an equilibrium to the tuple $\{k, d, z, \hat{\delta}\}$ using:

1. Equation 2: $p = 1$
2. Equation 10: $l = k - z$
3. Banks invest all money balances: $e = z$
4. Lemma 2: $1 + i_d = \frac{\hat{\delta}(1+i_l)(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3 d^0}$
5. The gross growth rate of money, $\mu$, the initial money supply $M_0$, and market clearing in money market Equation 31: $\phi = \frac{z}{\mu^t M_0}$

Moreover, given the conjecture $\mathbb{E}[\delta](1 + i_l) > 1$, banks invest in loans. Consequently, for the loan market to clear, entrepreneurs have to borrow. Therefore, the optimal loan size is given by Equation 13, which using $p = 1$ and $z + l = k$ becomes:

$$f'(k) = 1 + i_l$$

Before characterizing the full equilibrium, I derive the marginal value of money for households in the DM. First, entrepreneurs take $i_l$ and $p$ as given. Using that $pk = z + l$ and $f'(k) = 1 + i_l$, from Equation 14 the marginal value of money for an entrepreneur is:

$$V_z^e(z) = \mathbb{E}[\delta] f'(k)$$

From Equation 27, the marginal value of money for a traditional banker is:

$$V_z^b(z) = \mathbb{E}[\delta](1 + i_l)$$

From Equation 28, the marginal value of money for an idle household depends on the idle household's deposit size. Lemma 2 states that there are two cases: either the household deposits efficiently and $d$ is independent of $z$ or the liquidity requirement binds, $d = z$. Differentiating gives the following marginal value of money:

$$V_z^i(z) = \begin{cases} (1 + i_l)\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)\right], & \text{if } z < d^* \\ 1, & \text{if } z \geq d^* \end{cases}$$

Combining the different marginal values of money according to Equation 9 and substituting in Equation 5, which holds with equality as $z > 0$, gives:

$$1 + i = \begin{cases} f'(k)\left(\mathbb{E}[\delta] - \sigma_3\zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)\right), & \text{if } z < d^* \\ (\sigma_1 + \sigma_2)\mathbb{E}[\delta] f'(k) + \sigma_3, & \text{if } z \geq d^* \end{cases}$$

where I used $e = z$ and Equation 13.

If $d = d^*$, then $f'(k)\left(\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)\right) = 1$. Substituting this into the first case shows that both cases coincide at this point. The right-hand

side of the money choice equation is therefore continuous in $z$, and I consider the two cases in turn. It is useful to define $\psi \equiv \frac{d}{z}$ as the share of real money balances deposited.

Consider the case in which the liquidity constraint binds $d = z$ and $\psi = 1$. From Equation 26, the break-even point is:

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2+\sigma_3} = \int_{\hat{\delta}}^{1}(\delta-\hat{\delta})dH(\delta)$$

As the RHS is decreasing from $\mathbb{E}[\delta]$ to zero, there is a unique solution, which I denote by $\bar{\delta}$ as in Assumption 1. The money choice equation is then:

$$1+i = f'(k)\left(\mathbb{E}[\delta]-\sigma_3\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^{1}(\delta-\bar{\delta})dH(\delta)\right)\right)$$

Loan market clearing, then gives that $\sigma_1 k = z$. From Lemma 2, this equilibrium case occurs if $z < d^*$:

$$f'(k)\left(\mathbb{E}[\delta]-\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^{1}(\delta-\bar{\delta})dH(\delta)\right)\right) > 1$$

Using the money choice equation, this gives:

$$i > \bar{i} = \frac{(1-\sigma_3)\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^{1}(\delta-\bar{\delta})dH(\delta)\right)}{\left(\mathbb{E}[\delta]-\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^{1}(\delta-\bar{\delta})dH(\delta)\right)\right)}$$

Next, suppose that $d = d^*$, so $\psi \in [0,1)$. Then the money choice equation is given by:

$$1+i = (\sigma_1+\sigma_2)\mathbb{E}[\delta]f'(k)+\sigma_3$$

Equation 26 then gives the following break-even point:

$$\frac{\sigma_2}{\sigma_2+\sigma_3\psi}\mathbb{E}[\delta] = \int_{\hat{\delta}}^{1}(\delta-\hat{\delta})dH(\delta)$$

This gives $\hat{\delta}$ as an increasing function of $\psi$, with $\psi = 0$ at $\hat{\delta} = 0$ and $\psi = 1$ at $\hat{\delta} = \bar{\delta}$. Lastly, $\psi$ is then given by Equation 25:

$$1 = f'(k)\left(\mathbb{E}[\delta]-\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^{1}(\delta-\hat{\delta})dH(\delta)\right)\right)$$

$$i = \frac{(1-\sigma_3)\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^{1}(\delta-\hat{\delta})dH(\delta)\right)}{\left(\mathbb{E}[\delta]-\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^{1}(\delta-\hat{\delta})dH(\delta)\right)\right)}$$

The RHS is increasing in $\hat{\delta}$ and thus increasing in $\psi$. At $\psi = 0$, the RHS is zero, while at $\psi = 1$ the RHS is $\bar{i}$. Therefore, for $i < \bar{i}$, there exists a unique $\psi$ that solves the equation.

Loan market clearing Eq. 30 gives $\sigma_1 k = (\sigma_1 + \sigma_2 + \sigma_3\psi)z$ and then $d$ is given by $d = \psi z$.

Finally, I verify my three conjectures: i) the price of money is independent of the current realization of $\delta$, ii) households have no incentive to deviate and monitor, and iii) $\mathbb{E}[\delta](1 + i_l) > 1$.

First, $z$ is independent of the realization of $\delta$, so $\phi = \frac{z}{\mu^t M_0}$ is also independent of $\delta$.

Second, households take prices and contracts as given. Consider a household who deviates and monitors. If they become an entrepreneur, they face the same problem as any other entrepreneur. If they become a banker, they can set up a shadow bank, but no sophisticated idle households exist to deposit in it, so they earn the outside option $\mathbb{E}[\delta](1 + i_l)$, as a traditional banker does. If they become idle, no shadow bank exists, so they can deposit only in a traditional bank on the same terms as other depositors.[13] Consequently, both traditional bankers and depositors face the same marginal value of money. As the marginal value of money for an entrepreneur is equal to or lower than that of a non-deviating entrepreneur, for a given nominal interest rate the deviating household chooses the same real money balances. Therefore, deviating yields a negative payoff of $-\gamma$ and no household wants to deviate.

Third, suppose that $\mathbb{E}[\delta]f'(k) \leq 1$. Then traditional banks would not invest in loans. Therefore, the supply of loans is zero and $l = 0$, $k = z$. Entrepreneurs invest at least until $\mathbb{E}[\delta]f'(k) = 1$, so if $\mathbb{E}[\delta]f'(k) > 1$, each entrepreneur strictly prefers to borrow, which contradicts the zero loan supply. Suppose instead that $\mathbb{E}[\delta]f'(k) = 1$, which implies that $k = k^*$. Then the marginal value of money equals 1 for every household, and Equation 5 requires $i = 0$, which contradicts $i > 0$. □

*Proof of Corollary 1.* With a uniform distribution, expected welfare (Equation 32) in a traditional banking economy is given by

$$\frac{(1-\beta)}{\beta}\mathcal{W} = \sigma_1 \left(\tfrac{1}{2}f(k) - k\right) - \zeta \tfrac{1}{2}f'(k)\left(\sigma_2 z + \sigma_3 d\right)\hat{\delta}^2$$

where I used that $\hat{\gamma} = 0$. From Proposition 2, for $i < \bar{i}$ the allocation with a uniform distribution is given by:

$$i = (1 - \sigma_3)(\mathbb{E}[\delta]f'(k) - 1) \tag{A3}$$

$$1 = \frac{1}{2}f'(k)\left[1 - \zeta\hat{\delta}\right] \tag{A4}$$

$$\sigma_1 k = (\sigma_1 + \sigma_2)z + \sigma_3 d \tag{A5}$$

$$\hat{\delta} = 1 - \sqrt{\frac{\sigma_2 z}{\sigma_2 z + \sigma_3 d}} \tag{A6}$$

From Equation A3, the amount of capital invested is independent of $\zeta$. Combining

[13] Even if the deviator can observe $\delta$, it is exposed to the verification cost. Verification is triggered if a single depositor asks for verification.

Equations A3 and A4 gives:

$$\zeta\hat{\delta} = \frac{i}{1+i-\sigma_3}$$

Denote by $q \equiv 1-\hat{\delta} = 1 - \frac{i}{\zeta(1+i-\sigma_3)}$, which is increasing in $\zeta$. Using Equations A5 and A6, I can write:

$$z = \frac{\sigma_1 q^2}{\sigma_1 q^2 + \sigma_2} k \tag{A7}$$

$$(\sigma_2 z + \sigma_3 d) = \frac{\sigma_2 z}{q^2} = \frac{\sigma_1 \sigma_2 k}{\sigma_1 q^2 + \sigma_2} \tag{A8}$$

Using these expressions in expected welfare gives:

$$\frac{(1-\beta)}{\beta}\mathcal{W} = \sigma_1 \left[\frac{1}{2}f(k) - k\right] - \frac{1}{2}f'(k)\sigma_1\sigma_2 k \left[\frac{i(1-q)}{(1+i-\sigma_3)(\sigma_1 q^2 + \sigma_2)}\right]$$

which is increasing in $q$ and thus increasing in $\zeta$. □

*Proof of Corollary 2.* From Propositions 1 and 2, the amount of capital invested is given by Equations 33, 35, and 39. For $i \leq \bar{i}$, this implies that:

$$\mathbb{E}[\delta]f'(k^B) - 1 = (\sigma_1 + \sigma_2)\left(\mathbb{E}[\delta]f'(k^T) - 1\right)$$

As $f'(k)$ is decreasing, this implies that $k^B > k^T$. Similarly, if $i > \bar{i}$, this implies that:

$$\mathbb{E}[\delta]f'(k^B) - \mathbb{E}[\delta]f'(k^T) = -\zeta\sigma_3 f'(k^T)\left[\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right]$$

As the RHS is strictly negative, $k^B > k^T$.

Using Equation 32, the expected welfare difference is

$$\Delta\mathcal{W} = \frac{(1-\beta)}{\beta}(\mathcal{W}^B - \mathcal{W}^T) = \sigma_1\left[\mathbb{E}[\delta]f(k^B) - k^B - (\mathbb{E}[\delta]f(k^T) - k^T)\right] \tag{A9}$$

$$+ \zeta f'(k^T)(\sigma_2 z^T + \sigma_3 d^T)\int_0^{\hat{\delta}} \delta dH(\delta) \tag{A10}$$

For $k < k^*$, $\mathbb{E}[\delta]f(k) - k$ is increasing in $k$. For $i > 0$, the amount of capital invested is below $k^*$ in both equilibria. Therefore, both bracketed terms are strictly positive and the welfare difference is positive. □

*Proof of Proposition 3.* This proof follows directly from the proof of Proposition 1 with a positive monitoring cost. □

*Proof of Corollary 3.* From the proof of Corollary 2, the welfare difference between the benchmark case and the traditional banking equilibrium is strictly positive for any nominal interest rate. As the shadow banking economy with commitment implements the benchmark allocation given that households monitor, the only difference is the

monitoring cost. Therefore, for any $i$ there exists an upper bound on $\gamma$ below which expected welfare is higher if all households commit to monitoring:

$$\gamma^C(i) = \beta\sigma_1\left(\mathbb{E}[\delta]f(k^B) - k^B - \mathbb{E}[\delta]f(k^T) + k^T\right) + \beta\zeta\alpha\sigma_1 f(k^T)\frac{(\sigma_2+\sigma_3\psi)}{\sigma_1+\sigma_2+\sigma_3\psi}\int_0^{\hat{\delta}}\delta dH(\delta) \tag{A11}$$

□

*Proof of Lemma 3.* The lemma consists of two results. I first construct the money choice equations for both types of households. Equation 14 gives the following marginal value of money:

$$V_z^{J,e}(z) = \mathbb{E}[\delta]f'(k^J)$$

where I used that $p = 1$. From Equations 27 and 20, the marginal value of money for a banker is:

$$V_z^{J,b}(z) = \mathbb{E}[\delta](1+i_l)$$

From Equation 28, the marginal value of money for an unsophisticated idle household depends on the idle household's deposit size. As in the proof of Proposition 2, differentiating gives the following marginal value of money:

$$V_z^{0,i}(z) = \begin{cases} (1+i_l)\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right], & \text{if } z \le d^* \\ 1, & \text{if } z > d^* \end{cases}$$

Equation 19 gives the marginal value of money for a sophisticated idle household:

$$V_z^{1,i}(z) = \mathbb{E}[\delta](1+i_l)$$

Combining according to Equation 9 and substituting into Equation 5 gives:

$$1 + i \ge \sigma_1\mathbb{E}[\delta]f'(k^1) + (\sigma_2+\sigma_3)\mathbb{E}[\delta](1+i_l) \tag{A12}$$

$$1 + i \ge \begin{cases} \sigma_1\mathbb{E}[\delta]f'(k^0) + \sigma_2\mathbb{E}[\delta](1+i_l) \\ \quad + \sigma_3(1+i_l)\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right], & \text{if } z < d^* \\ \sigma_1\mathbb{E}[\delta]f'(k^0) + \sigma_2\mathbb{E}[\delta](1+i_l) + \sigma_3, & \text{if } z > d^* \end{cases} \tag{A13}$$

which bind with equality if $z^J > 0$. At least one of the two has to bind with equality. Otherwise, $k^0 = k^1 = 0$, which is impossible due to the Inada conditions.

I claim that $l^0 > l^1 = 0$. Suppose not, so that $l^1 > 0$. Then sophisticated entrepreneurs borrow, so $f'(k^1) = (1+i_l)$, and money earns the expected loan rate in every state for a sophisticated household.

Since deposits and cash earn strictly less than the expected loan rate, the marginal value of money for an unsophisticated household is strictly below $\mathbb{E}[\delta](1+i_l)$, the value for sophisticated households. Hence, $z^0 = 0$ and unsophisticated households bring no money. This rules out $l^0 = 0$ as it would imply that $k^0 = z^0 = 0$, which contradicts the Inada conditions. If $l^0 > 0$, both types borrow at the same loan rate, so $k^1 = k^0 = k$. The expected utility of monitoring gross of their monitoring cost is:

$$\begin{aligned} W^1 &= \tau - \mu z^1 + \beta \left[\sigma_1 \mathbb{E}[\delta] f(k) - \sigma_1 \mathbb{E}[\delta](1+i_l) l^1 + (\sigma_2 + \sigma_3)\mathbb{E}[\delta](1+i_l) z^1\right] + \beta W(0) \\ &= \tau + \sigma_1 \beta \mathbb{E}[\delta] \left(f(k) - f'(k)k\right) + \beta W(0) \end{aligned}$$

where I used that $l^1 = k - z^1$ from Equation 10 and that $\frac{\mu}{\beta} = 1 + i$. Similarly, the expected utility of not monitoring is:

$$W^0 = \tau + \sigma_1 \beta \mathbb{E}[\delta](f(k) - f'(k)k) + \beta W(0)$$

Using Equation 8 gives $\hat{\gamma} = 0$. Since $G(0) = 0$, no household monitors, which contradicts $l^1 > 0$. Hence, $l^1 = 0$, which implies that $k^1 = z^1$. Consequently, from loan market clearing Equation 30, I find that $l^0 > 0$. The second result, $z^1 > z^0 \geq 0$, follows because $z^0 = k^0 - l^0 < k^0$, while $z^1 = k^1$ and $f'(k^1) \leq f'(k^0)$. □

*Proof of Proposition 4.* First, I can reduce an equilibrium to the tuple $\{k^0, d^0, z^1, z^0, \hat{\delta}, \hat{\gamma}\}$ using:

1. Equation 2: $p = 1$
2. Lemma 3: $l^1 = 0$, $l^0 > 0$
3. Equation 10: $l^0 = k^0 - z^0$
4. Banks invest all money balances: $e^J = z^J$
5. Lemma 2: $1 + i_d = \frac{\hat{\delta}(1+i_l)(\sigma_2 e^0 + \sigma_3 d^0)}{\sigma_3 d^0}$
6. Shadow bank deposit repayment function satisfies Equation 17:

$$\frac{\sigma_3}{\sigma_2} z \theta(\delta) \leq \delta(1+i_l)\left(\frac{\sigma_3}{\sigma_2} z + e\right), \forall \delta \in [0,1],$$

$$\mathbb{E}[\delta](1+i_l)\left(\frac{\sigma_3}{\sigma_2} z + e\right) - \mathbb{E}\left[\theta(\delta)\right] \frac{\sigma_3}{\sigma_2} z = \mathbb{E}[\delta](1+i_l) e$$

7. Equation 13: $1 + i_l = f'(k^0)$.
8. The gross growth rate of money, $\mu$, the initial money supply $M_0$, and market clearing in money market Equation 31: $\phi = \frac{G(\hat{\gamma}) z^1 + (1 - G(\hat{\gamma})) z^0}{\mu^t M_0}$

This proof consists of two parts. First, I solve the model for a given $\hat{\gamma}$. Second, I analyze the fixed point equation for $\hat{\gamma}$. Following the steps in the proof of Lemma 3, I

find Equations A12 and A13:

$$1+i=\sigma_1\mathbb{E}[\delta]f'(z^1)+(\sigma_2+\sigma_3)\mathbb{E}[\delta]f'(k^0) \tag{A14}$$

$$1+i\geq f'(k^0)\left[\mathbb{E}[\delta]-\sigma_3\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right] \tag{A15}$$

where I used that $f'(k^0)=1+i_l$ and $k^1=z^1$, and in the case of partial deposits, that Equation 25 gives $d=d^*$ . The second equation binds if $z^0>0$. Therefore, there are two potential cases: either $z^0>0$ or $z^0=0$. I solve each case separately below.

First, suppose that $z^0>0$. Then $z^1$ and $k^0$ are given by:

$$1+i=f'(k^0)\left[\mathbb{E}[\delta]-\sigma_3\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right] \tag{A16}$$

$$f'(z^1)=f'(k^0)\left[1-\frac{\sigma_3}{\mathbb{E}[\delta]\sigma_1}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right] \tag{A17}$$

Denote by $\rho\equiv\frac{z^1}{k^0}$. Using the properties of homogeneous functions, $\rho$ is given by:

$$\rho=\left[1-\frac{\sigma_3}{\mathbb{E}[\delta]\sigma_1}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]^{\frac{1}{(\alpha-1)}} \tag{A18}$$

where $\rho\geq 1$ as $\sigma_3\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)<\sigma_1\mathbb{E}[\delta]$ and $\alpha\in(0,1)$.

Second, define $\psi\equiv\frac{d^0}{z^0}\in[0,1]$ as the share of real money balances deposited by unsophisticated households. Following the steps of the proof of Proposition 2, there are two cases: the liquidity constraint binds and $\psi=1$ or the constraint is slack and $\psi\in[0,1)$. First, consider the case in which the liquidity constraint binds, $d^0=z^0$ and $\psi=1$. From Equation 26, the break-even point is:

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2+\sigma_3}=\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)$$

As the RHS is decreasing from $\mathbb{E}[\delta]$ to zero, there is a unique solution, which I denote again by $\bar{\delta}$. The money choice equation is then:

$$1+i=f'(k^0)\left(\mathbb{E}[\delta]-\sigma_3\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^1(\delta-\bar{\delta})dH(\delta)\right)\right)$$

From Lemma 2, this equilibrium case occurs if $z^0\leq d^*$:

$$f'(k^0)\left(\mathbb{E}[\delta]-\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta)+\mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^1(\delta-\bar{\delta})dH(\delta)\right)\right)\geq 1$$

Using the money choice equation, this gives:

$$i \geq \bar{i} = \frac{(1-\sigma_3)\zeta\left(\int_0^{\bar{\delta}} \delta dH(\delta) + \mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^1 (\delta - \bar{\delta})dH(\delta)\right)}{\left(\mathbb{E}[\delta] - \zeta\left(\int_0^{\bar{\delta}} \delta dH(\delta) + \mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^1 (\delta - \bar{\delta})dH(\delta)\right)\right)} \tag{A19}$$

Next, suppose that $z^0 > d^*$ so $\psi \in [0,1)$. Equation 26 then gives the following break-even point:

$$\frac{\sigma_2}{\sigma_2 + \sigma_3\psi}\mathbb{E}[\delta] = \int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta) \tag{A20}$$

This gives $\hat{\delta}$ as an increasing function of $\psi$, with $\hat{\delta} = 0$ at $\psi = 0$ and $\hat{\delta} = \bar{\delta}$ at $\psi = 1$. Lastly, $\psi$ is then given by Equation 25:

$$\begin{aligned} 1 &= f'(k^0)\left(\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)\right) \\ i &= \frac{(1-\sigma_3)\zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)}{\left(\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)\right)} \end{aligned} \tag{A21}$$

where I used the money choice equation. The RHS is increasing in $\hat{\delta}$ and thus increasing in $\psi$. At $\psi = 0$, the RHS is zero, while at $\psi = 1$ the RHS is $\bar{i}$. Therefore, for $i < \bar{i}$, there exists a unique $\psi$ that solves the equation.

Lastly, the real money balances of the unsophisticated households are given by loan market clearing (Eq. 30), $l^1 = 0$, and $l^0 = k^0 - z^0$:

$$z^0 = \frac{\sigma_1 k^0 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2 + \sigma_3)z^1}{(\sigma_1 + \sigma_2 + \sigma_3\psi)} \tag{A22}$$

Consequently, $\{k^0, \rho, z^1, \psi, \hat{\delta}\}$ are independent of $\hat{\gamma}$, while only $z^0$ depends on $\hat{\gamma}$. For $z^0 > 0$ to hold, I find an upper bound on $\hat{\gamma}$ given by:

$$\bar{\gamma} = G^{-1}\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2 + \sigma_3)\rho}\right] \tag{A23}$$

Next, suppose that $z^0 = 0$. Then Equation 30 gives $z^1(\sigma_2 + \sigma_3)\frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))} = \sigma_1 k^0$, or $k^0 = \frac{(\sigma_2+\sigma_3)G(\hat{\gamma})}{\sigma_1(1-G(\hat{\gamma}))}z^1$. Then using the properties of a homogeneous function and the money choice equation for sophisticated households gives:

$$1 + i = \mathbb{E}[\delta]f'(z^1)\left[\sigma_1 + (\sigma_2 + \sigma_3)^{\alpha}\sigma_1^{1-\alpha}\left(\frac{(1-G(\hat{\gamma}))}{G(\hat{\gamma})}\right)^{1-\alpha}\right] \tag{A24}$$

As $f''(k) < 0$, $z^1$ is decreasing in $\hat{\gamma}$. If $z^0 = 0$, then in the traditional banking problem both the liquidity constraint and the non-negativity constraint bind. As an idle household, an unsophisticated household decides how much of an infinitesimal extra

unit of money to deposit. If it deposits only part of it, the marginal value of money equals the value of money in the DM. Otherwise, it equals the expected marginal return on deposits. The marginal value of money for an unsophisticated idle household is therefore given by Lemma 2, while that of an entrepreneur or banker is the expected loan rate. Implicitly, I evaluate the marginal value of money for households along the symmetric path, that is, the household is offered the contract of Lemma 2 at the equilibrium share $\psi$, which is the limit of the interior case as the money balances of the unsophisticated households, and hence the equity of traditional bankers, shrink to zero.[14]

As in the previous case, the marginal value of money is:

$$f'(k^0)\left[\mathbb{E}[\delta]-\zeta\sigma_3\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]$$

where $\hat{\delta}=\bar{\delta}$ and otherwise solves the partial deposit condition if $i<\bar{i}$. Unsophisticated households then carry no money if the marginal value of money does not exceed the cost of carrying money, $1+i$:

$$1+i\geq f'(k^0)\left[\mathbb{E}[\delta]-\zeta\sigma_3\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right] \tag{A25}$$

Using Equation A12, this is equivalent to:

$$f'(z^1)\geq f'(k^0)\left[1-\frac{\sigma_3}{\sigma_1\mathbb{E}[\delta]}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]$$

$$\left[\frac{\sigma_1(1-G(\hat{\gamma}))}{(\sigma_2+\sigma_3)G(\hat{\gamma})}\right]^{\alpha-1}\geq\left[1-\frac{\sigma_3}{\sigma_1\mathbb{E}[\delta]}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]=\rho^{\alpha-1}$$

where $\rho$ is defined in the case $z^0>0$. Since $\alpha-1<0$, this is equivalent to $\frac{\sigma_1(1-G(\hat{\gamma}))}{(\sigma_2+\sigma_3)G(\hat{\gamma})}\leq\rho$. Rewriting gives:

$$\hat{\gamma}\geq G^{-1}\left[\frac{\sigma_1}{\sigma_1+(\sigma_2+\sigma_3)\rho}\right]=\bar{\gamma} \tag{A26}$$

As $\rho$ is independent of $\hat{\gamma}$ and only depends on the nominal interest rate and the verification cost, there is a single threshold $\bar{\gamma}$ that separates the two cases. In addition, at $\hat{\gamma}=\bar{\gamma}$, both cases yield the same allocation.

The last step is to find $\hat{\gamma}$ using Equation 8:

$$\hat{\gamma}=\beta\left[-(1+i)(z^1-z^0)+\mathbb{E}[V^1(z^1)]-\mathbb{E}[V^0(z^0)]\right] \tag{A27}$$

[14] Because the break-even point depends only on the ratio $\frac{d^0}{e^0}$, the marginal value of money is not continuous at $d^0=e^0=0$. A single household that carries money and understands that traditional banks hold no equity faces $\hat{\delta}=1$ and earns $(1-\zeta)\mathbb{E}[\delta](1+i_l)$ per unit deposited. This lowers its marginal value of money, so in this case $z^0=0$ would be optimal under a weaker condition. As every equilibrium characterized below remains an equilibrium under the alternative selection, this selection is conservative: it can only add pure shadow banking equilibria.

where I used that $\frac{\mu}{\beta} = 1 + i$ and that $z^J = z^J_{+1}$ in a stationary equilibrium.

First, suppose that $\hat{\gamma} \geq \bar{\gamma}$. In this case, $z^0 = 0$ and an unsophisticated household borrows to finance its entire investment, while a sophisticated household invests their money balances as an entrepreneur and invests them (indirectly) in loans as a banker or idle household. Therefore, I have that:

$$V^1(z^1) = \sigma_1 \mathbb{E}[\delta] f(z^1) + (\sigma_2 + \sigma_3)\mathbb{E}[\delta](1 + i_l)z^1$$
$$V^0(0) = \sigma_1 \mathbb{E}[\delta]\Big(f(k^0) - (1 + i_l)k^0\Big)$$

Combining with Equation A24, $1 + i_l = f'(k^0)$, and using the properties of a homogeneous function gives:

$$\begin{aligned}\hat{\gamma} &= \beta\sigma_1 \mathbb{E}[\delta]\left[f(z^1) - f(k^0)\right](1 - \alpha) \\ &= \beta\sigma_1(1 - \alpha)\mathbb{E}[\delta] f(z^1)\left[1 - \left(\frac{(\sigma_2 + \sigma_3)G(\hat{\gamma})}{\sigma_1(1 - G(\hat{\gamma}))}\right)^{\alpha}\right]\end{aligned} \tag{A28}$$

For $\hat{\gamma} > \bar{\gamma}$, $f(z^1)$ is positive and decreasing in $\hat{\gamma}$, while the term in brackets is decreasing in $\hat{\gamma}$, strictly positive at $\bar{\gamma}$ and negative for $G(\hat{\gamma}) > \sigma_1$. Therefore, the RHS is decreasing on $(\bar{\gamma}, G^{-1}(\sigma_1))$, exactly zero at $G^{-1}(\sigma_1)$, and negative afterwards.

Second, suppose that $\hat{\gamma} < \bar{\gamma}$. In this case, both $z^1 > 0$ and $z^0 > 0$. It is useful to analyze the expected utility of monitoring and non-monitoring separately. The expected utility of monitoring gross of the monitoring cost is:

$$W^1 = -\mu z^1 + \beta\mathbb{E}[V^1(z^1)] = -\mu z^1 + \beta\mathbb{E}[\delta]\left[\sigma_1 f(z^1) + (\sigma_2 + \sigma_3)(1 + i_l)z^1\right] \tag{A29}$$

Using Equation A12, $\frac{\mu}{\beta} = 1 + i$, and the properties of a homogeneous function, this simplifies to:

$$W^1 = \sigma_1\beta\mathbb{E}[\delta](1 - \alpha)f(z^1)$$

Next, the expected utility of a unsophisticated household is:

$$\begin{aligned}W^0 = -\mu z^0 + \beta\mathbb{E}[V^0(z^0)] &= -\mu z^0 + \beta\left[\sigma_1\mathbb{E}[\delta](f(k^0) - (1 + i_l)l^0)\right] + \sigma_2\beta\mathbb{E}[\delta](1 + i_l)z^0 \\ &+ \beta\sigma_3\left((1 - \psi)z^0 + \mathbb{E}[\delta](1 + i_l)\psi z^0 - f'(k^0)\zeta\int_0^{\hat{\delta}}\delta dH(\delta)z^0\frac{(\sigma_2 + \sigma_3\psi)}{\sigma_3}\right)\end{aligned}$$

Using that $l^0 = k^0 - z^0$ and $\frac{\mu}{\beta} = 1 + i$ gives:

$$\begin{aligned}W^0 &= \beta\sigma_1\mathbb{E}[\delta]f(k^0)(1 - \alpha) \\ &+ \beta z^0\underbrace{\left(\mathbb{E}[\delta]f'(k^0)(\sigma_1 + \sigma_2 + \sigma_3\psi) + \sigma_3(1 - \psi) - (1 + i) - \zeta f'(k^0)\int_0^{\hat{\delta}}\delta dH(\delta)(\sigma_2 + \sigma_3\psi)\right)}_{\Theta}\end{aligned}$$

First, I simplify $\Theta$. Using Equation A16, it simplifies to:

$$\begin{aligned}\Theta = &-(\mathbb{E}[\delta]f'(k^0)-1)\sigma_3(1-\psi)\\ &+f'(k^0)\zeta\left[\sigma_3\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)-(\sigma_2+\sigma_3\psi)\int_0^{\hat{\delta}}\delta dH(\delta)\right]\end{aligned}$$

If $i \geq \bar{i}$, $\psi = 1$, $\Theta$ is:

$$\Theta = f'(k^0)\zeta\left[\sigma_3\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)-(\sigma_2+\sigma_3)\int_0^{\hat{\delta}}\delta dH(\delta)\right]$$

Otherwise, $\psi$ is given by Equation 25:

$$\mathbb{E}[\delta]f'(k^0)-1 = f'(k^0)\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)$$

Substituting into $\Theta$ simplifies it to:

$$\Theta = f'(k^0)\zeta\left[\sigma_3\psi\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)-(\sigma_2+\sigma_3\psi)\int_0^{\hat{\delta}}\delta dH(\delta)\right]$$

$$\Theta = \zeta f'(k^0)(\sigma_2+\sigma_3\psi)\left[\frac{\sigma_3\psi}{\sigma_2+\sigma_3\psi}\left(\int_0^{\hat{\delta}}\delta dH(\delta)+\mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)-\int_0^{\hat{\delta}}\delta dH(\delta)\right]$$

$$\Theta = \zeta f'(k^0)(\sigma_2+\sigma_3\psi)\hat{\Theta}$$

Therefore, the first term vanishes either way, and I can use the last equation for $\Theta$.

Before continuing, it is useful to prove that $\Theta > 0$. The sign is determined by $\hat{\Theta}$. Using Equation 26, I can write:

$$\frac{\sigma_3\psi}{\sigma_2+\sigma_3\psi} = 1-\frac{\sigma_2}{\sigma_2+\sigma_3\psi} = 1-\frac{\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)}{\mathbb{E}[\delta]}$$

Next, splitting $\mathbb{E}[\delta] = \int_0^{\hat{\delta}}\delta dH(\delta)+\int_{\hat{\delta}}^1\delta dH(\delta) = \int_0^{\hat{\delta}}\delta dH(\delta)+\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)+(1-H(\hat{\delta}))\hat{\delta}$. Substituting back and using some algebra, $\hat{\Theta}$ can be written as:

$$\hat{\Theta} = \frac{\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)}{\mathbb{E}[\delta]}\left[\int_0^{\hat{\delta}}\delta dH(\delta)(\mathcal{H}(\hat{\delta})\hat{\delta}-1)+\hat{\delta}^2h(\hat{\delta})\right]$$

Therefore, $\hat{\Theta}$ is strictly positive if the first term in brackets is positive. If $\mathcal{H}(\hat{\delta})\hat{\delta} \geq 1$, this is always satisfied. If $\mathcal{H}(\hat{\delta})\hat{\delta} < 1$, I use that $\int_0^{\hat{\delta}}\delta dH(\delta) \leq \hat{\delta}H(\hat{\delta})$ to find the following inequality:

$$\int_0^{\hat{\delta}}\delta dH(\delta)\left(\mathcal{H}(\hat{\delta})\hat{\delta}-1\right)+\hat{\delta}^2h(\hat{\delta}) \geq \hat{\delta}H(\hat{\delta})(\mathcal{H}(\hat{\delta})\hat{\delta}-1)+\hat{\delta}^2\mathcal{H}(\hat{\delta})(1-H(\hat{\delta})) = \hat{\delta}(\mathcal{H}(\hat{\delta})\hat{\delta}-H(\hat{\delta}))$$

As the hazard function is increasing, $\mathcal{H}(s) \leq \mathcal{H}(\hat{\delta}), \forall s \in [0, \hat{\delta}]$. Integrating gives:

$$-\ln(1 - H(\hat{\delta})) = \int_0^{\hat{\delta}} \mathcal{H}(s)ds \leq \mathcal{H}(\hat{\delta})\hat{\delta}$$

Consequently, the inequality holds if $-\ln(1 - H(\hat{\delta})) - H(\hat{\delta}) > 0$, which holds for $H(\hat{\delta}) \in (0, 1)$ as $i > 0$ implies that $\hat{\delta} > 0$. Therefore, $\hat{\Theta}$ is strictly positive.

It is then useful to note that $\Theta$ is:

$$\Theta = \zeta f'(k^0)(\sigma_2 + \sigma_3\psi)\frac{\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)}{\mathbb{E}[\delta]}\left[\int_0^{\hat{\delta}} \delta dH(\delta)(\mathcal{H}(\hat{\delta})\hat{\delta} - 1) + \hat{\delta}^2 h(\hat{\delta})\right]$$

$$\Theta = \zeta f'(k^0)(\sigma_2 + \sigma_3\psi)\hat{\Theta} > 0$$

As $z^0 = \left[\frac{\sigma_1 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)\rho}{(\sigma_1+\sigma_2+\sigma_3\psi)}\right] k^0$, the expected utility of a unsophisticated household simplifies to:

$$W^0 = \beta\sigma_1\mathbb{E}[\delta]f(k^0)(1-\alpha) + \beta\alpha\zeta f(k^0)\left[\sigma_1 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)\rho\right]\frac{(\sigma_2+\sigma_3\psi)}{(\sigma_1+\sigma_2+\sigma_3\psi)}\hat{\Theta}$$

Using $W^1$ and $W^0$, the monitoring threshold equation for $\hat{\gamma} < \bar{\gamma}$ is given by:

$$\begin{aligned}\hat{\gamma} &= \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^0)(\rho^\alpha - 1) \\ &\quad - \zeta\beta\alpha f(k^0)\hat{\Theta}\left[\frac{\sigma_2+\sigma_3\psi}{(\sigma_1+\sigma_2+\sigma_3\psi)}\right]\left(\sigma_1 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)\rho\right)\end{aligned}$$

The RHS is linear in $\frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}$ and strictly increasing on $(0, \bar{\gamma})$ as $\hat{\Theta} > 0$. Combining with Equation A28 gives the following continuous fixed point equation:

$$\hat{\gamma} = \Xi(\hat{\gamma}) = \begin{cases} \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^0)(\rho^\alpha - 1) \\ \quad - \beta\zeta\alpha f(k^0)\hat{\Theta}\left[\frac{\sigma_2+\sigma_3\psi}{(\sigma_1+\sigma_2+\sigma_3\psi)}\right]\left(\sigma_1 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)\rho\right), & \text{if } \hat{\gamma} < \bar{\gamma} \\ \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(z^1)\left[1 - \left(\frac{(\sigma_2+\sigma_3)G(\hat{\gamma})}{\sigma_1(1-G(\hat{\gamma}))}\right)^\alpha\right], & \text{if } \hat{\gamma} \geq \bar{\gamma} \end{cases} \tag{A30}$$

Since $\Xi$ is continuous and $\Xi(G^{-1}(\sigma_1)) = 0 < G^{-1}(\sigma_1)$, the intermediate value theorem implies that there exists at least one solution if $\lim_{\hat{\gamma}\to 0^+} \Xi(\hat{\gamma}) > 0$. Evaluating the limit gives:

$$\Xi(0) = \sigma_1\beta(1-\alpha)\mathbb{E}[\delta]f(k^0)(\rho^\alpha - 1) - \sigma_1\beta\zeta\alpha f(k^0)\hat{\Theta}\left[\frac{\sigma_2+\sigma_3\psi}{(\sigma_1+\sigma_2+\sigma_3\psi)}\right]$$

It is convenient to denote by $\nu(\hat{\delta}) = \int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)$, such that $\rho$ is given by:

$$\rho = \left[1 - \frac{\sigma_3\zeta}{\sigma_1\mathbb{E}[\delta]}\nu(\hat{\delta})\right]^{\frac{1}{\alpha-1}}$$

To prove that $\Xi(0) > 0$, I start by proving that the first term has a lower bound. By Bernoulli's inequality, $(1-y)^{-p} \geq 1+py$ for $y \in [0,1)$ and $p > 0$, with strict inequality for $y > 0$. Therefore, the term $\rho^{\alpha} - 1$ satisfies:

$$\left[1 - \frac{\sigma_3 \zeta}{\sigma_1 \mathbb{E}[\delta]} \nu(\hat{\delta})\right]^{-\frac{\alpha}{1-\alpha}} - 1 > \frac{\alpha}{1-\alpha} \frac{\sigma_3 \zeta}{\sigma_1 \mathbb{E}[\delta]} \nu(\hat{\delta})$$

Substituting into the first term gives:

$$\sigma_1(1-\alpha)\beta \mathbb{E}[\delta] f(k^0)(\rho^{\alpha} - 1) > \sigma_1 \beta f(k^0) \alpha \zeta \frac{\sigma_3}{\sigma_1} \nu(\hat{\delta})$$

Next, the term $(\sigma_2 + \sigma_3 \psi)\hat{\Theta}$ is bounded from above:

$$(\sigma_2 + \sigma_3 \psi)\hat{\Theta} = \sigma_3 \psi \nu(\hat{\delta}) - (\sigma_2 + \sigma_3 \psi) \int_0^{\hat{\delta}} \delta dH(\delta) < \sigma_3 \psi \nu(\hat{\delta})$$

Evaluating at the bounds gives:

$$\Xi(0) > \sigma_1 \beta \alpha \zeta f(k^0) \nu(\hat{\delta}) \frac{\sigma_3}{\sigma_1} \left[1 - \frac{\sigma_1 \psi}{\sigma_1 + \sigma_2 + \sigma_3 \psi}\right] > 0$$

which is strictly positive as $\psi \in [0,1]$ and $\sigma_1 \psi < \sigma_1 + \sigma_2 + \sigma_3 \psi$.

Finally, I have to check the following conjectures: i) the price of money is independent of the realization of $\delta$, ii) the amount of capital invested is below the first best, such that $\mathbb{E}[\delta](1+i_l) > 1$, and iii) a sophisticated entrepreneur wants to invest rather than to deposit.

As $\hat{\gamma}$, $z^1$ and $z^0$ only depend on the expectation of next period's $\delta$ and not on its current realization, it follows from $\phi = \frac{G(\hat{\gamma})z^1 + (1-G(\hat{\gamma}))z^0}{\mu^t M}$ $\phi$ is independent of the current state.

Second, $1 + i_l = f'(k^0)$, therefore the conjecture holds if $k^0 < k^*$. As $k^1 \geq k^0$ in equilibrium, it is sufficient to check that $k^1 < k^*$, that is, $\mathbb{E}[\delta] f'(k^1) > 1$. Suppose instead that $\mathbb{E}[\delta] f'(k^1) \leq 1$. Then Equation A12 implies $\mathbb{E}[\delta] f'(k^0) \geq \frac{1+i-\sigma_1}{1-\sigma_1}$. In both cases, Equation A13 implies that $1 + i \geq (\sigma_1 + \sigma_2)\mathbb{E}[\delta] f'(k^0) + \sigma_3$. Combining the two inequalities gives $(1+i)(\sigma_1 - \sigma_3) \leq (\sigma_1 - \sigma_3)$ or $i \leq 0$, a contradiction.

Third, I have to show that in any equilibrium a sophisticated entrepreneur wants to invest rather than to deposit in either traditional or shadow banks, and under which conditions this conjecture holds. Suppose not, and a sophisticated entrepreneur deposits. Their marginal value of money is then the expected loan rate and the money choice equation becomes:

$$1 + i = \mathbb{E}[\delta](1 + i_l)$$

All sophisticated households earn the expected loan rate on their money balances. Consequently, the expected utility of monitoring is:

$$W^1 = -\mu z^1 - \gamma + \beta \mathbb{E}[\delta](1+i_l) z^1 + \tau + \beta W(0) = \tau + \beta W(0) - \gamma$$

as $\frac{\mu}{\beta} = 1 + i$. A unsophisticated household, by contrast, obtains $\tau + \beta W(0) + \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^0)$ by carrying no real money balances and borrowing as an entrepreneur. Monitoring is therefore strictly worse than not monitoring, and there is no equilibrium in which a sophisticated entrepreneur deposits. Consequently, in equilibrium the following inequality has to hold:

$$\mathbb{E}[\delta]f(z^1) > \mathbb{E}[\delta]f'(k^0)z^1$$

From Equations A12 and A15, I have that:

$$\left(\frac{z^1}{k^0}\right)^{\alpha-1} = \frac{f'(z^1)}{f'(k^0)} = \rho^{\alpha-1} \geq 1 - \zeta\frac{\sigma_3}{\sigma_1\mathbb{E}[\delta]}\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)$$

with equality if $z^0 > 0$. Using the properties of a homogeneous function, the required inequality is equivalent to $\frac{f'(z^1)}{f'(k^0)} > \alpha$, which holds if

$$\alpha < \left[1 - \frac{\sigma_3\zeta}{\sigma_1\mathbb{E}[\delta]}\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right)\right]$$

The RHS is decreasing in both $\hat{\delta}$ and $\zeta$. Since $\hat{\delta} \leq \bar{\delta}$, the RHS is strictly larger than $1 - \frac{\sigma_3}{\sigma_1}$ for all $i$ and $\zeta$. Therefore, a sufficient condition is:

$$\alpha \leq 1 - \frac{\sigma_3}{\sigma_1}$$

Lastly, as the traditional bank contract offers a lower expected return than the shadow bank deposit contract, an entrepreneur also refuses to deposit in a traditional bank. □

*Proof of Proposition 7.* The fixed point equation is given by A30, which can be written as:

$$\mathcal{F}(\hat{\gamma}) = \hat{\gamma} - \Xi(\hat{\gamma}) = 0$$

which is continuous in $\{i, \zeta, \hat{\gamma}\}$ from Proposition 4 and the continuity of $\mathcal{D}(i, \zeta)$ below. By Corollary 5:

- $\Xi(\hat{\gamma})$ is strictly decreasing on $(\bar{\gamma}, \tilde{\gamma})$,
- $\Xi(\hat{\gamma})$ is strictly positive on $[\bar{\gamma}, \tilde{\gamma})$,
- $\Xi(\hat{\gamma})$ is increasing and strictly convex on $(0, \bar{\gamma})$,
- $\lim_{\hat{\gamma}\to 0}\Xi(\hat{\gamma}) > 0$ and $\lim_{\hat{\gamma}\to\tilde{\gamma}} = 0$,

which implies that there are two cases:

1. If $\mathcal{F}(\bar{\gamma}) > 0$, there is a unique fixed point on $(0, \bar{\gamma})$ and none on $[\bar{\gamma}, \Gamma)$.
2. If $\mathcal{F}(\bar{\gamma}) \leq 0$, there is a unique fixed point on $[\bar{\gamma}, \tilde{\gamma})$ and either zero or two fixed points on $(0, \bar{\gamma})$.

Therefore, it is sufficient to analyze $\mathcal{F}(\hat{\gamma})$ at $\bar{\gamma}$:

$$\mathcal{F}(\bar{\gamma};i,\zeta) = G^{-1}\left[\frac{\sigma_1}{\sigma_1+(\sigma_2+\sigma_3)\rho}\right] - \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^0)(\rho^\alpha-1)$$

where the RHS is a function of $\zeta$ and $i$.

Denote $\hat{\rho} = \frac{k^0}{k^*}$. From Equation A16 and the definition of $k^*$, this is given by:

$$\hat{\rho} = \left[\frac{1+i}{1-\frac{\sigma_3\zeta}{\mathbb{E}[\delta]}\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)}\right]^{-\frac{1}{1-\alpha}}$$

This implies that $f(k^0) = \hat{\rho}^\alpha f(k^*)$. Equation A18 gives:

$$\rho = \left[1-\frac{\sigma_3}{\mathbb{E}[\delta]\sigma_1}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]^{\frac{1}{(\alpha-1)}}$$

Define $\mathcal{D}(i,\zeta) = \frac{\sigma_3}{\mathbb{E}[\delta]}\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)$. Then, the fixed point can be written as

$$\begin{aligned}\mathcal{F}(\bar{\gamma};i,\zeta) = G^{-1}&\left[\frac{\sigma_1}{\sigma_1+(\sigma_2+\sigma_3)\left(1-\frac{\mathcal{D}}{\sigma_1}\right)^{\frac{1}{(\alpha-1)}}}\right] \\ &- \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^*)(1+i)^{-\frac{\alpha}{1-\alpha}}\left[\left(\frac{1-\mathcal{D}}{1-\frac{\mathcal{D}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\mathcal{D})^{\frac{\alpha}{(1-\alpha)}}\right]\end{aligned} \tag{A31}$$

Consequently, all dependence on $\zeta$ arises through $\mathcal{D}$, which depends on $\zeta$ directly and through $\hat{\delta}$.

From Proposition 4, for $i \geq \bar{i}$, the break-even point is $\bar{\delta}$ given by:

$$\mathbb{E}[\delta]\frac{\sigma_2}{\sigma_2+\sigma_3} = \int_{\bar{\delta}}^1(\delta-\bar{\delta})dH(\delta)$$

which is independent of $\zeta$. In this case, $\mathcal{D}(i,\zeta)$ is given by:

$$\mathcal{D}(i,\zeta) = \frac{\sigma_3}{\mathbb{E}[\delta]}\zeta\left(\int_0^{\bar{\delta}}\delta dH(\delta) + \mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^1(\delta-\bar{\delta})dH(\delta)\right) = \sigma_3\zeta\bar{\mathcal{D}}$$

If $i < \bar{i}$, the break-even point is given by Equation A21

$$i = \frac{(1-\sigma_3)\zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)}{\left(\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)\right)}$$

$$\frac{i}{1+i-\sigma_3} = \frac{\zeta}{\mathbb{E}[\delta]}\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta}) dH(\delta)\right)$$

As at $i = \bar{i}$, $\hat{\delta} = \bar{\delta}$, I can write $\mathcal{D}(i,\zeta)$ as:

$$\mathcal{D}(i,\zeta) = \sigma_3 \min\left[\frac{i}{1+i-\sigma_3}, \zeta\bar{\mathcal{D}}\right]$$

which is continuous in $(i,\zeta)$. Consequently, I find that:

$$\frac{\partial \mathcal{D}}{\partial \zeta} = \begin{cases} 0, & \text{if } i < \bar{i}(\zeta) \\ \sigma_3\bar{\mathcal{D}}, & \text{if } i \geq \bar{i}(\zeta) \end{cases}.$$

Moreover, $\mathcal{F}(\bar{\gamma}; i, \zeta)$ is decreasing in $\mathcal{D}$, as the first term is decreasing and the term in square brackets increases with $\mathcal{D}$. Consequently, $\mathcal{F}(\bar{\gamma}; i, \zeta)$ is weakly decreasing in $\zeta$.

Since $\mathcal{D} \leq \sigma_3\zeta\bar{\mathcal{D}}$ and $(1+i)^{-\frac{\alpha}{1-\alpha}} < 1$, $\mathcal{F}(\bar{\gamma}; i, \zeta)$ is decreasing in $\mathcal{D}$. Therefore, I can find the following lower bound independent of $i$ on $\mathcal{F}(\bar{\gamma}; i, \zeta)$;

$$\underline{\mathcal{F}(\bar{\gamma};\zeta)} = G^{-1}\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2+\sigma_3)\left[1 - \frac{\sigma_3\zeta\bar{\mathcal{D}}}{\sigma_1}\right]^{\frac{1}{(\alpha-1)}}}\right]$$
$$- \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^*)\left[\left(\frac{1-\sigma_3\zeta\bar{\mathcal{D}}}{1-\frac{\sigma_3\zeta\bar{\mathcal{D}}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\sigma_3\zeta\bar{\mathcal{D}})^{\frac{\alpha}{(1-\alpha)}}\right] \tag{A32}$$

which holds for all $i > 0$. Consequently, $\zeta \to 0$ gives the following limit of the uniform lower bound:

$$\lim_{\zeta\to 0} \underline{\mathcal{F}}(\bar{\gamma};\zeta) = G^{-1}(\sigma_1)$$

Next, I look at the uniform bounds of $\mathcal{F}(\bar{\gamma}; i, \zeta)$ for $i$. Since $\mathcal{F}(\bar{\gamma}; i, \zeta)$ is decreasing in $\mathcal{D}$,

and for all $i > 0$, $\mathcal{D} \leq \frac{\sigma_3 i}{1+i-\sigma_3}$, I find the following lower bound:

$$\underline{\mathcal{F}(\bar{\gamma};i)}_1 = G^{-1}\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2+\sigma_3)\left[1-\frac{\sigma_3 i}{\sigma_1(1+i-\sigma_3)}\right]^{\frac{1}{(\alpha-1)}}}\right]$$
$$-\beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^*)\left[\left(\frac{1-\frac{\sigma_3 i}{1+i-\sigma_3}}{1-\frac{\frac{\sigma_3 i}{1+i-\sigma_3}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\frac{\sigma_3 i}{1+i-\sigma_3})^{\frac{\alpha}{(1-\alpha)}}\right]$$
$$\lim_{i\to 0}\underline{\mathcal{F}(\bar{\gamma};i)}_1 = G^{-1}(\sigma_1) > 0$$

Similarly, for all $i > 0$, $\mathcal{D} \leq \sigma_3\bar{\mathcal{D}}$ and I find a second lower bound:

$$\underline{\mathcal{F}(\bar{\gamma};i)}_2 = G^{-1}\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2+\sigma_3)\left[1-\frac{\sigma_3\bar{\mathcal{D}}}{\sigma_1}\right]^{\frac{1}{(\alpha-1)}}}\right]$$
$$-\beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^*)(1+i)^{\frac{-\alpha}{1-\alpha}}\left[\left(\frac{1-\sigma_3\bar{\mathcal{D}}}{1-\frac{\sigma_3\bar{\mathcal{D}}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\sigma_3\bar{\mathcal{D}})^{\frac{\alpha}{(1-\alpha)}}\right]$$
$$\lim_{i\to\infty}\underline{\mathcal{F}(\bar{\gamma};i)}_2 = G^{-1}\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2+\sigma_3)\left[1-\frac{\sigma_3}{\sigma_1}\bar{\mathcal{D}}\right]^{\frac{1}{(\alpha-1)}}}\right] > 0$$

Consequently, there exists a non-empty compact interval $I = [i_1, i_2]$ independent of $\zeta$, such that for $i \notin I$ $\mathcal{F}(\bar{\gamma};i,\zeta) > 0$, $\forall\zeta \in (0,1)$. As $\mathcal{F}(\bar{\gamma};i,\zeta)$ is continuous in $i$, by Weierstrass theorem, $\mathcal{F}(\bar{\gamma};i,\zeta)$ attains a minimum on $I$:

$$\underline{\mathcal{M}}(\zeta) = \min_{i\in I}\mathcal{F}(\bar{\gamma};i,\zeta)$$

Consequently, $\mathcal{F}(\bar{\gamma};i,\zeta) > 0$ for all $i \in (0,\infty)$ if and only if $\underline{\mathcal{M}}(\zeta) > 0$.

As $I$ is a non-empty, compact set independent of $\zeta$ and $\mathcal{F}(\bar{\gamma};i,\zeta)$ is jointly continuous in $\zeta$ and $i$, $\underline{\mathcal{M}}(\zeta)$ is continuous in $\zeta$ by the maximum theorem, see e.g. Theorem 3.6 in Stokey and Lucas Jr. (1989). Define $\zeta^*$ as follows:

$$\zeta^* = \inf_{\zeta\in(0,\bar{\zeta})}\{\underline{\mathcal{M}}(\zeta) \leq 0\}$$

with $\zeta^* = \bar{\zeta}$ if the set is empty.

As $\mathcal{F}(\bar{\gamma};i,\zeta)$ is weakly decreasing in $\zeta$, so is $\underline{\mathcal{M}}(\zeta)$. As $\underline{\mathcal{M}}(\zeta) \geq \underline{\mathcal{F}(\bar{\gamma};\zeta)}$, $\underline{\mathcal{F}(\bar{\gamma};\zeta)}$ is continuous in $\zeta$ and $\lim_{\zeta\to 0}\underline{\mathcal{F}(\bar{\gamma};\zeta)} = G^{-1}(\sigma_1) > 0$, there exists a $\zeta_0$, such that $\underline{\mathcal{M}}(\zeta) > 0$ for $\zeta < \zeta_0$ and $\zeta^* > 0$. Consequently, $\underline{\mathcal{M}}(\zeta) > 0$ for $\zeta < \zeta^*$, and $\underline{\mathcal{M}}(\zeta) \leq 0$ for $\zeta \geq \zeta^*$ if $\zeta^* < \bar{\zeta}$.

This leads back to the two cases:

1. If $\zeta < \zeta^*$, the first case arises, and for all $i > 0$ there is a unique fixed point on $(0, \bar{\gamma})$, such that there exists only a single mixed banking equilibrium and no pure shadow banking equilibrium.

2. If $\zeta \geq \zeta^*$, define $\mathcal{I}(\zeta) = \{i > 0 : \mathcal{F}(\bar{\gamma}; i, \zeta) \leq 0\}$. By the continuity of $\underline{\mathcal{M}}(\zeta)$, the minimum is non-positive and attained in $\mathcal{I}(\zeta)$. Therefore, $\mathcal{I}(\zeta)$ is non-empty. It is closed by the continuity of $\mathcal{F}(\bar{\gamma}; i, \zeta)$, $\mathcal{I}(\zeta) \subset I \subset (0, \infty)$, and as $\mathcal{F}_\zeta(\bar{\gamma}; i, \zeta) \leq 0$, it is expanding in $\zeta$. Therefore, for $i \in \mathcal{I}(\zeta)$ there is a unique fixed point on $[\bar{\gamma}, \tilde{\gamma})$ and there exists a pure shadow banking equilibrium, while there could be either zero or two fixed points on $(0, \bar{\gamma})$. For $i \notin \mathcal{I}(\zeta)$, there is a unique fixed point on $(0, \bar{\gamma})$ and there exists a single mixed banking equilibrium.

□

*Proof of Corollary 6.* Using Equation 53, define:

$$\mathcal{F}(\hat{\gamma}, i) = \hat{\gamma} - \Xi(\hat{\gamma}, i)$$

As $\zeta < \zeta^*$, the equilibrium with monitoring is a unique mixed banking equilibrium, so $\hat{\gamma} < \bar{\gamma}$ and from Equation A30:

$$\begin{aligned}\Xi(\hat{\gamma}, i) = &\underbrace{\beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^0)(\rho^\alpha - 1)}_{\mathcal{A}(i)} \\ &\underbrace{- \beta\zeta\alpha f(k^0)\hat{\Theta}\left[\frac{\sigma_2 + \sigma_3\psi}{(\sigma_1 + \sigma_2 + \sigma_3\psi)}\right]\left(\sigma_1 - \frac{G(\hat{\gamma})}{(1 - G(\hat{\gamma}))}(\sigma_2 + \sigma_3)\rho\right)}_{\mathcal{B}(i,\hat{\gamma})}\end{aligned}$$

where $\hat{\Theta} = \left[\frac{\sigma_3\psi}{\sigma_2+\sigma_3\psi}\left(\int_0^{\hat{\delta}} \delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})dH(\delta)\right) - \int_0^{\hat{\delta}} \delta dH(\delta)\right]$.

From the proof of Proposition 7, $\zeta < \zeta^*$ implies $\mathcal{F}(\bar{\gamma}, i) > 0$ and by Corollary 5 $\Xi$ is convex on $(0, \bar{\gamma})$. so $\mathcal{F}$ is concave there. Since $\mathcal{F}$ is concave, $\mathcal{F}(\hat{\gamma}, i) = 0$ and $\bar{\gamma}$ is independent of $i$, I can show that:

$$\mathcal{F}(\hat{\gamma}, i) \geq \frac{\mathcal{F}(\bar{\gamma}, i)}{\bar{\gamma} - \hat{\gamma}} > 0$$

This allows me to use the implicit function theorem, to find that:

$$\frac{d\hat{\gamma}}{di} = -\frac{\mathcal{F}_i}{\mathcal{F}_{\hat{\gamma}}} = \frac{\Xi_i(\hat{\gamma}, i)}{1 - \Xi_{\hat{\gamma}}(\hat{\gamma}, i)}$$

where $1 - \Xi_{\hat{\gamma}} > 0$. Therefore, the sign depends of $\frac{d\hat{\gamma}}{di}$ is the derivative of $\Xi$ with respect to $i$ at fixed $\hat{\gamma}$.

First, consider $i \approx 0$. I start by finding the dependence on $i$ of the different equilibrium allocations $\{k^0, \hat{\delta}, \psi, \rho\}$ . As $\bar{i} > 0$, for $i$ close to zero there is partial depositing in

the traditional banking sector. From Equation 43, $k^0$ is given by:

$$\mathbb{E}[\delta]f'(k^0) = \frac{1+i-\sigma_3}{(1-\sigma_3)}. \tag{A33}$$

Combining Equations A18, A20, and A21 gives:

$$\rho = \left[1 - \frac{\sigma_3 i}{\sigma_1(1+i-\sigma_3)}\right]^{\frac{1}{(\alpha-1)}} \tag{A34}$$

$$\psi = \frac{\sigma_2}{\sigma_3}\left[\frac{\mathbb{E}[\delta]}{\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)} - 1\right] \tag{A35}$$

$$i = \frac{(1-\sigma_3)\zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)}{\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)} \tag{A36}$$

Consequently, I find $\rho(i)$, $\psi(i)$, $k^0(i)$, and $\hat{\delta}(i)$ as functions of $i$. To use the implicit function theorem around $i=0$, I require that $h(0)>0$, which I assume. Differentiating $\{\psi, k^0, \rho, \hat{\delta}\}$ with respect to $i$ gives:

$$\frac{dk^0}{di} = \frac{1}{(1-\sigma_3)\mathbb{E}[\delta]f''(k^0)} \tag{A37}$$

$$\frac{d\rho}{di} = \frac{\rho^{2-\alpha}}{(1-\alpha)}\left[\frac{\sigma_3(1-\sigma_3)}{\sigma_1(1+i-\sigma_3)^2}\right] \tag{A38}$$

$$\frac{d\hat{\delta}}{di} = \frac{\left[\mathbb{E}[\delta] - \zeta\left(\int_0^{\hat{\delta}}\delta dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)\right)\right]^2}{\mathbb{E}[\delta](1-\sigma_3)\zeta\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)(\mathcal{H}'(\hat{\delta})\hat{\delta}+\mathcal{H}(\hat{\delta}))} \tag{A39}$$

$$\frac{d\psi}{di} = \frac{\sigma_2}{\sigma_3}\frac{\mathbb{E}[\delta]}{\mathbb{E}[\delta-\hat{\delta}|\delta\geq\hat{\delta}]\int_{\hat{\delta}}^1(\delta-\hat{\delta})dH(\delta)}\frac{d\hat{\delta}}{di} \tag{A40}$$

First, note that at $i=0$ both $\frac{d\hat{\Theta}}{d\psi}$ and $\frac{d\hat{\Theta}}{d\hat{\delta}}\frac{d\hat{\delta}}{di}$ are zero, such that $\hat{\Theta}(0)=0$. Consequently, $\mathcal{B}(0,\hat{\gamma})=0$ and $\mathcal{B}_i(0,\hat{\gamma})=0$. Moreover, $\mathcal{A}(0)=0$ as $\rho=1$ and thus $\hat{\gamma}(0)=0$. The equilibrium equation solves $\mathcal{F}(\hat{\gamma},i)=0$, which is differentiable near $(0,0)$, with $\mathcal{F}(0,0)=0$ and $\frac{\partial\mathcal{F}}{\partial\hat{\gamma}}(0,0)=1$. Consequently, by the implicit function theorem, $\hat{\gamma}(i)$ is continuously differentiable near $i=0$. Differentiating gives:

$$\frac{d\hat{\gamma}}{di}(0) = \left.\frac{d\Xi(\hat{\gamma},i)}{di}\right|_{i=0} = \beta\alpha\mathbb{E}[\delta]f(k^*)\frac{\sigma_3}{(1-\sigma_3)}$$

which is strictly positive. By continuity of $\frac{d\hat{\gamma}}{di}>0$, there exists a nominal interest rate $i^1$, such that $\frac{d\hat{\gamma}}{di}>0$ for all $i\in(0,i^1)$.

For $i>\bar{i}$, the unsophisticated idle household fully deposits and $d=z$, which implies that $\psi=1$ independent of $i$. The break-even point is given by Equation 52 and is also independent of $i$. Therefore, $\hat{\Theta}$ is independent of $i$. From Equation A18, $\rho$ is independent

of $i$. Therefore, $\Xi(\hat{\gamma}, i)$ only depends on $i$ through $f(k^0)$ and we can write $\Xi(\hat{\gamma}, i)$ as:

$$\Xi(\hat{\gamma}, i) = f(k^0)\Psi(\hat{\gamma})$$

where $\Psi$ is independent of $i$. Since from Corollary 5 $\Xi(\hat{\gamma}, i) > 0$ on $(0, \hat{\gamma})$, $f(k^0) > 0$, it follows that $\Psi(\hat{\gamma}) > 0$. Implicitly differentiating Equation 48 gives:

$$\frac{dk^0}{di} = \frac{1}{f''(k^0)\left(\mathbb{E}[\delta] - \sigma_3\zeta\left(\int_0^{\hat{\delta}} \delta\, dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^1 (\delta - \hat{\delta})\, dH(\delta)\right)\right)} < 0$$

where I used that $\hat{\delta}$ is independent of $i$, that $f''(k^0) < 0$, and the term in brackets is strictly positive. This implies that $\Xi_i < 0$ and that $\frac{d\hat{\gamma}}{di} < 0$ for $i > \bar{i}$. □

*Proof of Corollary 7.* If $\gamma \sim U(0, \Gamma)$, then I have that $G^{-1}(x) = \Gamma x$. From the proof of Proposition 7, if $\mathcal{F}(\bar{\gamma}; i, \zeta) > 0$ for all $i \in (0, \infty)$ and $\zeta \in (0, \bar{\zeta})$, then $\zeta^* = 1$. Using a uniform distribution in the lower bound given on $\mathcal{F}(\bar{\gamma}; i, \zeta)$ in Equation A32 gives:

$$\underline{\mathcal{F}(\bar{\gamma}; \zeta)} = \Gamma\left[\frac{\sigma_1}{\sigma_1 + (\sigma_2 + \sigma_3)\left[1 - \frac{\sigma_3\zeta\bar{\mathcal{D}}}{\sigma_1}\right]^{\frac{1}{(\alpha-1)}}}\right] - \beta\sigma_1(1-\alpha)\mathbb{E}[\delta]f(k^*)\left[\left(\frac{1-\sigma_3\zeta\bar{\mathcal{D}}}{1-\frac{\sigma_3\zeta\bar{\mathcal{D}}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\sigma_3\zeta\bar{\mathcal{D}})^{\frac{\alpha}{(1-\alpha)}}\right] \tag{A41}$$

From the proof of Proposition 7, the RHS is weakly decreasing in $\zeta$. Consequently, if $\underline{\mathcal{F}(\bar{\gamma}; 1)} > 0$, $\zeta^* = 1$. Rewriting gives the following bound on $\Gamma$:

$$\Gamma \geq \beta(1-\alpha)\mathbb{E}[\delta]f(k^*)\left[\left(\frac{1-\sigma_3\bar{\mathcal{D}}}{1-\frac{\sigma_3\bar{\mathcal{D}}}{\sigma_1}}\right)^{\frac{\alpha}{1-\alpha}} - (1-\sigma_3\bar{\mathcal{D}})^{\frac{\alpha}{(1-\alpha)}}\right]\left[\sigma_1 + (\sigma_2+\sigma_3)\left[1-\frac{\sigma_3\bar{\mathcal{D}}}{\sigma_1}\right]^{\frac{1}{(\alpha-1)}}\right]$$

$$\Gamma \geq \bar{\Gamma}$$

□

*Proof of Lemma 4.* From Propositions 2 and 5, $k^0$ and $\hat{\delta}$ are the same for both economies. Similarly, from the proofs of Propositions 2 and 4, idle unsophisticated households deposit the same share of money balances, $\psi = \frac{d}{z}$. However, the amount of money balances differs. In the traditional banking equilibrium, all households carry $z^T = \frac{\sigma_1 k^0}{(\sigma_1+\sigma_2+\sigma_3\psi)}$, while from Equation A22 unsophisticated households in the mixed banking equilibrium carry $z^0 = \frac{\sigma_1 k^0 - \frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)z^1}{(\sigma_1+\sigma_2+\sigma_3\psi)} = z^T - \frac{\frac{G(\hat{\gamma})}{(1-G(\hat{\gamma}))}(\sigma_2+\sigma_3)z^1}{(\sigma_1+\sigma_2+\sigma_3\psi)}$.

Substituting into expected welfare (Equation 32) gives:

$$(1-\beta)\mathcal{W}^M = \beta(1-G(\hat{\gamma}))\left(\sigma_1\left(\mathbb{E}[\delta]f(k^T)-k^T\right)-\zeta f'(k^0)(\sigma_2+\sigma_3\psi)z^T\int_0^{\hat{\delta}}\delta dH(\delta)\right)$$
$$+\beta G(\hat{\gamma})\left[\sigma_1\left(\mathbb{E}[\delta]f(k^1)-k^1\right)+\zeta f'(k^0)\frac{(\sigma_2+\sigma_3\psi)}{(\sigma_1+\sigma_2+\sigma_3\psi)}(\sigma_2+\sigma_3)z^1\int_0^{\hat{\delta}}\delta dH(\delta)\right]$$
$$-\int_0^{\hat{\gamma}}\gamma dG(\gamma)$$

Using that $(1-\beta)\mathcal{W}^T = \beta\sigma_1\left(\mathbb{E}[\delta]f(k^T)-k^T\right)-\zeta\beta f'(k^0)(\sigma_2+\sigma_3\psi)z^T\int_0^{\hat{\delta}}\delta dH(\delta)$, this gives:

$$(1-\beta)(\mathcal{W}^M-\mathcal{W}^T) = \beta\int_0^{\hat{\gamma}}\sigma_1\left[\mathbb{E}[\delta]f(z^1)-z^1-(\mathbb{E}[\delta]f(k^0)-k^0)\right]dG(\gamma)$$
$$+\beta\int_0^{\hat{\gamma}}\zeta f'(k^0)\left(\sigma_2+\sigma_3\frac{d^0}{z^0}\right)\int_0^{\hat{\delta}}\delta dH(\delta)\left(\frac{\sigma_1k^0+(\sigma_2+\sigma_3)z^1}{\sigma_1+\sigma_2+\sigma_3\frac{d^0}{z^0}}\right)dG(\gamma)$$
$$-\int_0^{\hat{\gamma}}\gamma dG(\gamma)$$

□

*Proof of Proposition 8.* As $\zeta<\zeta^*$, the equilibrium with monitoring is a unique mixed banking equilibrium for all $i>0$ and $\hat{\gamma}\in(0,\bar{\gamma})$. As $\Xi$ is increasing on $(0,\bar{\gamma})$, I have that $\Xi(0)<\hat{\gamma}<\Xi(\bar{\gamma})$. Using Lemma 4, the expected welfare difference is given by: $\Delta(i)=(1-\beta)(\mathcal{W}^M-\mathcal{W}^T)=\int_0^{\hat{\gamma}}\beta S-\gamma dG(\gamma)$, where $S$ is given by:

$$S=\sigma_1\left[\mathbb{E}[\delta]f(z^1)-z^1-(\mathbb{E}[\delta]f(k^0)-k^0)\right]$$
$$+\zeta f'(k^0)\left(\sigma_2+\sigma_3\frac{d^0}{z^0}\right)\int_0^{\hat{\delta}}\delta dH(\delta)\left(\frac{\sigma_1k^0+(\sigma_2+\sigma_3)z^1}{\sigma_1+\sigma_2+\sigma_3\frac{d^0}{z^0}}\right)$$

Using the proof of Proposition 4, especially $\rho k^0=z^1$ and $\psi=\frac{d^0}{z^0}$, this simplifies to:

$$\Delta(i) = G(\hat{\gamma})\beta\sigma_1\left[\mathbb{E}[\delta](\rho^\alpha-1)f(k^0)-(\rho-1)k^0\right]$$
$$+G(\hat{\gamma})\beta\zeta\alpha f(k^0)\left(\sigma_1+(\sigma_2+\sigma_3)\rho\right)\int_0^{\hat{\delta}}\delta dH(\delta)\left(\frac{\sigma_2+\sigma_3\psi}{\sigma_1+\sigma_2+\sigma_3\psi}\right)-\int_0^{\hat{\gamma}}\gamma dG(\gamma)$$
$$=G(\hat{\gamma})\beta S-\int_0^{\hat{\gamma}}\gamma dG(\gamma)$$

To show the first part of the proposition, I continue in three steps. First, I evaluate $S(i)$ and $S'(i)$. Second, I evaluate $\hat{\gamma}(i)$, and $\hat{\gamma}'(i)$. Finally, I show that $\lim_{i\to0}\Delta(i)=\lim_{i\to0}\Delta'(i)=0$ and that $\lim_{i\to0}\Delta''(i)<0$.

Equations A33, A18, A20, A21, and A37-A40 give $\{k,\rho,\hat{\delta}\psi\}$ and their partial derivatives with respect to $i$ At $i=0$, $k^0=k^*$, $\hat{\delta}=0$, $\psi=0$, and $\rho=1$, therefore $S(0)=0$. As the derivative with respect to $\hat{\delta}$, $\psi$, $\rho$, or $k^0$ of the second term is either multiplied by $\hat{\delta}$

or $\int_0^{\hat{\delta}} \delta dH(\delta)$ at $i = 0$ the second term is zero. Differentiating the first term gives:

$$\sigma_1 \left( \mathbb{E}[\delta](\rho^\alpha - 1) f'(k^0) - (\rho - 1) \right) \frac{dk}{di} + \sigma_1 \left( \alpha \mathbb{E}[\delta] f(k^0) \rho^{\alpha-1} - k^0 \right) \frac{d\rho}{di}$$

At $i = 0, \rho = 1$, and the first term is zero, while the second term is $\sigma_1(\mathbb{E}[\delta]\alpha f(k^*) - k^*)\frac{d\rho}{di}$ As $\mathbb{E}[\delta] f'(k^*) = 1$, the second is zero due to the properties of a homogeneous function. Consequently, $S(0) = S'(0) = 0$.

Second, I analyze $\hat{\gamma}(i)$. For $i < \bar{i}$, Equation A30 gives $\hat{\gamma}$ as:

$$0 = \mathcal{F}(\hat{\gamma}; i) = \hat{\gamma}(i) - \underbrace{\beta\sigma_1(1-\alpha)\mathbb{E}[\delta] f(k^0(i))(\rho(i)^\alpha - 1)}_{\mathcal{A}(i)}$$
$$\underbrace{+ \beta\zeta\alpha f(k^0(i))\hat{\Theta}(i) \left[ \frac{\sigma_2 + \sigma_3\psi(i)}{(\sigma_1 + \sigma_2 + \sigma_3\psi(i))} \right] \left( \sigma_1 - \frac{G(\hat{\gamma}(i))}{(1 - G(\hat{\gamma}(i)))}(\sigma_2 + \sigma_3)\rho(i) \right)}_{\mathcal{B}(i)}$$

where $\hat{\Theta}(i) = \frac{\sigma_3\psi(i)}{\sigma_2+\sigma_3\psi(i)} \left( \int_0^{\hat{\delta}(i)} \delta dH(\delta) + \mathcal{H}(\hat{\delta}(i))\hat{\delta}(i) \int_{\hat{\delta}(i)}^1 (\delta - \hat{\delta}(i)) dH(\delta) \right) - \int_0^{\hat{\delta}(i)} \delta dH(\delta)$. As in the proof of Corollary 6 at $i = 0$ both $\frac{d\hat{\Theta}}{d\psi}$ and $\frac{d\hat{\Theta}}{d\hat{\delta}}\frac{d\hat{\delta}}{di}$ are zero, such that $\hat{\Theta}(0) = 0$. Consequently, $\mathcal{B}(0) = 0$ and $\mathcal{B}'(0) = 0$. Moreover, $\mathcal{A}(0) = 0$ as $\rho = 1$ and thus $\hat{\gamma}(0) = 0$. The equilibrium equation solves $\mathcal{F}(\hat{\gamma}, i) = 0$, which is differentiable near $(0, 0)$, with $\mathcal{F}(0, 0) = 0$ and $\frac{\partial \mathcal{F}}{\partial \hat{\gamma}}(0, 0) = 1$. Consequently, by the implicit function theorem, $\hat{\gamma}(i)$ is continuously differentiable near $i = 0$. Differentiating gives: $\frac{d\hat{\gamma}}{di}(0) = \beta\alpha\mathbb{E}[\delta] f(k^*)\frac{\sigma_3}{(1-\sigma_3)}$, which is strictly positive.

Finally, using that the expected welfare difference is:

$$\Delta(i) = \int_0^{\hat{\gamma}(i)} \beta S(i) - \gamma dG(\gamma)$$

First, as $\hat{\gamma}(0) = 0$, $\Delta(0) = 0$. Differentiating expected welfare then gives:

$$\Delta'(i) = g(\hat{\gamma}(i))(\beta S(i) - \hat{\gamma}(i))\frac{d\hat{\gamma}}{di}(i) + \beta S'(i) G(\hat{\gamma})$$

As $S'(0) = S(0) = \hat{\gamma}(0) = 0$, also $\Delta'(0) = 0$. Using that $\Delta'(0) = 0$, the second derivative can be written as:

$$\Delta''(0) = \lim_{i \to 0} \frac{\Delta'(i)}{i} = -g(0) \left[ \frac{d\hat{\gamma}}{di}(0) \right]^2$$

As $g(0) > 0$, this term is strictly negative. Consequently, there exists an $i_1$ near zero, such that there is an open interval $(0, i_1)$, where $\Delta'(i) < 0$ and $\Delta(i) < 0$.

To prove the second part of the proposition, note that $S$ does not depend on $\hat{\gamma}$. Therefore, I can formulate the following lower bound on the welfare difference:

$$(1 - \beta)(\mathcal{W}^M - \mathcal{W}^T) \geq G(\hat{\gamma})\beta S - \int_0^{\hat{\gamma}} \gamma dG(\gamma) \geq G(\hat{\gamma})(\beta S - \Xi(\hat{\gamma}))$$

where I used Equation A30. Consequently, if $\beta S > \Xi(\hat{\gamma})$, the welfare difference is positive. From Proposition 5, if $i > \bar{i}$, $\psi = 1$, and $\hat{\delta} = \bar{\delta}$, where $\bar{\delta}$ is:

$$\frac{\sigma_2}{\sigma_2 + \sigma_3}\mathbb{E}[\delta] = \int_{\bar{\delta}}^{1} (\delta - \bar{\delta}) dH(\delta)$$

Consequently, both $\psi$ and $\hat{\delta}$ are independent of $i$. Moreover, from Equation A18 $\rho$ is also independent of $i$:

$$\rho = \left[1 - \frac{\sigma_3}{\mathbb{E}[\delta]\sigma_1}\zeta\left(\int_0^{\bar{\delta}} \delta dH(\delta) + \mathcal{H}(\bar{\delta})\bar{\delta}\int_{\bar{\delta}}^{1}(\delta - \bar{\delta})dH(\delta)\right)\right]^{\frac{1}{(\alpha-1)}} = \bar{\rho}$$

For tractability, let $\nu \equiv \frac{\sigma_3}{\mathbb{E}[\delta]}\zeta\left(\int_0^{\hat{\delta}} \delta\, dH(\delta) + \mathcal{H}(\hat{\delta})\hat{\delta}\int_{\hat{\delta}}^{1}(\delta - \hat{\delta})\, dH(\delta)\right)$. Using the properties of a homogeneous function on Equation 48, I can write $k^0 = \frac{\alpha\mathbb{E}[\delta]f(k^0)(1-\nu)}{(1+i)}$. Substituting into $\beta S - \Xi(\hat{\gamma})$ and simplifying gives the following lower bound:

$$\beta S - \Xi(\hat{\gamma}) \geq \beta\alpha\mathbb{E}[\delta]f(k^0)\left[\sigma_1\left(\bar{\rho}^{\alpha} - 1 - \frac{(1-\nu)(\bar{\rho}-1)}{(1+i)}\right)\right.$$
$$\left. + \zeta\frac{\int_0^{\bar{\delta}}\delta dH(\delta)}{\mathbb{E}[\delta]}(\sigma_2+\sigma_3)(\sigma_1 + (\sigma_2+\sigma_3)\bar{\rho})\right]$$

A sufficient condition for a positive sign is that the term in brackets is positive. As this term is increasing in $i$, I find the following bound on the nominal interest rate:

$$\hat{i}(\zeta) = \frac{\sigma_1(1-\nu)(\bar{\rho}-1)}{\sigma_1(\bar{\rho}^{\alpha}-1) + \zeta\frac{\int_0^{\bar{\delta}}\delta dH(\delta)}{\mathbb{E}[\delta]}(\sigma_2+\sigma_3)(\sigma_1+(\sigma_2+\sigma_3)\bar{\rho})} - 1$$

Therefore, if $i \geq \bar{i}^{W}(\zeta) = \max\{\bar{i}, \hat{i}(\zeta)\}$, expected welfare in the mixed banking equilibrium is strictly higher than in the traditional banking equilibrium. □

*Proof of Corollary 8.* As $\zeta < \zeta^*$, there is a unique mixed banking equilibrium with threshold $\hat{\gamma} \in (0, \bar{\gamma})$. Define

$$\Lambda(i) \equiv \beta S(i) - \Xi(\hat{\gamma}(i)) = \beta S(i) - \hat{\gamma}(i)$$
$$\Delta(i) \equiv (1-\beta)(\mathcal{W}^M - \mathcal{W}^T) = G(\hat{\gamma}(i))\beta S(i) - \int_0^{\hat{\gamma}(i)} \gamma dG(\gamma)$$

where the second expression follows from Lemma 4.

Both $\Lambda(i)$ and $\Delta(i)$ are continuous in $i$ on $(0, \infty)$. The allocation is given by Equations A17-A21 in each regime, and both coincide at the threshold $i = \bar{i}$. Hence, $S(i)$ is continuous in $i$. Moreover, $\hat{\gamma}(i)$ is the unique zero of $\mathcal{F}(\hat{\gamma}; i) = \hat{\gamma} - \Xi(\hat{\gamma}(i); i)$ on $(0, \bar{\gamma})$, where $\mathcal{F}(\hat{\gamma})$ is continuous in both $i$ and $\hat{\gamma}$ and changes sign at $\hat{\gamma}(i)$, so $\hat{\gamma}(i)$ is also continuous.

From the proof of Proposition 8, $S(0) = S'(0) = 0$, $\hat{\gamma}(0) = 0$, and $\hat{\gamma}'(0) > 0$. Hence,

$\Lambda(i) < 0$ for $i \approx 0$, while from the same proof $\beta S(i) - \Xi(\hat{\gamma}(i)) > 0$ for all $i > \bar{i}^W(\zeta)$. As for $i \approx 0$ $\Lambda(i) < 0$ and for $i > \bar{i}^W(\zeta)$ $\Lambda(i) > 0$, by the intermediate value theorem there exists a nominal interest rate $i^\dagger$, where $\Lambda(i^\dagger) = 0$ and $\Lambda(i) < 0$ for all $i < i^\dagger$. At $i^\dagger$, $\beta S = \hat{\gamma}$, so that

$$\Delta(i^\dagger) = G(\hat{\gamma})\hat{\gamma} - \int_0^{\hat{\gamma}} \gamma dG(\gamma) = G(\hat{\gamma})\left(\hat{\gamma} - \mathbb{E}[\gamma|\gamma \leq \hat{\gamma}]\right) > 0$$

By continuity of $\Delta$, there exists a range of nominal interest rates in $(0, i^\dagger)$, where $\Delta(i) > 0$. On this range, $\Lambda(i) < 0$ and the shadow banking system is too large, and expected welfare in the mixed banking equilibrium is higher than in the traditional banking equilibrium. □

*Proof of Proposition 9.* I start by showing that $k^{0,S} \geq k^T$. Using Equation A25 and Proposition 2, $z^0 = 0$ requires that

$$f'(k^T) \geq f'(k^{0,S})$$

As $f'(k)$ is decreasing, this implies that $k^{0,S} \geq k^T$. Next, I show that $k^{1,S} > k^B$. Combining Equations A24 and 33 gives:

$$\frac{f'(k^B)}{f'(k^{1,S})} = \sigma_1 + (\sigma_2 + \sigma_3)^\alpha \sigma_1^{1-\alpha} \left(\frac{(1 - G(\hat{\gamma}))}{G(\hat{\gamma})}\right)^{1-\alpha}$$

where $\hat{\gamma} \in (\bar{\gamma}, \tilde{\gamma})$, $G(\hat{\gamma}) < \sigma_1$, implies that the RHS is bigger than one. Therefore, $k^B < k^{1,S}$, while from the conjecture in the proof of Proposition 4, $k^B < k^{1,S} < k^*$.

In the pure shadow banking equilibrium, $z^0 = 0$ and there is no verification. Using Equation 32, the difference in expected welfare is given by:

$$\begin{aligned}(1-\beta)(\mathcal{W}^S - \mathcal{W}^T) = {} & \beta\sigma_1 G(\hat{\gamma}) \left[\mathbb{E}[\delta]f(k^{1,S}) - k^{1,S} - \mathbb{E}[\delta]f(k^T) + k^T\right] \\ & + \sigma_1\beta(1 - G(\hat{\gamma})) \left[\mathbb{E}[\delta]f(k^{0,S}) - k^{0,S} - \mathbb{E}[\delta]f(k^T) + k^T\right] \\ & + \beta\zeta\alpha\sigma_1 f(k^T)\frac{(\sigma_2 + \sigma_3\psi)}{\sigma_1 + \sigma_2 + \sigma_3\psi}\int_0^{\hat{\delta}} \delta dH(\delta) - \int_0^{\hat{\gamma}} \gamma dG(\gamma)\end{aligned}$$

Using that $\mathbb{E}[\delta]f(k) - k$ is increasing on $(0, k^*)$, that $G(\hat{\gamma}) < 1$, and that $\int_0^{\hat{\gamma}} \gamma dG(\gamma) \leq \hat{\gamma}G(\hat{\gamma})$, I can formulate the following lower bound on the expected welfare difference:

$$\begin{aligned}\underline{\Delta} = {} & \beta G(\hat{\gamma})\sigma_1 \left(\mathbb{E}[\delta]f(k^B) - k^B - \mathbb{E}[\delta]f(k^T) + k^T\right) \\ & + \beta G(\hat{\gamma})\zeta\alpha f(k^T)\sigma_1\frac{(\sigma_2 + \sigma_3\psi)}{\sigma_1 + \sigma_2 + \sigma_3\psi}\int_0^{\hat{\delta}} \delta dH(\delta) - \hat{\gamma}G(\hat{\gamma}) \\ = {} & G(\hat{\gamma})(\gamma^C(i) - \hat{\gamma})\end{aligned}$$

where $\gamma^C(i)$ comes from Equation A11 in Corollary 3.

Finally, from Equation A30, $\Xi(\hat{\gamma}) = \hat{\gamma}$, and from Corollary 5, $\Xi(\hat{\gamma})$ is decreasing on $[\bar{\gamma}, \tilde{\gamma})$. Consequently, a sufficient condition for the expected welfare difference to be

positive is $\Xi(\bar{\gamma}) < \gamma^C(i)$. □

## B Deposit Insurance

In the main text, unsophisticated idle households rely on a costly state verification technology to mitigate the information friction. In this appendix, I show that this formulation is equivalent to a model in which a government operates a particular actuarially fair deposit insurance scheme. Following Altermatt and Wang (2024), suppose that the government operates a deposit insurance scheme that receives a deposit premium from banks when they do not default and pays households the promised value on deposits when the bank defaults. In case of default, the insurance is financed by lump-sum taxation, while the deposit premia are rebated to households through lump-sum transfers. The fact that the government operates an actuarially fair insurance scheme means that the expected net transfer of this scheme is zero. Second, as the government has less expertise in operating a bank and collecting loan income, I assume that default destroys a share $\zeta$ of the bank's income.

Now I solve for the deposit contract between a traditional bank and its depositors. Recall that there is a share $\sigma_2$ of bankers and a share $\sigma_3$ of depositors. As before, I can restrict attention to the situation in which each bank only invests in loans. Let $\kappa$ denote the deposit premium per unit of deposits. The bank's expected profit is:

$$\mathbb{E}[\Pi] = \int_{\hat{\delta}}^{1} \delta(1+i_l)\frac{(\sigma_2 e + \sigma_3 d)}{\sigma_2} - (1+i_d+\kappa)\frac{\sigma_3}{\sigma_2} d dH(\delta)$$

where the break-even point is now denoted by $\hat{\delta} = \frac{(1+i_d+\kappa)\sigma_3 d}{(1+i_l)(\sigma_2 e+\sigma_3 d)}$. As in the general model, the participation constraint of banks binds (Eq. 24 and Assumption 1), such that $\mathbb{E}[\Pi] = \mathbb{E}[\delta](1+i_l)e$.

Because banks are homogeneous and exposed to the same aggregate risk, all banks are either solvent or default at the same time. Consider the deposit insurer in case of bank default. In this case, the insurer pays out $(1+i_d)\sigma_3 d$ to depositors and receives the residual loan income of each bank $(1-\zeta)\delta(1+i_l)(\sigma_2 e+\sigma_3 d)$. In contrast, when banks are solvent, the insurer receives $\kappa\sigma_3 d$ in deposit premia. As banks default for $\delta < \hat{\delta}$, the expected profit of this scheme is:

$$\mathbb{E}[\pi] = \int_0^{\hat{\delta}} (1-\zeta)\delta(1+i_l)(\sigma_2 e+\sigma_3 d)dH(\delta) + (1-H(\hat{\delta}))\kappa\sigma_3 d - H(\hat{\delta})(1+i_d)\sigma_3 d = 0$$

which equals zero as the scheme is fair. Rewriting the zero-profit condition gives:

$$(1+i_d)\sigma_3 d = \int_0^{\hat{\delta}} (1-\zeta)\delta(1+i_l)(\sigma_2 e+\sigma_3 d)dH(\delta) + (1-H(\hat{\delta}))(1+i_d+\kappa)\sigma_3 d$$

Using the zero-profit condition, the expected utility of an unsophisticated idle household can be written as:

$$\begin{aligned} V^{0,i}(z) &= z - d + (1+i_d)d + W(0) \\ &= (1-H(\hat{\delta}))(1+i_d+\kappa)d + \int_0^{\hat{\delta}} \delta(1-\zeta)(1+i_l)\frac{(\sigma_2 e+\sigma_3 d)}{\sigma_3} dH(\delta) + z - d + W(0) \end{aligned}$$

The deposit contract maximizes the expected utility of the idle household subject to its liquidity constraint and the banker's participation constraint. Instead of solving for $1 + i_d$, it is useful to rewrite the problem using the break-even point:

$$\max_{d \in [0,z]} \left( (1 - H(\hat{\delta}))\hat{\delta} + (1 - \zeta) \int_0^{\hat{\delta}} \delta dH(\delta) \right) (1 + i_l) \frac{(\sigma_2 e + \sigma_3 d)}{\sigma_3} + z - d + W(0)$$

subject to:

$$\mathbb{E}[\delta]\sigma_2 e = \int_{\hat{\delta}}^{1} (\delta - \hat{\delta}) dH(\delta)(\sigma_2 e + \sigma_3 d)$$

This problem is equivalent to the problem solved in the proof of Lemma 2. Therefore, the costly state verification technology is equivalent to an economy in which the government operates a particular actuarially fair deposit insurance scheme and default destroys a share of the bank's loan income.